MICROBUBBLE ACOUSTIC CAVITATION AND SOFT MATTER



Ali Rezaei

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Akoestische microbel cavitatie in zachte materie

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CHAPTER 1

Introduction

The human ability to understand and engage with the environment, utilizing the senses of taste, vision, touch, smell, and hearing, plays a crucial role in shaping our responses to different situations. Our capacity to process and control sensory information allows us to navigate surroundings and react appropriately, such as wearing warm clothing when cold. Despite the remarkable evolution of human skills, some aspects of nature remain beyond our sensory reach. However, through tool-making, humans have overcome many of these limitations. Each tool addresses specific needs, collectively enhancing our adaptive capabilities.

The identification of diseases, as well as their causes and treatments, has always been a concern for humanity. In some cases, early diagnosis is essential, as conditions like blood vessel aneurysms can persist without clinical symptoms. Therefore, developing a real-time, affordable, safe, and robust imaging device becomes crucial. Centuries of scientific research have revealed the limitations of optical imaging in medical applications, leading to the exploration of alternatives such as magnetic resonance imaging (MRI) and X-ray imaging. However, these alternatives present drawbacks like a lack of real-time imaging, high costs, and potential health hazards.

1.1 Ultrasound contrast imaging

Ultrasound imaging has proven to be a powerful tool for deep tissue visualization. However, it faces limitations in imaging blood flow since red blood cells are weak scatterers. The advent of ultrasound contrast agents (UCAs) marks a significant advancement, introducing microbubbles (MBs) with radii ranging from 1 to 5 μm [1]. These MBs exhibit dramatically enhanced ultrasound scattering due to both the acoustic mismatch between blood and gas, and the compressibility of the gas core that enables volumetric oscillation. The most common type of ultrasound contrast agent (UCA) consists of phospholipid-coated microbubbles. A protective phospholipid shell prevents bubble dissolution, and the inclusion of a heavy, inert gas such as C_4F_{10} enhances stability within the bloodstream [2]. These bubbles can be produced by tip sonication, or in microfluidic chips, such as flow-focusing chips [3], [4].

The viscoelastic phospholipid shell strongly influences the dynamics of the microbubble. Numerous studies propose models that explain the impact of phospholipids on the bubble dynamics[5], [6]. When exposed to ultrasound pressure, these microbubbles undergo harmonic oscillations similar to a mass-spring system, resonating at medical ultrasound frequencies in the MHz range. Controlling the complete parameter regime of acoustic pressure and frequency allows for revealing, different bubble behaviors, which we classify into linear features like said harmonic oscillator and associated resonance behavior, and nonlinear phenomena such as higher harmonics, sub-harmonics, and jetting phenomena near surfaces [7], [8].

For ultrasound imaging, low pressures are employed to avoid the destruction of UCAs in the bloodstream. Upon exposure to ultrasound, the scattering cross section of microbubbles is up to nine orders of magnitude stronger than those of red blood cells, significantly enhancing the visibility of blood flow [9]. This technique aids in diagnosing various diseases, by monitoring the perfusion (or lack thereof) of organs, such as the heart, liver, kidney and the brain [10]. The small size of microbubbles allows them to travel through the microvasculature, and recording their acoustic responses helps the creation of detailed maps of microvascular structures in tissues [11], [12].

While UCAs have been available for forty years, ongoing research continues to enhance their efficacy and expand their potential field of use, for example, in therapeutic applications such as a local controlled release of drugs. Here, efficacy and control of the delivery mechanism may necessitate the use of monodisperse bubbles,

all of the same size, and where their natural frequency is tuned to the driving frequency. Currently, microbubble suspensions used in medicine are polydisperse: they consist of bubbles with varying sizes, which can lead to an inhomogeneous acoustic response. Monodisperse microbubbles, on the other hand, with a narrow size distribution obtained by filtration, centrifugation or sorting [13], as well as those produced in microfluidics by flow focusing, produce a more uniform backscattered signal for imaging and controlled mechanical action for the therapeutic application or intervention[14], [15]. For example, monodisperse UCAs can generate a more uniform shear stress on neighboring cells, which optimizes drug uptake in strategies relying on sonoporation and sonoprinting [16], [17]. They may also improve our capacity to exploit small (nonlinear) aspects of bubble scattering that have a specific link to quantities of diagnostic value, e.g. for local blood pressure sensing [18], [19].

1.2 Cavitation in viscoelastic materials

Cavitation used in a medical application can occur at a range of pressures. As an example, at high pressures it is used to break kidney stones. When the bubble collapses, next to mechanical action it also releases a shockwave, and the combined action can fracture the stone. At lower pressures, with milder oscillations, bubbles can be driven while preventing its dissolution. An example of this is the use of UCAs to visualize the blood pool. While these bubble dynamics have been extensively studied in water [20] and near rigid materials [21], one area in studying cavitation that is less investigated is the impact of the viscoelasticity of the surrounding medium, given the complex behavior of viscoelastic materials under shear stress. Factors such as the relaxation time, i.e. the time a material takes to respond to stress, as well as yield stress, i.e. the point beyond which tissue ruptures, play a significant role. Additionally, the nonlinear behavior of tissue and other variables contribute to the complexity of the system. This is a vast and complex subject that requires further exploration, especially when microbubbles oscillating in the MHz regime are the source of the stress since the behavior of the material at these strain rates is both largely unknown and challenging to measure.

Several studies have attempted to put together these puzzle pieces and have explored models to describe the viscous and elastic behavior of tissues, including the Kelvin-Voigt, Neo-Hookean, and Chasset-Thirion models [17]. However, these models have limitations and their validity is debatable, especially at high strain rates [22].

The Kelvin-Voigt model is a convenient framework frequently used to describe the deformation of solid-like materials in a linear viscoelastic regime. In this model, a parallel spring and dash-pot, or damper, represent the elastic and viscous behavior of the tissue, as shown in Figs. 1.1a and b. We utilize this model to introduce the effect of viscoelasticity on the behavior of microbubbles in a viscoelastic medium [23].

The dynamics of a spherically symmetric gas bubble in an incompressible and uniform medium are described by a Rayleigh-Plesset-type equation, Eq. 1.1 [24]. The equation incorporates the ideal gas law, the variable surface tension of the phospholipid microbubble, and the viscoelasticity of the surrounding medium:

$$\rho_L\left(R\ddot{R}+\frac{3}{2}\dot{R}^2\right)=\left(P_0+\frac{2\sigma(R_0)}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\gamma}-P_0-P_{\mathrm{ac}}- \frac{2\sigma(R)}{R}-\frac{4\kappa_s\dot{R}}{R^2}-\frac{4\mu\dot{R}}{R}-\frac{4G}{3}\left(\frac{R^3-R_0^3}{R^3}\right). \tag{1.1}$$

Here, ρ_L is the density of the surrounding medium. R represents the radius of the bubble, with $\dot{R}$ and $\ddot{R}$ the first and second derivatives of the radius with respect to time, respectively. R_0 is the radius of the bubble at rest. P_0 denotes the atmospheric pressure, and P_{ac} represents the acoustic driving pressure, γ is the polytropic exponent of the gas. $\sigma(R_0)$ is the size-dependent surface tension of a phospholipid-coated microbubble at rest. $\sigma(R)$ is the surface tension of a phospholipid-coated microbubble as described by Marmottant et al. [6], and κ_s represents the dilatational viscosity of the bubble shell [25]. The last two pressure terms represent the contribution of the viscoelasticity of the medium on the microbubble dynamics, described by the shear modulus G and the medium viscosity μ based on a linear Kelvin-Voigt viscoelastic description [22].

Figure 1.1a shows the schematic of a phospholipid-coated microbubble in such a viscoelastic medium. Upon exposure to ultrasound with a frequency close to resonance, this bubble will oscillate and apply a stress to the surrounding medium. Figure 1.1b shows the schematic of the linear Kelvin-Voigt model. We have solved Eq. 1.1 for a $R_0 = 2.0$ μm bubble driven at a frequency $f_0 = 1.8$ MHz with a pressure $P_{ac} = 100$ kPa in a media with G = 0 and 200 kPa. Figure 1.1c presents the resulting radius vs. time curves. The strongly reduced radial excursion for a bubble in a medium with G= 200 kPa is a direct consequence of medium elasticity by offering more mechanical resistance to bubble expansion (a lower gain of the system)

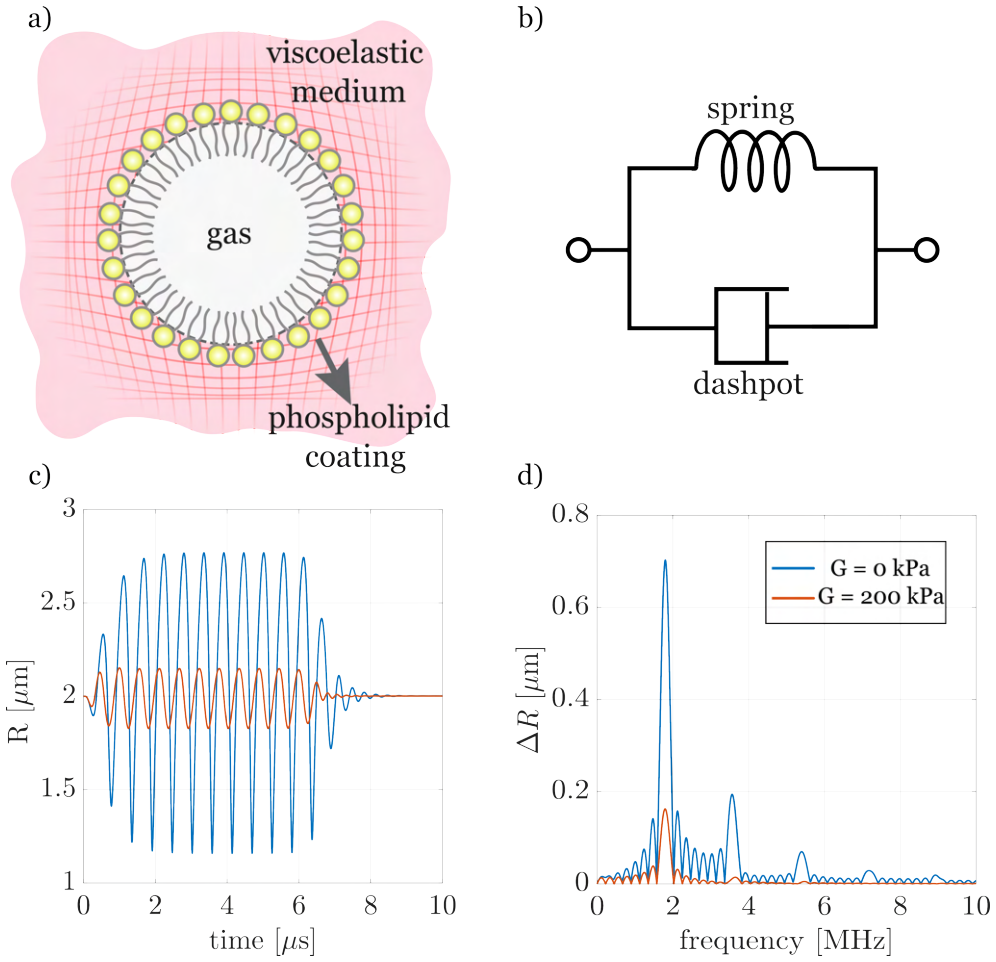


Figure 1.1: a) Schematic of a phospholipid-coated microbubble in a viscoelastic medium. Upon exposure to ultrasound, the bubble deforms the material. b) Schematic of the linear Kelvin-Voigt model. Elastic (spring) and viscous (dashpot) components are in a parallel configuration. c) Radius vs. time curves for the bubble with $R_0 = 2.0$ μm. The blue curve is for a coated microbubble with $\sigma(R_0)$= 20 mN/m that is driven at its resonance frequency (f_0= 1.8 MHz) by P_{ac} = 100 kPa in water G= 0 kPa. d) Frequency spectra of the graphs in panel c.

and indirectly by shifting the bubble's resonance frequency as compared to water. The frequency spectrum in Fig. 1.1d confirms that the oscillations are damped by the viscoelastic medium. They also show that the higher harmonics are weaker in the medium with a high shear modulus. As a result of the increased stiffness that leads

to a higher natural frequency of the bubble presenting a mismatch with the driving frequency, which then directly impacts the amplitude of oscillation and resonance behavior including the generation of harmonics.

Understanding the interplay between phospholipid coating and the viscoelasticity of the medium is essential to improve and enhance the efficacy of UCAs for medical applications because foremost the features such as resonance frequency are a function of both medium and bubble shell viscoelasticity. This knowledge helps to further optimize current UCAs and to develop faster, more robust, and effective methods for biomedical applications.

This thesis aims to explore the influence of viscoelasticity on microbubble dynamics. Through a series of experiments, we delve into the rheology of the surrounding viscoelastic gel using microbubbles and explore the impact of MHz strain rate on the elastic moduli of media. The interplay between the shell and viscoelastic medium will be investigated first through simulations using existing models for the dynamics of microbubbles and viscoelastic medium. We then investigate the scattered signal from microbubbles suspended in viscoelastic media with different shear moduli. The impact of microchannel confinement on the bubble dynamics such as resonance frequency will also be presented.

1.3 A GUIDE THROUGH THE THESIS

In this thesis, our objective is to evaluate the interaction between the phospholipid shell of the microbubbles and the viscoelastic medium. We conducted a series of experiments and numerical simulations to examine the response of phospholipid-coated contrast agents to the ultrasound field in viscoelastic media.

The second chapter focuses on measuring the rheology of a viscoelastic gel using microbubbles. We quantify the response of different elastic moduli of the surrounding medium under the strain exerted by the bubbles. In this chapter we compare our results and those obtained from conventional rotational rheometers, at low strain rates.

The third chapter is dedicated to investigating the interplay between the microbubble shell and the viscoelastic properties of the surrounding medium through simulations. We perform numerical simulations of the radial dynamics of microbubbles by solving a Rayleigh-Plesset-type equation for bubbles in a viscoelastic medium, and subsequently compute the scattered pressure using mass and momen-

tum conservation. Detailed discussions will cover the impact of the shell, medium viscoelasticity, and driving pressure on the complex interplay between the medium and the bubble dynamics. These simulations provide valuable insights to guide the experiments described in Ch. 4.

In the fourth chapter, our focus shifts to experimental measurements of the scattered signal from microbubbles, utilizing a calibrated multi-element transducer. This section will demonstrate the influence of the elastic modulus on the resonance frequency of the microbubbles.

In the fifth chapter, we investigate the acoustic cavitation of microbubbles within the confinement of microchannels. This investigation is particularly interesting, as microbubbles exhibit a distinctly different behavior as compared to bubbles in the free field. We explore the impact of parameters such as microchannel diameter and viscoelastic properties of the enclosing medium on the resonance behavior of the microbubbles.

Finally, in the sixth chapter, we present the conclusions and potential outcomes of this thesis. In the outlook section, we discuss the less studied aspects and highlight areas worth exploring to further enrich our understanding of the physical aspects of microbubble acoustic cavitation and its impact on biomedical applications.

CHAPTER 2

Rheology of viscoelastic media at MHz strain rates measured using ultrasound-driven microbubbles[1]

ABSTRACT

Ultrasound (US)-driven microbubbles play a crucial role in various biomedical applications, including contrast-enhanced US imaging, US-triggered drug delivery, and US-mediated therapies. The dynamics of these ultrasound contrast agents (UCAs) depend on their intrinsic nonlinear resonance behavior, as well as on the nonlinear viscoelastic properties of their stabilizing shells. In addition, microbubble dynamics are influenced by the viscoelastic properties of the surrounding medium which, for biomedical application, may be tissues. While the physical properties of soft tissues are widely available for strain rates in the range of 1 to 10 s^{-1}, they are largely unknown for strain rates of 10^6 s^{-1} at which the microbubbles are typically driven. Here, we make use of our knowledge of (nonlinear) bubble dynamics to characterize the rheology of polyacrylamide (PAM) hydrogels for strain rates exceeding 10^6

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s^{-1}. The narrow resonance curves of single coated microbubbles in a PAM hydrogel, obtained through high-speed imaging, were compared to a Rayleigh-Plesset-type model. This comparison yielded shear moduli estimates of 70 $\pm$ 20 kPa for soft (6.4%) and 270 $\pm$ 40 kPa for stiff (16%) PAM hydrogels. The shear viscosity was estimated to lie between 6 and 9 mPa.s. These results show a new approach to MHz frequency rheology of viscoelastic media, which is essential for a range of industrial and biomedical applications.

2.1 INTRODUCTION

Ultrasound contrast agents (UCAs) are widely used in medical imaging to enhance contrast and improve medical diagnosis, e.g. for organ perfusion imaging or to monitor hypervascularization in and around tumors [26], [27]. Additionally, UCAs are intensively investigated for their ultrasound-mediated therapeutic potential. Microbubbles can indeed increase the permeation of endothelial layers, resulting in the local delivery of drugs or genes through pore formation [28], [29].

Ultrasound contrast microbubbles have a typical size in the range of 1 to 5 μm in radius. They are filled with a low-solubility high-molecular weight gas, such as sulfur hexafluoride (SF_6), perfluorocarbon (C_3F_8, or C_4F_{10}), and are stabilized by a shell (most often phospholipid) that reduces the interfacial pressure of the microbubbles. Consequently, this shell hinders the diffusion of gas from the bubble core into the surrounding medium [8], which otherwise leads to rapid bubble dissolution.

Upon exposure to ultrasound, microbubbles undergo nonlinear volumetric oscillations which are key to their superior scattering cross-section. Furthermore, when the driving frequency is tuned to the eigenfrequency of the bubbles, they experience a mechanical resonance with larger bubble oscillations and higher scattering, as well as the generation of nonlinear harmonics, that thus further enhances contrast generation. This unique behavior is routinely exploited during clinical examinations, both in linear and nonlinear contrast imaging modes.

The ultrasound-driven response of UCAs is very well understood in the free field, including the effects of the phospholipid coating. These dynamics have been measured through both acoustic attenuation and high-speed imaging experiments [30]–[32]. The bubble behavior is well described by Rayleigh-Plesset-type models [8]. In most biomedical applications, however, bubbles are not oscillating in a free field environment and the presence of neighboring viscoelastic tissue has a crucial effect on their dynamics. This includes both their linear resonance behavior and their nonlinear dynamics [22]. Specifically, it has already been shown that elasticity of the medium can increase the resonance frequency of bubbles, while the additional viscous dissipation can lead to overdamped dynamics [33], [34]. It is thus essential to accurately describe the behavior of microbubbles in viscoelastic materials to control and optimize the therapeutic potential of these agents.

The oscillations of microbubbles in tissues have previously been modeled by adding a linear Kelvin-Voigt (K-V) viscoelastic model or a nonlinear Neo-Hookean

elastic model to the Rayleigh-Plesset-type equation [17]. Originally, Allen and Roy have developed a framework to compute the oscillations of bubbles in arbitrary linear [35] and nonlinear [23] viscoelastic media by solving the bubble dynamics equation in conjunction with a stress tensor analysis. More recently, Oratis et al. [36] have developed a unifying framework that allows for a straightforward numerical integration based on the relaxation function of linear viscoelastic media. One major issue that remains is that the use of any of these models requires a priori knowledge of the viscoelastic properties of tissue, which is generally not the case. More specifically, it is well known that the response of a viscoelastic material strongly depends on shear rate [37], [38]. However, classical rheology measurements can reach a shear rate up to 10^3 s^{-1}, which is 3 orders of magnitude lower than typically encountered at ultrasound driving frequencies [37].

Earlier experimental studies have employed a range of methods to investigate the rheology of viscoelastic materials at high strain rates [39], [40]. Bertin et al. [41] utilized atomic force microscopy to study the rheological properties of a medium by observing the displacement of a particle at strain rates of hundreds s^{-1}. Jamburidze et al. [33] measured the elastic moduli at a strain rate of 10^4 s^{-1} by fitting the linear oscillation amplitude and resonance frequency of 150 μm bubbles to the resonance curves extracted from high-speed recordings. These studies have offered deeper insights into medium rheology by pushing farther the boundary of measurable elastic moduli in terms of strain rates. However, none so far have reached strain rates of 10^6 s^{-1} [40], [42], which are relevant for biomedical ultrasound. Rheological measurement can be pushed to higher strain rates through the use of smaller microbubbles contained in the medium while being driven by ultrasound. However, such small microbubbles must be stabilized by a shell and the measured response is thus a combination of (known) shell rheology and medium rheology.

In this chapter, we use ultra-high-speed imaging of phospholipid-coated microbubbles to extract the shear moduli of a polyacrylamide hydrogel in the 10^6 s^{-1} range. Exploiting the narrowband resonance of microbubbles, we can use Rayleigh-Plesset type equation modified by a linear Kelvin-Voigt viscoelastic model to extract the local (in terms of strain rates) storage and loss moduli of the viscoelastic material in the range of 10^6 to 10^7 s^{-1}.

2.2 BUBBLE DYNAMICS MODEL

The Rayleigh-Plesset (RP) equation of motion for a spherically symmetric gas bubble in an incompressible and uniform medium, in the linear regime of deformation of a viscoelastic medium (traceless stress tensor) can be expressed as follows [35], [43]:

$$\rho_m\left(R\ddot{R}+\frac{3}{2}\dot{R}^2\right)=P_g(R)-P_\infty-\frac{2\sigma}{R}+3\int_R^\infty\frac{\tau_{rr}}{r}dr. \tag{2.1}$$

Here, R represents the radius of the bubble, and $\dot{R}$ and $\ddot{R}$ are the first and second derivatives of R with respect to time, respectively. ρ_m is the density of the medium and $P_g(R)$ is the pressure inside the bubble, and P_∞ is the pressure in the liquid far away from the bubble. σ represents the surface tension at the gas-liquid interface while τ_{rr} represents the radial normal stress component, and r is the radial distance from the center of the bubble [22], [44], [45].

The modeled system is represented in Fig. 2.1a. The pressure in the bubble, $P_g(R)$, can be expressed using a polytropic law [46]:

$$P_g(R)=\left(P_0+\frac{2\sigma}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\gamma}. \tag{2.2}$$

Here γ is the polytropic exponent of the gas. The vapor pressure is neglected as it is only a small fraction ($\sim 2\%$) of the total gas pressure. The bubbles considered are small compared to the ultrasound wavelength and the pressure in the medium far from the bubble wall, P_∞, consists of the atmospheric pressure, P_0, and the acoustic driving pressure, P_{ac}. The effect of the phospholipid coating is taken into account by adding a size-dependent surface tension $\sigma(R)$ following the Marmottant model [6], and a dissipation term ($4\kappa_s\dot{R}/R^2$) where κ_s is the dilatational viscosity of the shell [25].

For an incompressible, homogeneous, and isotropic viscoelastic medium with a small and linear deformation, the Kelvin-Voigt model gives the relation: $\tau_{rr} = 2(G\varepsilon_{rr}+\mu\dot{\varepsilon}_{rr})$. Here, G represents the shear modulus and $\varepsilon_{rr} = -2(R^3-R_0^3)/3r^3$ is the shear strain. μ and $\dot{\varepsilon}_{rr} = -2R^2\dot{R}/r^3$ are the shear viscosity and strain rates, respectively [47]. Substituting τ_{rr} in Eq. 2.1 leads to [22]:

$$\rho_m \left(R\ddot{R} + \frac{3}{2}\dot{R}^2 \right) = \left(P_0 + \frac{2\sigma(R_0)}{R_0} \right) \left(\frac{R_0}{R} \right)^{3\gamma} - P_0 - P_{ac} - \frac{2\sigma(R)}{R} - \frac{4\kappa_s \dot{R}}{R^2} - \frac{4\mu\dot{R}}{R} - \frac{4G}{3} \left(\frac{R^3 - R_0^3}{R^3} \right). \tag{2.3}$$

Note that a linear model was used to construct a term that appears to be nonlinear ($\sim 1/R^3$). This term is thus only valid if the small deformation condition is met, in other words, this model should only be used in the linear elastic regime of the medium.

The specific choice of a Kelvin-Voigt model may seem arbitrary. However, in the limit of a linearized model i.e. a first-order viscoelastic description, the contribution of the viscoelastic medium will always add two linear damping and elastic contributions, that are both represented in Eq. 2.3. Although the Kelvin-Voigt model may not be valid at all timescales (strain rates), it can be used to describe any arbitrary (linear) medium in an infinitesimal frequency band. This is the strength of the approach, which remains valid as long as the strain rate bandwidth is much narrower than the inverse of the relaxation timescale of the material.

The complex modulus is defined as $G^*(\omega) = G'(\omega) + iG''(\omega)$ and describes the entire viscoelastic behavior of the material. $G'(\omega)$ is the storage modulus which represents the solid-like (elastic) response of the material, $G''(\omega)$ is the loss modulus which represents liquid-like (viscous) properties of the material and ω is the angular frequency (strain rate). In the Kelvin-Voigt model, $G = G'$ and $G'' = \mu\omega$ [48].

In the linear bubble oscillation regime Eq. 2.3 can be linearized, which leads to an expression for the eigenfrequency of the bubble [8]:

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi R_0} \sqrt{\frac{1}{\rho_m} \left(3\gamma P_0 + (3\gamma - 1)\frac{2\sigma(R_0)}{R_0} + \frac{4\chi_{eff}}{R_0} + 4G \right)}, \tag{2.4}$$

with $\sigma(R_0)$ the surface tension of the bubble at rest and χ_{eff} the effective surface elasticity of the shell [31] This effective elasticity is finite for infinitesimally small oscillations, and gradually decreases to 0 for larger amplitudes of oscillation. For reference, the shell elasticity has a minor impact on the resonance frequency for amplitudes of oscillations above approximately 25% of the resting radius, but must still be considered to accurately extract the viscoelastic parameters of the medium. Linearizing Eq. 2.3 also provides an expression for the damping, which includes viscous dissipation at the interface and shell dissipation, which is dominant over the other

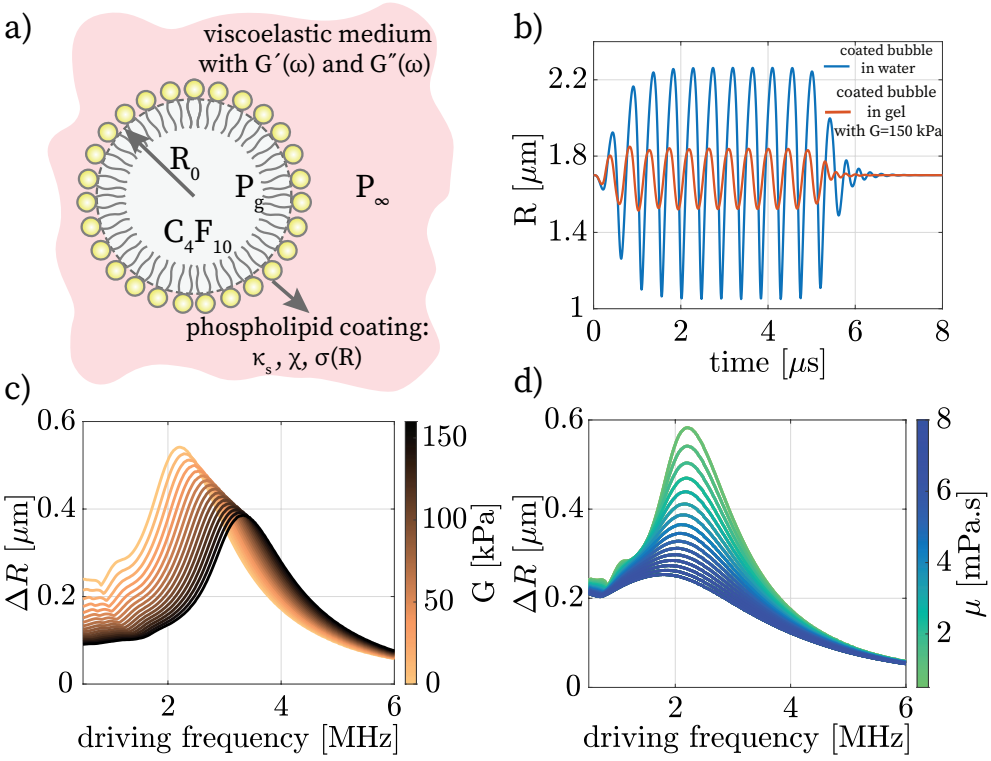


Figure 2.1: (a) Schematic and relevant parameters for a phospholipid-coated microbubble in a viscoelastic medium. (b) Radial response for a coated bubble in water ($\mu = 1$ mPa.s) and in a viscoelastic medium (G = 150 kPa and $\mu = 1$ mPa.s). (c) Resonance curves of a coated microbubble in media with a viscosity of $\mu = 1$ mPa.s and G ranging from 0 to 150 kPa. (d) Resonance curves of a coated microbubble for media with $G = 0$ Pa and μ ranging from 1 to 8 mPa.s.

damping terms, i.e. acoustic reradiation and thermal damping [49], [50].

As an illustration of the effect of viscoelasticity, Fig. 2.1b compares the solution of Eq. 2.3 for a coated bubble in water to that in hydrogel with a modulus of 150 kPa, which is the upper range that can be expected for soft tissue, at low strain rates [51], [52]. The bubble with $R_0 = 1.7$ μm is driven at a frequency of 2.2 MHz (its resonance frequency in water) with a 15-cycle, 100 kPa ultrasound pulse tapered with a 2-cycle hyperbolic tangent (tanh) envelope. One effect of medium elasticity is a decrease in the bubbles excursion, which is due to the increased medium stiffness both directly by offering more mechanical resistance to bubble expansion (a lower gain of the

system) and indirectly by shifting the bubble's resonance frequency as compared to water [17]. The resonance curves obtained by numerical integration of Eq. 2.3 are further presented in Figs. 2.1c and d for a varying shear modulus and viscosity, respectively, at an acoustic driving pressure of 100 kPa. They show the independent effect of the elastic modulus, affecting the resonance frequency, and of the viscosity, affecting the damping of the system, in the context of the Kelvin-Voigt model.

2.3 MATERIALS AND METHODS

2.3.1 Experimental procedure

Microbubble production

Phospholipid-coated microbubbles were produced using a microfluidic platform built in-house [15]. The machine contains a flow-focusing chip to generate highly reproducible mono-sized microbubbles. In the chip, a gas inflow is focused through a narrow orifice using two orthogonal liquid flows containing phospholipids, which generates bubbles through hydrodynamic instability [2]. The phospholipid formulation contained a 9:1 molar ratio of DSPC and DPPE-PEG 5000, with a concentration of 12.5 mg/mL in Isoton (Beckman Coulter, ISOTON II Diluent). The bubbles were filled with a gas mixture of 15 v% of C_4F_{10} gas in CO_2 as detailed by Segers et al. [53]. Bubble production rate were on the order of one million bubbles per second. Two bubble populations were produced with mean radii of 1.7 and 2.3 μm, and with a polydispersity index on the order of 5% [15].

Viscoelastic phantom preparations

The phospholipid-coated microbubbles were embedded in polyacrylamide (PAM) hydrogels. These hydrogels are commonly used for their viscoelastic properties that resemble those of soft tissues, at low strain rates [54], [55]. In addition, PAM presents multiple advantages. First, they have excellent optical transparency, which makes them suitable for optical characterization, e.g., through ultra-high-speed imaging. Second, their similarity to tissue extends to their acoustic properties, which makes the acoustic excitation of microbubbles straightforward with standard immersion transducers. Finally, their rheological properties can be tuned by adjusting the acrylamide concentration which allows to vary the stiffness of the hydrogel.

Samples are prepared using the recipe outlined in Table 2.1 [55]. When the monomer acrylamide (Acrylamide/bis 19:1, 40% W/v, SERVA) undergoes polymerization with APS (Ammonium Persulfate, ACS reagent $\geq$ 98.0%, Sigma-Aldrich), the polymer forms a cross-linked network, which increases the stiffness of the hydrogel. Microbubbles are added to the acrylamide solution before curing. The rise of the microbubbles to the surface due to buoyancy is prevented by stirring the solution until solidification begins. TEMED (N, N, N', N'-Tetramethylethylenediamine, ReagentPlus®, 99%) is added as a catalyst to speed up the reaction. Since the polymerization is exothermic, the concentration of TEMED was reduced as compared to standard recipes. In combination with a cold bath, this allows to keep the temperature below 35°C so as not to damage the coated microbubbles. The native acrylamide/bis solution was diluted with MilliQ water to create 6.4% and 16% PAM hydrogels, referred to as the soft and stiff hydrogels, respectively.

Table 2.1: Formulation for the preparation of a 10 mL polyacrylamide hydrogel.

Materials	Soft hydrogel (6.4% PAM)	Stiff hydrogel (16% PAM)
40% acrylamide (mL)	1.6	4
APS (mg)	14	14
TEMED (μL)	4	4
MilliQ water (mL)	8.4	6

The mechanical properties of these hydrogels at low strain rates ($< 16\ s^{-}1$) were evaluated using a rotational rheometer (MCR 502e, Anton-Paar) set-up in a parallel plate configuration. To ensure proper contact between the hydrogel and the platform, the samples were cured directly on the rheometer platform. The top parallel plate was lowered to form a gap of 0.5 mm. After curing, a frequency sweep test was performed with 5% strain to measure the storage and loss moduli.

Ultra-high-speed imaging experiments

The dynamics of acoustically driven phospholipid-coated microbubbles in a soft and stiff PAM hydrogel were captured using ultra-high-speed imaging (Shimadzu HPV-X2) at a frame rate of 10 million frames per second, see Fig. 2.2. A 12-cycle ultrasound pulse, tapered with a 2-cycle tanh envelop was transmitted from a single-element

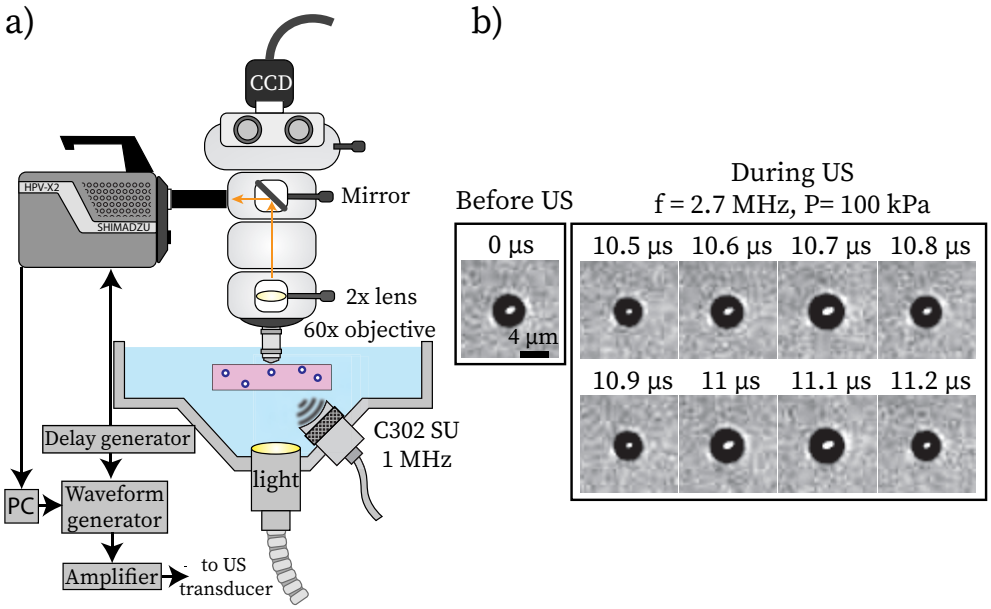


Figure 2.2: (a) Schematic of the experimental setup. (b) Snapshots of an ultra-high-speed recording of an oscillating microbubble with an initial radius of $R_0 = 1.7$ μm in a soft hydrogel with a shear modulus of $G(\omega \to 0) \approx 70$ kPa. The interframe time is 100 ns.

immersion transducer (2.25 MHz, C305 SU, Olympus) driven by a programmable arbitrary waveform generator (8026, Tabor Electronics Ltd.) connected to a high-power RF amplifier (VBA100-200, Vectawave). The transducer was calibrated using an optical fiber hydrophone (Precision Acoustics, UK) to maintain a constant pressure at the focus for each transmit frequency [56]. Fig. 2.2a shows the experimental setup and Fig. 2.2b shows snapshots of a typical high-speed recording.

The radius-time curves of oscillating microbubbles in the hydrogel were obtained by analyzing the high-speed recordings with a custom edge detection MATLAB® script [56]. The frequency content of the microbubble oscillations was determined by performing a fast Fourier transform (FFT) on the radius-time curves in MATLAB.

2.3.2 Numerical methods

Equation 2.3 was numerically integrated using the `ode45` solver in MATLAB®. Specifically, we have computed the radial dynamics of a microbubble with a radius

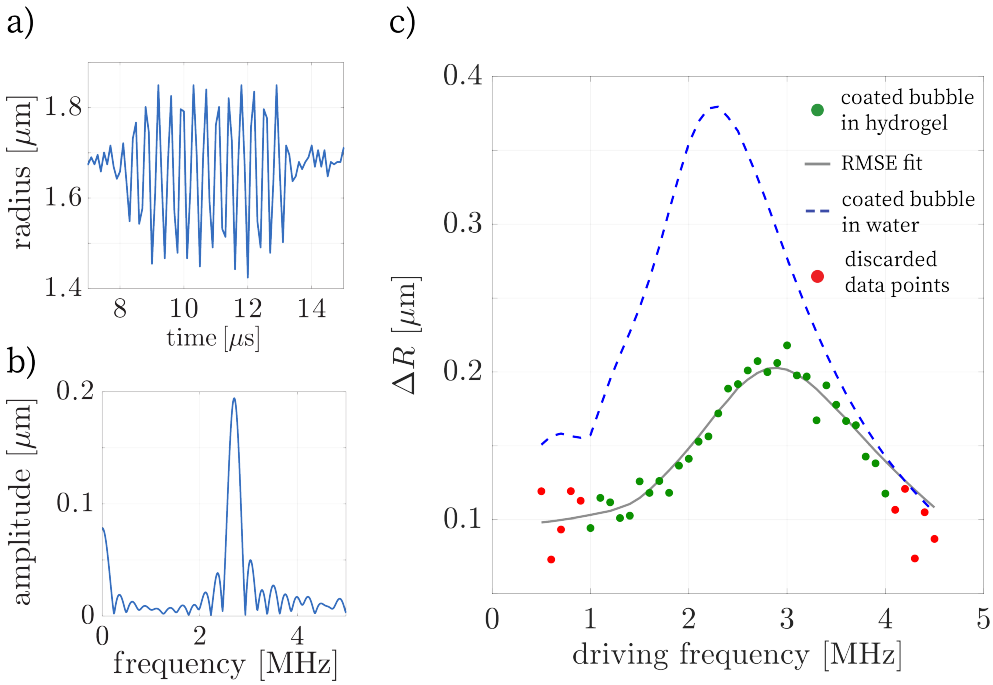


Figure 2.3: (a) Bubble radius versus time curve, extracted from a high-speed recording. The bubble was driven at a frequency of 2.7 MHz and at a pressure of 100 kPa. Panel (b) shows the Fast Fourier Transform of the data presented in (a). In (c), the resonance curve obtained from 41 high-speed recordings is shown with a best fit to the experimental data obtained through RMSE minimization. In this figure, the best-fit parameters were $G \approx 70$ kPa, $\mu = 5.6$ mPa.s, and $P_{ac} = 100$ kPa. The blue dotted line represents the simulated resonance curve for the very same bubble in water.

of 1.7 and 2.3 μm to match the experiments, in response to the 15-cycle ultrasound pulse. The amplitude and the frequency content of the bubble oscillations were determined using a Fast Fourier Transform (FFT) algorithm. The resonance curves are reconstructed using the amplitude of the fundamental response at each driving frequency while keeping the pressure constant. The model was solved for a range of acoustic pressures, shear moduli, and viscosities to generate an atlas of the bubble response. The specific parameters are provided in Table 2.2. This atlas was later used for fitting the experimental data. The best fit within this atlas was determined by finding the minimum of the Root Mean Squared Error (RMSE) between the set of simulations and the experimental resonance curves.

The local driving pressure P_{ac} is not known precisely in the hydrogel due to attenuation and reflections during transmission through the hydrogel and due to calibration uncertainty. The local driving pressure can change the resonance behavior of the microbubble, as the surface elasticity of the coating (that adds stiffness to the system) becomes less influential for higher pressures [57]. If the driving pressure would be higher than expected from the calibration, the microbubble resonance frequency will be underestimated, which in the fitting procedure would be compensated for by lowering G and μ. The pressure loss is also higher in the stiff hydrogel due to a larger acoustical impedance mismatch and thus larger reflection at the water-hydrogel interface. These sources of potential error were accounted for by also fitting the local acoustic pressure.

Table 2.2: Simulation parameters for creating the atlas of the bubble response.

Parameter	Soft hydrogel (6.4% PAM) (min:step:max)	Stiff hydrogel (16% PAM) (min:step:max)
Radius R_0 (μm)	1.7	2.3
Acoustic pressure P_{ac} (kPa)	30:5:120	30:5:120
Shear modulus G (kPa)	30:5:150	100:5:300
Viscosity μ (mPa.s)	3.0:0.1:10.0	3.0:0.1:12.0

The fitting was applied to the experiments performed in soft and stiff hydrogels. The initial radius of the microbubbles was taken from the bubble sizes determined in the experiment, R_0 =1.7 μm and R_0 =2.3 μm, in soft and stiff hydrogels, respectively. The other parameters are fixed: $P_0 = 10^5$ Pa, $\rho_m = 10^3$ kg/m^3, $\gamma = 1.07$ for C_4F_{10} and $\sigma(R >> R_0) = 72$ mN/m. The shell parameters for our typical microbubbles are $\kappa_s = 6 \times 10^{-9}$ kg/s, $\chi = 0.5$ N/m and $\sigma(R_0)$ = 20 mN/m [25], [57]. The fitting method was replicated for $R_0 \pm 5\%$, to estimate the sensitivity of the method to the size distribution and in optical sizing.

2.4 RESULTS

Fig. 2.3a shows an example of a typical R(t) curve. The corresponding frequency spectrum is shown in Fig. 2.3b. The obtained resonance curve constructed from 41 individual recordings of the very same bubble at different frequencies is plotted in Fig. 2.3c. Fig. 2.3c also shows an example of the best fit to the experimental data for

a bubble of radius R_0 = 1.7 μm in a soft hydrogel (gray line). The filled circles represent the experimental data. Red circles represent the 10% of data from each side of the resonance curve that were discarded due to the weak transducer response on the high- and low-frequency side of its characteristics, which results in a large uncertainty on the transmit pressure and in distorted waveforms. A calculated resonance curve for a 1.7 μm coated bubble in water is also shown (blue dashed line). In comparison to the same bubble in water, the resonance frequency of the bubble in the viscoelastic medium increases by 26% and the radial excursion amplitude decreases by 47%.

The values of shear modulus (G) and viscosity (μ) resulting from the RMSE fit for both hydrogels are plotted in Fig. 2.4. The figure comprises four panels, where G and μ are plotted as a function of the transmitted acoustic pressure for both soft and stiff hydrogel. In all panels of Fig. 2.4, the uncertainty in determining R_0 optically is depicted by the red and blue dashed lines, for a bubble 5% larger and 5% smaller than optically measured, respectively. Only the shear modulus of the soft hydrogel is significantly affected by this uncertainty, see Fig. 2.4a.

The results from the fitted pressures suggest a lower pressure compared to the reference calibration: 17% lower for the soft hydrogel and between 40% and 60% lower for the stiff hydrogel. The higher mismatch for stiffer hydrogel is expected. Literature on the acoustic characterization of PAM hydrogels suggests that the observed values for the local pressure are within the expected range [58].

To evaluate the importance of inaccuracies in the fitted acoustic pressure amplitude, the fitting procedure was repeated while imposing a pressure that was increased and decreased by 10% compared to the initial fit as this lies within the experimental calibration accuracy. These results are represented by the error bars in Fig. 2.4. This verification is important as the higher oscillation amplitude leads to a lower effective surface elasticity inducing, which also shifts the resonance frequency.

Figures 2.4a and b show that for the soft hydrogel, G increases as a function of pressure, although this shift is on the order of a 10% error on the acoustic pressure. For the stiff hydrogel, this relationship is not consistent and within the standard deviation of all data sets. It thus shows no obvious dependency on pressure. Although some minor effects of nonlinear viscoelasticity cannot be completely ignored, the weak dependency on pressure supports our linear viscoelastic approximation.

For direct comparison, these hydrogels were also measured at low shear rates (<16 s^{-1}) using a rotational rheometer. Figure 2.5 provides the result for a 5% strain

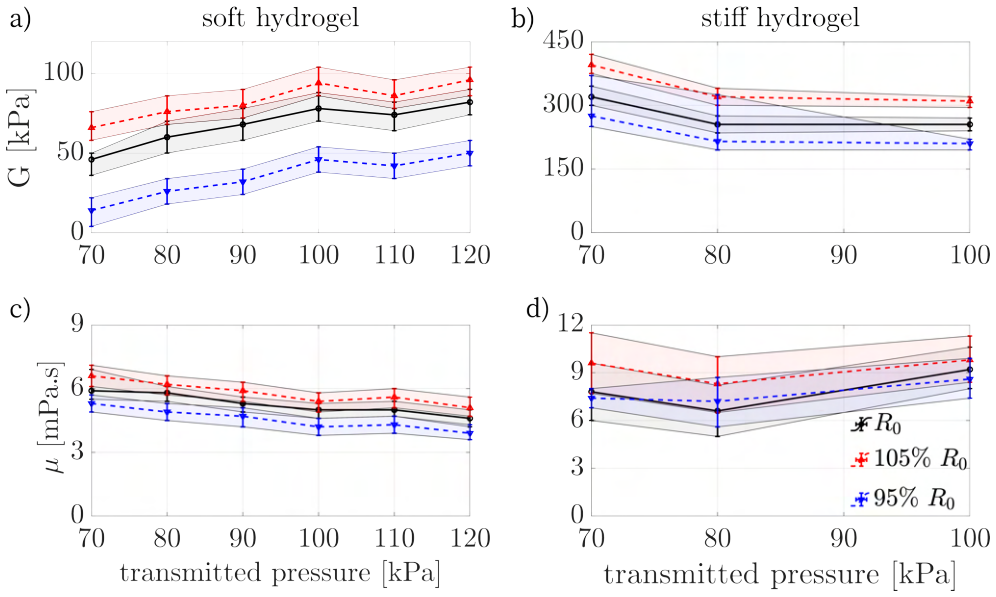


Figure 2.4: (a) and (b) Extracted values of G for the soft and stiff hydrogels, respectively, using RMSE minimization. (c) and (d), viscosity as a function of the driving pressure for both hydrogels. The error bars (and shadowed region) represent the results for the same bubble but for $\pm 10\%$ of the acoustic pressure to assess the effect of calibration uncertainties. Due to the sizing error of microbubble G and μ are also calculated for radii with $\pm 5\%$ change. These are shown by the red and blue dashed lines.

for the soft and stiff hydrogels. The mean values of the storage modulus are G'= 74 kPa and 144 kPa, respectively. The blue solid lines represent the expected G' behavior within a Kelvin-Voigt model. Finally, the red dashed lines represent the expected G'' for a viscosity equal to that of water which is the minimum loss modulus that we can expect.

Figure 2.5 shows that the values of G' in the 10^6 s^{-1} rate range are of the same order of magnitude as those obtained in the low-frequency range for these specific hydrogels. The horizontal error bars in this figure show the bandwidth of the bubble response, over which the elastic moduli have been estimated. The vertical error bars represent the deviations across the different driving pressures used in the experiments [33], [59].

Although G' does not show a frequency dependency, $G'' = \mu\omega$ is a function of frequency and is therefore 3 orders of magnitude larger at low-strain rates. At high

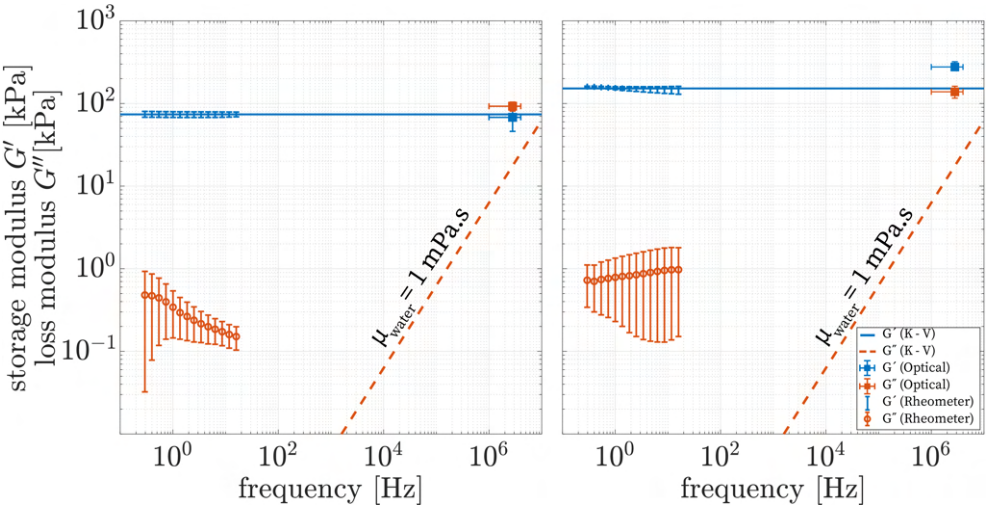


Figure 2.5: A comparison between the data obtained at low strain rates ($<16\ s^{-1}$) using a rotational rheometer and at high strain rates ($>10^6\ s^{-1}$) using ultra-high-speed imaging of the microbubble oscillations. The horizontal error bars of optical measurements suggest that the G' and G'' are measured across a range of frequencies by utilizing the resonance curves of the microbubbles. The vertical error bars originate from the standard deviation between all the measurements.

frequencies, G'' exhibits behavior similar to that of water.

2.5 Discussion and conclusion

A precise determination of the bubble size R_0 is important since it bears an intimate relation with the frequency of maximum response. If R_0 is underestimated, the resonance frequency and viscous damping terms will be overestimated. This will lead to an underestimation of G, as expected from Eq. 2.4, where $f_0 \sim \frac{\sqrt{G}}{R_0}$, and to an underestimation of μ since the damping (derived from Eq. 2.3) has the shape $\zeta \sim \frac{\kappa_s}{R_0^3}$. A similar observation can be made in Fig. 2.4 where the fit parameters are displayed for sizes of $1.05R_0$ and $0.95R_0$. This effect is stronger than that of the 10% uncertainty of the driving pressure.

The initial radius R_0 is defined as the radius of the bubble measured at the start of the experiments, assuming that the system is perfectly at rest and not pre-constrained. However, the resting radius of the bubble slightly decreases during the

measurement. There is thus stress that builds up on the bubble. This stress can be expressed as an additional pressure term $P_{stress} = \frac{4}{3}G\frac{R_0^3(t>0)-R_0^3(t=0)}{R_0^3(t>0)}$. For instance, in the soft hydrogel measurements, the initial radius was approximately 1.73 μm at the first measurement ($P_{ac} = 70$ kPa), whereas it was approximately 1.67 μm at the last measurement ($P_{ac} = 120$ kPa). While in principle it is easy to incorporate this concept into the simulations, it does require highly accurate absolute sizing data, which in our case was limited by the optical resolution of the high-speed imaging system [60].

Each bubble is exposed to 41 ultrasound pulses for each driving frequency, as shown in Fig. 2.3c. It is therefore possible that the composition of the bubble gas core changes over time as a result of diffusive processes, e.g., that C_4F_{10} is partly replaced by air, thereby changing the polytropic exponent γ, see Eq. 2.2. A 10% decrease in γ will yield a prediction error of about 7 kPa for G, based on Eq. 2.4, notably from the first term. However, this potential error is less than 10%, even in the soft hydrogel.

The measurement concept proposed here makes use of the resonant response of microbubbles. The resonance is essential to be able to accurately fit the data and extract the viscoelastic parameters of the hydrogel. Thus, this method cannot be applied to viscoelastic media that feature a strongly increasing loss modulus at high strain rates, like e.g. PDMS, since this leads to overdamped bubble dynamics [61]. Likewise, it is not a priori clear whether all tissues will allow resonant oscillations in the MHz range [36].

In this study, we have used optical recordings since these provide a direct comparison to developed models. Soft tissues, however, do not offer optical access. Instead, we may use the resonance curves obtained from acoustic attenuation or acoustic scattering of suspended monodisperse bubbles to study the viscoelastic contributions [32].

Using both experimental and numerical methods, we have successfully measured the storage and loss moduli of hydrogels at MHz strain rates employing phospholipid-coated microbubbles. The approach and findings of this work aid in the characterization of viscoelastic materials, which is necessary for the development and safety of new therapeutic applications making use of ultrasound and bubbles.

2.6 Acknowledgment

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CHAPTER 3

Modeling nonlinear scattering of phospholipid-coated microbubbles in elastic media[1]

ABSTRACT

Microbubble contrast agents are well-known in the biomedical field for their ability to enhance the contrast in ultrasound imaging and to facilitate targeted drug delivery within the body. Typically introduced into the vasculature, these bubbles are strongly influenced by the viscoelastic properties of surrounding soft tissues. In this chapter, we simulate the scattered pressure from a single phospholipid microbubble in a uniform viscoelastic medium. We modify the Rayleigh-Plesset equation to incorporate a viscoelastic model representing the behavior of the phospholipid shell, combined with a Kelvin-Voigt model to represent the effect of the viscoelastic properties of tissue on the bubble dynamics. Our simulations aim to address the complex relationship between the linear viscoelasticity of the medium and the phospholipid shell. Here, we investigate how this interplay influences the bubble dynamics as

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a function of driving pressure, bubble radius, and medium elastic modulus. Results show that the frequency of maximum scatter response increases with the elastic modulus of the medium. For example, the radial excursion of a bubble with a radius of 2.0 μm decreased by 45% in a medium with an elastic modulus of 200 kPa compared to water, while the scattered pressure increased by approximately 50%. In contrast, the change in amplitude of the scattered second-harmonic as a function of elastic modulus is negligible: simulations predict a increase of only 5% for a 2.0 μm bubble when increasing the medium elasticity from 0 to 200 kPa at a driving pressure of 120 kPa. Subharmonics generated at the resonance frequency (TR) are stronger or comparable to the subharmonics generated at twice the resonance frequency in the pressure range considered. This is in sharp contrast with the resonance behavior of a bubble in water. These results bear consequences for strategies that exploit fundamental, harmonic, and subharmonic bubble contrast.

3.1 INTRODUCTION

The healthcare sector is eagerly searching for imaging modalities that are both safe and effective for diagnostic and therapeutic purposes. Ultrasound (US) imaging has proven to be a reliable method for quantifying (microvascular) blood flow and is being extensively investigated for therapeutic applications in lithotripsy [62], histotripsy [63], blood-brain barrier opening [64], [65], or immunotherapy [66]. However, this modality often relies on the use of contrast agents to visualize the blood pool or to induce the therapeutic effects. In contrast-enhanced imaging, tracking contrast intensity or speckle movement offers spatio-temporally resolved information about organ perfusion [67]–[69], the state of stenotic lesions [70], aneurysm rupture risks [71], or tumor malignancy [72]. In therapy, bubbles are the main mediators for therapeutic action on tissues by exerting localized mechanical stress on cells [73].

Ultrasound contrast agents consist of microbubbles that typically have a radius of 1 to 5 μm. To prevent the rapid dissolution of the gas core, the bubble is encapsulated in a biocompatible protective shell, typically consisting of phospholipids. Furthermore, the bubbles are often filled with a heavy, low-solubility, and inert gas, such as C_4F_{10}, to prolong their circulation time [8]. When exposed to ultrasound in the MHz frequency range, these microbubbles undergo volumetric oscillations and exhibit an acoustic scattering cross-section that is nine orders of magnitude larger than that of the similarly sized red blood cells. Bubbles can be driven into resonance near their natural frequency, which further increases their scattering cross-section by one order of magnitude [8], [9].

The scattering strength of the microbubbles is key to their widespread use as ultrasound contrast agents. Furthermore, bubble dynamics are nonlinear, and the resulting harmonic components of the acoustic echo can be utilized to further enhance the contrast-to-tissue ratio in ultrasound imaging by suppressing the fundamental tissue echoes. These nonlinearities arise from both intrinsic bubble dynamics and extrinsic contributions, e.g., the (nonlinear) mechanics of the coating, and translate into the generation of second harmonics [57], ultra-harmonics [74], and subharmonics [75].

Since understanding and predicting these nonlinearities is a prerequisite for the effective use of bubbles, many studies have sought to model bubble dynamics, initially for the first-generation ultrasound contrast agents such as Albunex and Optison [76]–[78]. Following this pioneering work, the introduction of so-called second-

generation microbubbles coated with phospholipids led to new modeling efforts [5], [6], [78], with a particular focus on the viscoelastic behavior of the lipid shell. Notably, Marmottant et al. [6] developed an ad-hoc model to describe the surface tension of the lipid-coated microbubble as a function of surface area, where the bubble surface can cyclically buckle upon compression and rupture upon expansion. In the intermediate state, the microbubble coating behaves elastically and is characterized by a finite surface elasticity. This highly nonlinear response during bubble vibrations strongly affects the bubble dynamics and results in a pressure-dependent resonance behavior with a higher frequency for low driving pressure amplitudes, and bears resemblence of that of a free gas bubble at higher pressures [25], [57]. At resonance, this may also lead to an increased nonlinearity of the scattering [79].

The dynamics of microbubbles is also influenced by the mechanical properties of the surrounding medium. The presence of a soft, viscoelastic medium significantly affects both the linear and nonlinear behavior of the bubbles [22], [43]. Prosperetti et al. studied the linear and nonlinear oscillations of gas bubbles in various media by introducing a new pressure term to the modified Rayleigh-Plesset equation to account for the non-Newtonian behavior of liquids and for mass exchange at the bubble interface [43]. A series of analytical and numerical studies have explored cavitation in soft materials. For example, Allen and Roy et al. [35], [80] have incorporated viscoelastic constitutive equations into the bubble dynamics equations, highlighting the impact of both linear and nonlinear medium deformations on microbubble dynamics. Gaudron et al. [17] have investigated the effects of nonlinear elasticity on microbubble collapse using large deformation elastic models. More recently, Oratis et al. [36] have proposed a generalized model that accounts for arbitrary linear viscoelastic models using the relaxation function. Since these models quickly become quite complex to handle, a simplified model, such as the linear Kelvin-Voigt model, is often used as an appropriate constitutive model [81]. It predominantly applies to solid-like materials in a narrow excitation bandwidth and, by extension, to soft tissues since they generally recover their original configuration after the stress is removed.

While the aforementioned models have shown potential in predicting experimental outcomes and improving measurement accuracy, most studies on bubbles in viscoelastic materials have focused on examining the effects of therapeutic ultrasound where bubbles are driven at high pressures, e.g., for high-intensity focused ultrasound (HIFU) applications [82], [62]. Only a few studies have investigated the

imaging regime which uses much lower driving pressures [33], [83]. Our understanding of the nonlinear behavior of bubbles in viscoelastic media is thus still incomplete, and in particular the role of the combined effects of shell and medium viscoelasticity on the bubble dynamics and the consequences on the scattered ultrasound signals are still largely unexplored.

Based on the well-established understanding of phospholipid shell behavior in Newtonian liquids [25], this study examines the interplay between the shell and a Kelvin-Voigt viscoelastic medium using numerical simulations. We aim to explore the linear and nonlinear scattered pressure of a phospholipid-coated microbubble as a function of the driving pressure and frequency of the ultrasound wave and as a function of the elastic modulus of the surrounding medium. The insights gained here will be particularly valuable for the next chapter, which strives to measure the pressure scattered by microbubbles embedded in polyacrylamide hydrogels of varying stiffness.

3.2 THEORY

3.2.1 Bubble dynamics

In the linear regime of deformation of a Kelvin-Voigt viscoelastic medium, the modified Rayleigh-Plesset (RP) equation of motion for a spherically symmetric phospholipid-coated gas bubble can be expressed as follows (see also Ch. 2) [22]:

$$\rho_m\left(R\ddot{R}+\frac{3}{2}\dot{R}^2\right)=\left(P_0+\frac{2\sigma(R_0)}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\gamma}-P_0-P_{ac}-\frac{2\sigma(R)}{R}-\frac{4\kappa_s\dot{R}}{R^2}-\frac{4\mu\dot{R}}{R}-\frac{4G}{3}\left(\frac{R^3-R_0^3}{R^3}\right). \tag{3.1}$$

Here, ρ_m is the density of the medium, and R is the radius of the bubble. The single and double overdots represent the first and second time derivatives of R, respectively. The first right-hand-side term describes the gas pressure inside the bubble, where γ is the polytropic exponent of the gas and R_0 is the initial bubble radius. P_{ac} and P_0 are the acoustic driving and ambient pressures, respectively. The surface tension of the microbubble $\sigma(R)$ is size-dependent. Here, we choose to use a parametric representation of $\sigma(R)$ obtained from attenuation experiments [25]. Finally, κ_s is the dilatational shell viscosity and the last two terms of Eq. 3.1 describe the surrounding

medium which is represented by a Kelvin-Voigt model [22]. Its viscosity and elastic modulus are denoted by μ and G, respectively.

Linearization of Eq.3.1, assuming small oscillation amplitudes around the equilibrium radius, i.e., using $R = R_0(1 + x(t))$, where $x(t) << 1$, yields

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = -\frac{P_{ac}}{\rho_m R_0^2}\sin(\omega t), \tag{3.2}$$

the classic equation for a driven harmonic oscillator.

$$\omega_0 = 2\pi f_0 = \frac{1}{R_0}\sqrt{\frac{1}{\rho_m}\left(3\gamma P_0 + (3\gamma - 1)\frac{2\sigma(R_0)}{R_0} + \frac{4\chi}{R_0} + 4G\right)} \tag{3.3}$$

is the natural angular frequency of the bubble. $\sigma(R_0)$ is the surface tension of the bubble at rest and χ is the shell elasticity [31]. In Eq. 3.2, ζ is the damping ratio that contains dissipative terms including acoustic re-radiation, medium viscosity, thermodynamic losses, and shell viscous dissipation. In the following we will simply refer to ζ as the total damping that encompasses all these term whose expression we do not further specify.

In linearized form and using the complex notation (represented by the underline), the oscillation amplitude is given by:

$$\left|\frac{\underline{x}}{\underline{P_{ac}}}\right| = \frac{\frac{1}{\rho_m R_0^2}}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \left(2\zeta\omega\omega_0\right)^2}}, \tag{3.4}$$

where the relative oscillation amplitude $x = real(\underline{x})$. The resonance frequency ω_r can then be inferred from finding the frequency of maximum oscillation amplitude from the derivative of Eq. 3.4, which then writes:

$$\omega_r = \omega_0\sqrt{1 - 2\zeta^2}. \tag{3.5}$$

Inserting Eq. 3.5 into Eq. 3.4 leads to the maximum oscillation amplitude at resonance:

$$max\left(\left|\frac{\underline{x}}{\underline{P_{ac}}}\right|\right) = \frac{1}{2\rho_m R_0^2 \omega_0^2 \zeta\sqrt{1 - \zeta^2}}, \tag{3.6}$$

which decays quadratically with the natural frequency of the bubbles. In addition, shell and viscous damping ratios vary with ω_0^{-1} [31]. Using Eq. 3.3, considering

a small damping ratio ($\zeta \ll 1$), the maximum oscillation amplitude scales with the inverse of ω_0, and:

$$max\left(\left|\frac{x}{\underline{P_{ac}}}\right|\right) \propto \frac{1}{R_0}\frac{\sqrt{\rho_m}}{\sqrt{3\gamma P_0 + (3\gamma-1)\frac{2\sigma(R_0)}{R_0} + \frac{4\chi}{R_0} + 4G}}, \tag{3.7}$$

which then decreases with the square root of the elastic modulus.

3.2.2 Scattered pressure

The scattered pressure from a microbubble can be obtained from the momentum conservation equation in a weakly compressible medium [24]:

$$P_s\left(r, t + \frac{r}{c}\right) = \frac{\rho_m R}{r}\left(2\dot{R}^2 + R\ddot{R}\right), \tag{3.8}$$

where c is the speed of sound. Linearization then yields $P_s(r, t + \frac{r}{c}) \approx \frac{\rho_m R_0^3}{r}\ddot{x}$. In the context of harmonic oscillations:

$$P_s\left(r, t + \frac{r}{c}\right) = -\omega^2 \frac{\rho_m R_0^3}{r} x. \tag{3.9}$$

Using the behavior of x described by Eq. 3.4:

$$\left|\frac{P_s}{\underline{P_{ac}}}\right| = \frac{R_0}{r}\frac{\omega^2}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \left(2\zeta\omega\omega_0\right)^2}}, \tag{3.10}$$

Calculating the maximum of Eq. 3.10 provides the frequency of maximum response in terms of scattering $\omega_{m,s}$, which is notably different from the frequency of maximum response for the oscillation amplitude ω_r:

$$\omega_{m,s} = \frac{\omega_0}{\sqrt{1-2\zeta^2}} \tag{3.11}$$

Inserting Eq. 3.11 into Eq. 3.10 leads to the maximum scatter response:

$$max\left(\left|\frac{P_s(r)}{\underline{P_{ac}}}\right|\right) = \frac{R_0}{2r\zeta\sqrt{1-\zeta^2}}. \tag{3.12}$$

Using the same arguments as for the amplitude of oscillations, and in the limit of

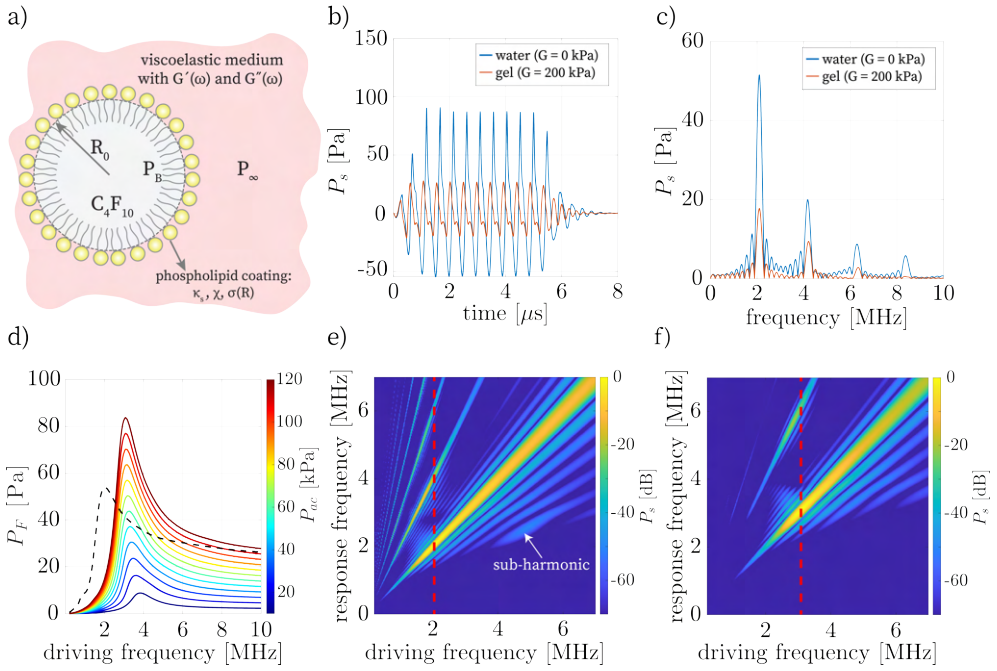


Figure 3.1: a) Schematics of the model: a phospholipid-coated microbubble is embedded in a viscoelastic medium characterized by an elastic modulus G and a viscosity μ. b) Pressure scattered by a bubble driven at 2 MHz for $R_0 = 2.0$ μm, $P_{ac} = 120$ kPa, and $G = 0$ (water) or 200 kPa (gel). c) Frequency spectra of the curves in b. d) Fundamental scattered pressure amplitude for $R_0 = 2.0$ μm and $G = 200$ kPa. The dashed line is the resonance curve for the same bubble in water with $P_{ac} = 120$ kPa. e) and f) Two-dimensional frequency spectral maps for $R_0 = 2.0$ μm, $P_{ac} = 120$ kPa, and $G = 0$ kPa (e) or 200 kPa (f). The red dashed lines marks the resonance frequency of the bubble.

small damping:

$$max\left(\left|\frac{P_s(r)}{P_{ac}}\right|\right) \propto \sqrt{3\gamma P_0 + (3\gamma - 1)\frac{2\sigma(R_0)}{R_0} + \frac{4\chi}{R_0} + 4G}. \tag{3.13}$$

Thus, in contrast to the maximum amplitude of oscillation, Eq. 3.7, that decreases with increasing medium stiffness, the maximum scattering response, Eq. 3.13 increases with medium stiffness with the square root of the elastic modulus.

Table 3.1: list of parameters and their values used to run the simulations.

parameter	name	value (min:step:max)	unit
R_0	initial bubble radius	(1.0:0.5:5.0)	μm
γ	polytropic exponent (gas)	1.07	
$\sigma(R_0)$	initial surface tension (shell)	20	mN/m
κ_s	dilatational viscosity (shell)	6×10^{-9}	kg/s
σ_w	surface tension (water)	72	mN/m
ρ_m	density (medium)	1000	kg/m^3
G	elastic modulus (medium)	(0:10:200)	kPa
μ	viscosity (medium)	2	mPa.s
P_0	atmospheric pressure	100	kPa
P_{ac}	driving pressure	(10:10:120)	kPa
f	driving frequency	(0.4:0.2:10.0)	MHz
N_{cy}	total number of cycles	30	
N_{env}	number of cycles tapering	10	
r	distance from bubble	1	cm

3.3 MATERIALS AND METHODS

A schematic of the modeled system is shown in Fig. 3.1a: a phospholipid-coated microbubble filled with C_4F_{10} is placed in an infinite viscoelastic medium with elastic modulus G and viscosity μ. The bubble is insonified by a pressure wave (parameters listed in Table 3.1), and the scattered pressure is evaluated at a distance of $r = 1$ cm.

3.3.1 Numerical procedures

We solve Eq. 3.1 in MATLAB® using the `ode45` ordinary differential equation solver for the parameter values presented in Table 3.1. The output, $R(t)$, and its derivatives $\dot{R}(t)$ and $\ddot{R}(t)$, are then fed into Eq. 3.8 to calculate the scattered pressure. In cases where `ode45` failed, the `ode15` solver was employed. Values and parameter ranges specified in Table 3.1 were chosen to align with the experiments conducted in Ch. 4. The driving acoustic wave is purely harmonic, 30 cycles in length, tapered with 10 cycles *tanh* envelope at both ends to avoid driving transient bubble oscillations. In the Kelvin-Voigt viscoelastic model, the elastic modulus and viscous contributions, G and μ, respectively, can be treated independently. This chapter focuses on the impact of the elastic modulus and the viscosity of the medium was therefore kept

constant at 2 mPa.s, which corresponds to the viscosity of water, adjusted by a factor of 2 to account for thermal damping[8]. Segers et al. have experimentally measured the surface area-dependent elasticity of the shell $\chi(A)$ from high-precision attenuation measurements on monodisperse microbubbles [25]. The surface tension curve was obtained by integrating the elasticity through $\sigma(A) = \int \frac{\chi(A)}{R} dA$, then averaging the measured $\sigma(A)$ curves by parametrization by a 10th degree polynomial fit in Matlab. Here we resample the surface tension curve to give $\sigma(R)$ that was then incorporated empirically into our simulations.

3.3.2 Data analysis

The data was analyzed using the Fast Fourier Transform (FFT) in MATLAB to extract the frequency spectrum of the bubble oscillations for each bubble size, driving frequency, driving pressure, and elastic modulus. The maximum scattered pressure amplitude at the fundamental frequency P_F around the driving frequency was then extracted from the FFT to create the fundamental resonance curves. The same strategy was used to extract the pressure amplitude P_{2H} of the pressure scattered at the second harmonic and P_{subH} of the pressure scattered at the subharmonic frequency.

Figure 3.1b shows the pressure $P_s(t)$ scattered by a microbubble with a radius R_0 = 2.0 μm, in response to a driving pulse (f = 2.1 MHz, P_{ac} = 120 kPa) and evaluated 1 cm away from the bubble. The bubble is either embedded in a medium with G = 0 kPa (water), or in a medium with G = 200 kPa. The frequency spectra of these two examples are shown in Fig. 3.1c. These spectra show the fundamental response at 2.1 MHz (at the driving frequency) and the higher harmonics near 4.2, 6.3, and 8.4 MHz.

Figure 3.1d shows the scatter response as a function of the driving frequency for the fundamental component in the spectrum of a bubble with R_0 = 2.0 μm in a medium with G = 200 kPa at a range of acoustic driving pressures. When driven at a driving pressure P_{ac} = 120 kPa, the resonance frequency in the elastic medium increases by 1.0 MHz compared to that in water (dashed black line). Fig. 3.1e and f show spectrograms of the response of the same bubble to a 120 kPa driving pulse in media with G = 0 kPa and G = 200 kPa, respectively. Each vertical line in the image is the spectrum of the bubble response to the ultrasound frequency on the horizontal axis. At this pressure range, the effect of the shell on the resonance frequency is minimal, and the changes observed can therefore be attributed to the elasticity

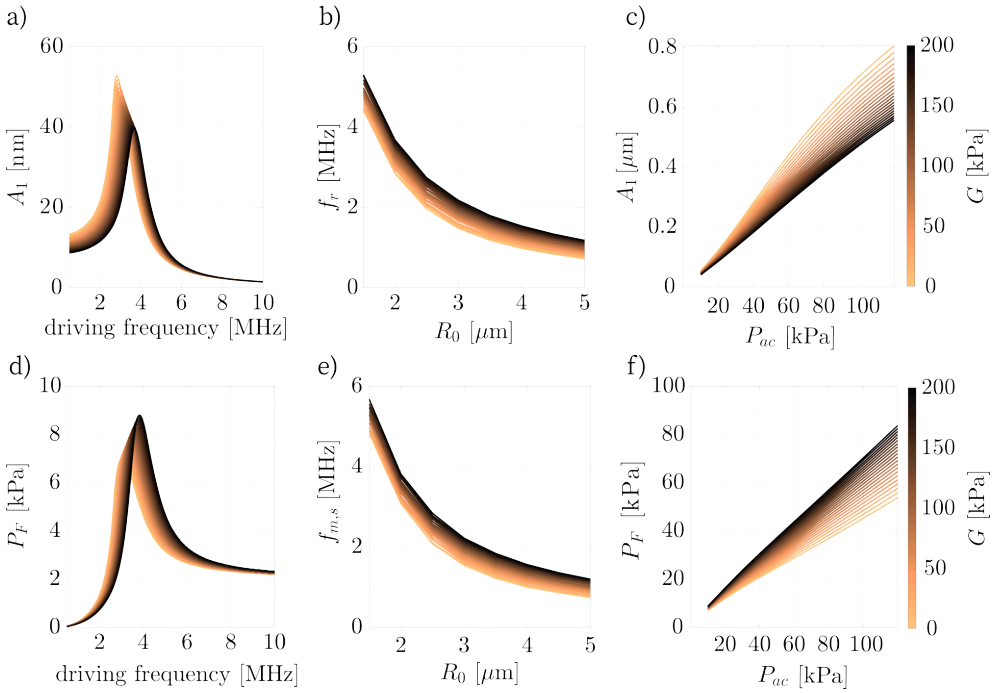


Figure 3.2: a) Amplitude of oscillation at the fundamental frequency A_1 as a function of driving frequency for $R_0 = 2.0$ μm and $P_{ac} = 10$ kPa. b) Resonance frequency f_r as a function of R_0 for varying elastic moduli G and $P_{ac} = 10$ kPa. c) Fundamental oscillation amplitude A_1 of $R(t)$ versus driving pressure P_{ac}. d) Fundamental scattered pressure amplitude P_F as a function of the driving frequency for $R_0 = 2.0$ μm and $P_{ac} = 10$ kPa. e) Frequency of maximum scatter as a function of R_0 for $P_{ac} = 10$ kPa and for varying elastic moduli G. f) Maximum scattered pressure as a function of driving pressure P_{ac} for various elastic moduli.

of the medium. With these acoustic parameter settings, the bubble oscillations are small enough to be consistent with the assumption of linear elasticity (see also the Discussion section) [17].

3.4 RESULTS

3.4.1 Fundamental response

Figure 3.2a shows the fundamental resonance curves in terms of oscillation amplitude for a $R_0 = 2.0$ μm microbubble embedded in media with elastic moduli ranging

from 0 to 200 kPa. The bubble is driven at a pressure of 10 kPa. The resonance curves are computed from solutions $R(t)$ of the modified Rayleigh-Plesset equation Eq. 3.1. It is observed that with increasing medium stiffness, the resonance frequency increases, while the oscillation amplitude decreases. This can be understood by plotting the resonance frequency f_r as a function of the resting radius of the bubble, Fig. 3.2b, also for a driving pressure of 10 kPa. It shows the combined effect of the inverse relationship between the eigenfrequency ω_0 and R_0, and that of the elastic modulus G that increases the eigenfrequency of the bubble, a feature that we predicted from the linearized Eqs. 3.3 and 3.5. While the resonance frequency increases with increasing G, Fig. 3.2c shows that increasing G leads to a lower oscillation amplitude, a direct consequence of the behavior expressed in the linearized equation Eq. 3.7.

For the scattered pressure the resonance behavior is different as can be inferred from Fig. 3.2d. Here the curves are computed by inserting the solutions $R(t)$ and their derivatives into Eq. 3.8. It is observed that with increasing medium stiffness, the frequency of maximum scatter response $f_{m,s}$ also increases, while other than the oscillation amplitude, the scattering increases. Figure 3.2e shows the frequency of maximum scatter response as a function of bubble size, which we understand, again from the linearized Eqs. 3.3, and from 3.11. Finally, plotting the maximum scatter response as a function of driving pressure gives Fig. 3.2f, which shows increasing scatter response with increasing medium stiffness, that nicely follows the trend predicted by the linearized Eq. 3.13.

In conclusion, and as expected from the linearized Eqs. 3.7 and 3.13, a higher medium stiffness leads to a higher resonance frequency and results in a lower amplitude of oscillations, but a larger scattered pressure.

Interplay between shell and elastic modulus

As already evidenced in Fig. 3.1d, for a higher driving pressure of 120 kPa, we observe very similar behavior. The resonance frequency shifts by 1.0 MHz towards higher frequencies when the elastic modulus is increased from 0 to 200 kPa and the maximum scattered pressure also increases with increasing elastic modulus. Notably, for an acoustic pressure of 120 kPa (Fig. 3.1b), a bubble in a medium with G = 200 kPa scatters 55% more compared to a medium with G = 0 kPa. However, this example of a bubble driven at a higher pressure entails a more complex interplay between the nonlinear viscoelastic shell and the medium stiffness, as a result of a

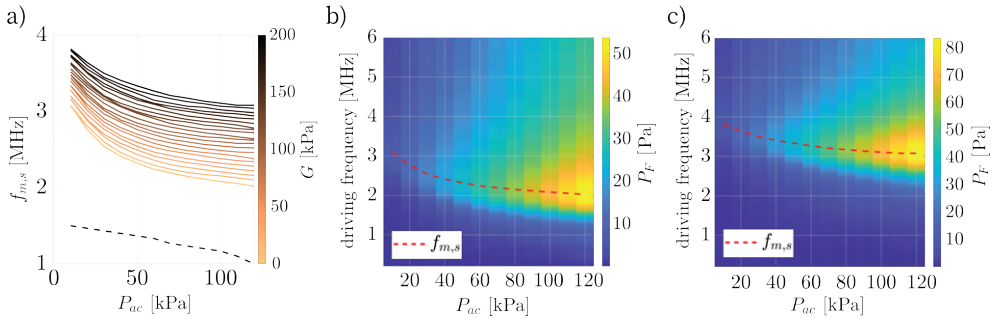


Figure 3.3: Interplay between phospholipid shell and viscoelastic medium for a bubble with $R_0 = 2.0$ μm. a) Resonance frequency as a function of driving pressure. The dashed black line corresponds to the resonance frequency of an uncoated bubble in free field. b) and c) 2D maps describing the amplitude of the scattered pressure as a function of driving frequency and driving pressure for $G = 0$ kPa and $G = 200$ kPa, respectively.

reduced effective shell stiffness where the bubble travels out of its elastic regime into the buckling and ruptured regimes [57]. The reduction in effective shell stiffness as the oscillation amplitude increases is further illustrated in Fig. 3.3.

Fig. 3.3a shows the frequency of maximum response as a function of driving pressure for a R_0 = 2.0 μm bubble for different elastic moduli. The bubble achieves its highest resonance frequency at the lowest pressure, where it oscillates purely in the elastic regime of the shell. The mediums elasticity also contributes to the overall stiffness of the system, and the effects of shell stiffness and medium elasticity are of similar magnitude for soft tissues. This is also evident from Fig. 3.3a, where the resonance frequency of an uncoated bubble (dotted line) is found to be at a very much lower frequency than for a coated bubble in water, which in turn is yet a fraction lower than the resonance frequency of a coated bubble in the elastic material with G = 200 kPa. At a driving pressure above 50 kPa, however, the effect of shell stiffness vanishes, and the primary factor contributing to the increase in resonance frequency is the elastic modulus G.

The spectrograms in Figs. 3.3b and c show the corresponding maximum scattered pressure amplitude as a function of driving frequency and pressure for a R_0 = 2.0 μm bubble in media with G = 0 kPa and G = 200 kPa, respectively. The frequency of maximum scattering for each driving pressure is indicated by the dashed line. The distinct difference between both maps highlight the combined effects of driving frequency, driving pressure, surface elasticity of the shell, and elasticity of

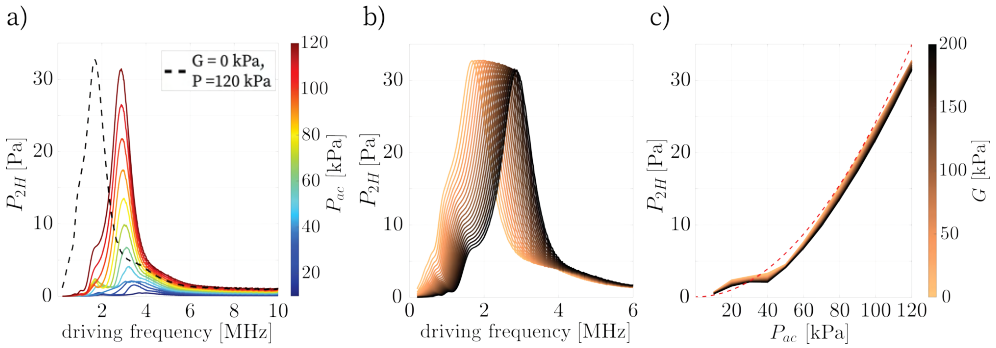


Figure 3.4: a) Maximum scattered pressure at the second harmonic for a bubble with $R_0 = 2.0$ μm as a function of driving frequency for different pressures in a medium with $G = 200$ kPa. The dashed line represents the resonance frequency of the same bubble in water at P_{ac}= 120 kPa. b) Maximum amplitude of the second harmonic for a bubble with $R_0 = 2.0$ μm in media with different elastic moduli (driven at P_{ac}= 120 kPa). c) Maximum amplitude of the second harmonic as a function of driving pressure for a $R_0 = 2.0$ μm bubble. The red dashed line is a guide to the eye showing a quadratic relation of the second harmonic scatter to the acoustic driving pressure

the surrounding medium on the scattered pressure.

3.4.2 Second-harmonic response

As evident from the ordinary differential equation Eq. 3.1, bubble oscillations are inherently nonlinear. Fig. 3.1c already showed the emergence of a second harmonic response, as well as that of higher-order harmonics.

Figure 3.4a, shows the maximum amplitude of the second harmonic for a $R_0 = 2.0$ μm bubble in a gel with a 200 kPa elastic modulus at different driving pressures. The amplitude of the pressure scattered at the second harmonic is highly dependent on pressure. Figure 3.4a also shows that the harmonics are not significant below a driving pressure of $P_{ac} \approx 40$ kPa. Figure 3.4b displays the scattered pressure amplitude at the second harmonic for different elastic moduli at a driving pressure of 120 kPa. The amplitude P_{2H} does not vary significantly with increasing elastic modulus, unlike the fundamental response. Specifically, the amplitude decreases by approximately 4% in a medium with $G = 200$ kPa compared to $G = 0$ kPa. Figure 3.4c shows the amplitude P_{2H} as a function of the driving pressure for increasing elastic moduli. It displays the expected nearly quadratic increase of the harmonic amplitudes above 40 kPa (indicated by the dotted guide to the eye) and it confirms the weak

dependency of P_{2H} on G.

3.4.3 Subharmonic response

Subharmonics can appear at two distinct locations in the frequency spectrum: they occur either when the bubble is insonified at its resonance frequency (the so-called TR subharmonic), or when it is driven at twice the resonance frequency, so-called T2R subharmonic [84], [85]. Figure 3.1e has already shown the appearance of a subharmonic response for a $R_0 = 2.0$ μm bubble driven at an acoustic pressure of 120 kPa in water. There, the T2R subharmonic is visible, but not the TR subharmonic that typically appears at pressures larger than 150 kPa, even for a free gas bubble.

Figure 3.5a shows the maximum amplitude of the subharmonic response, P_{subH}, for a $R_0 = 2.0$ μm bubble suspended in water for acoustic pressures ranging from 10 to 120 kPa. At low pressures, the subharmonic scattering is weak. However, when driven at higher pressures, e.g., $P_{ac} > 50$ kPa, the T2R subharmonic rises sharply. Like in Fig. 3.1e, the T2R subharmonics appear around 4.8 MHz, while the resonance frequency of the bubble is 2.0 MHz (see e.g., Fig. 3.2b. This mismatch is expected and has been repeatedly discussed before [86], [87]. Throughout the observed pressure range, the TR subharmonics remains low, below 0.5 Pa.

Figure 3.5b shows the same information as Fig. 3.5a, for the very same bubble suspended in a medium with $G = 200$ kPa. While, at low driving pressure amplitude, Figs. 3.5a and b show comparable subharmonic responses, the T2R subharmonic rises more slowly as a function of the driving pressure in the elastic material. Furthermore, while the TR subharmonic was not visible for the bubble immersed in water, it now reaches an amplitude comparable to that of the T2R subharmonic.

The subharmonic scattered pressure generated by the bubble in the elastic material is approximately half that generated by the bubble in water. To quantify TR and T2R subharmonic behavior, we have plotted the ratio between TR to T2R ratio as a function of G for all driving pressures in Fig. 3.5c. At pressures below 50 kPa, the TR subharmonic pressure amplitude is comparable to T2R subharmonic pressure amplitude, as shown before. However, at pressures above 50 kPa, the ratio increases monotonically as a function of G.

Figure 3.5d shows the maximum subharmonic scattering amplitude as a function of the driving frequency for a $R_0 = 2.0$ μm microbubble driven at 120 kPa, and for various elastic moduli. Mirroring the fundamental scattering, the TR subharmonic

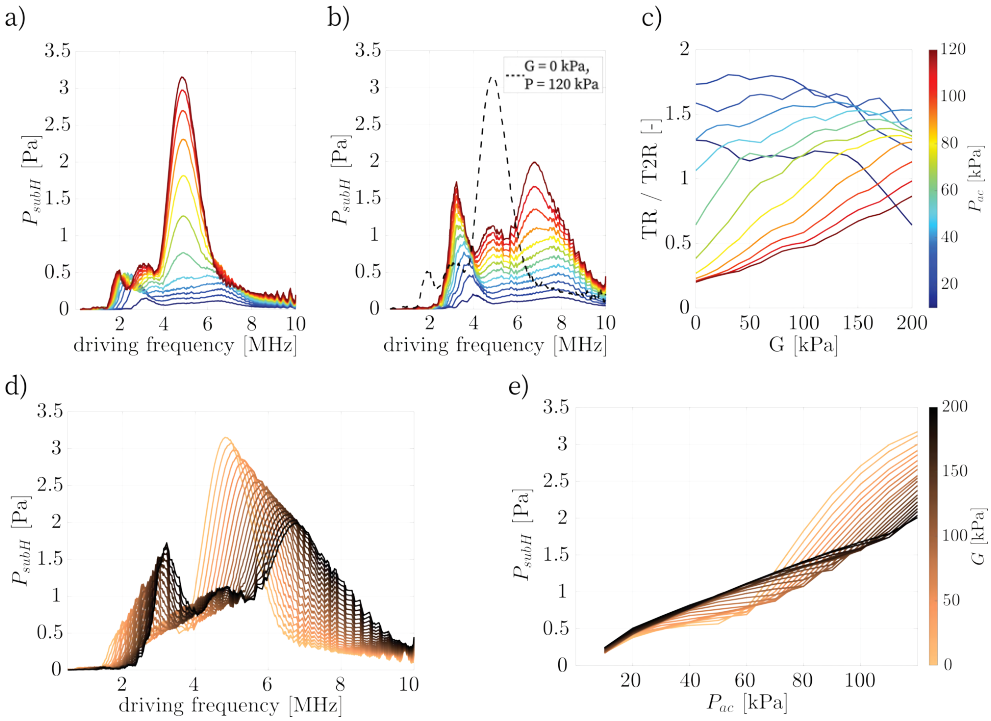


Figure 3.5: Maximum amplitude of the subharmonic for a bubble with $R_0 = 2.0$ μm as a function of driving frequency in media with a) $G = 0$ kPa and b) $G = 200$ kPa. c) Ratio of the TR to T2R subharmonic amplitudes. d) Amplitude of the subharmonic response as a function of the frequency in media with elastic moduli ranging from 0 to 200 kPa, for a bubble with $R_0 = 2.0$ μm driven at 120 kPa. e) Maximum amplitude of P_{subH} as a function of driving pressure.

response increases with increasing medium stiffness until becoming comparable to the T2R subharmonic for $G = 200$ kPa. Interestingly, the T2R subharmonic amplitude decreases by about 35% when the elastic modulus increases from 0 to 200 kPa, in sharp contrast with both the fundamental and the TR subharmonic scattering.

The maximum amplitude of subharmonic scattering as a function of driving pressure is plotted in Fig. 3.5e. It shows a clear (and well-known) threshold for the emergence of subharmonics visible for $G = 0$ kPa [50]. In the elastic material we do not observe a clear threshold but rather a transition going from a TR subharmonic-dominated regime towards one where ultimately the T2R subharmonic grows faster.

3.5 Discussion

For simplicity, we used a Kelvin-Voigt linear viscoelastic model, which is valid for narrowband excitation, and as long as the medium remains within a linear deformation regime. Gaudron et al. [17] compared the linear Kelvin-Voigt model with non-linear models, e.g., Neo-Hookean, showing minimal deviations between these models when bubble expansion reaches $(R_{max} - R_0)/R_0 = \Delta R/R_0 = 0.5$ and compression up to $\Delta R/R_0 = -0.3$. Since, in our study the relative bubble expansion is $\Delta R/R_0 = 0.4$ at maximum, we conclude that the Kelvin-Voigt model is a valid approach. Nonlinear scattering in viscoelastic media at larger amplitudes is, however, an interesting avenue to explore.

The calculations presented in this chapter focus on a single microbubble. However, medical applications typically require a suspension of bubbles, which can never be perfectly monodisperse. A statistical analysis based on typical, or a range of size distributions of bubbles would provide insight into the nonlinear scattering of a suspension as compared to that of a single bubble, and thereby shed more light on the practical relevance of the results reported in this chapter.

In Ch. 2, we took an inverse approach, using microbubbles as probes to measure the elasticity of a tissue phantom, though this was constrained by optical limitations. These results, especially if combined with the statistical approach described above, pave the road towards the viscoelastic characterization of opaque tissues in the MHz range through acoustic scattering. The strong dependence of subharmonic scattering to the medium elasticity would furthermore allow for using nonlinear imaging, possibly using the TR to T2R ratio in order to measure the tissue elasticity with a high signal-to-background, or contrast-to-tissue, ratio.

3.6 Conclusion

In conclusion, the scattering behavior of microbubbles is significantly influenced by the stiffness of the surrounding medium. Stiffer media result in higher scattered pressures at the fundamental frequency. For a 2.0 μm bubble, this increase is calculated to be 50% when the medium elasticity rises from 0 to 200 kPa under a driving pressure of 120 kPa. The shift in resonance frequency induced by the shell and soft tissues is of the same order; however, the effect of the shell diminishes at driving pressures above 70 kPa, while the influence of medium elasticity persists. Under the

same conditions, the second harmonic amplitude changes by 5% which is relatively small. Subharmonic scattering spectra are heavily affected by the presence of a viscoelastic medium. Interestingly, while TR subharmonics become stronger in stiffer media, the T2R subharmonic response decreases with increasing elastic modulus, highlighting the complex influence of the surrounding medium.

3.7 Acknowledgment

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CHAPTER 4

Ultrasound contrast microbubble scattering in viscoelastic media [1]

ABSTRACT

Ultrasound contrast microbubbles are known for their ability to enhance contrast in ultrasound imaging and for offering new avenues for localized drug delivery. These microbubbles exhibit a volumetric resonance in response to ultrasound excitation, and within viscoelastic media such as perfused tissues, this resonance depends on the viscoelastic properties of the surrounding medium. In this study, we present a microbubble scattering spectroscopy method to investigate the behavior of phospholipid-coated microbubbles in polyacrylamide hydrogels with 3 different elastic moduli. Our results show that the resonance frequency of 2.7 μm radius coated bubbles increases by 42%, 77%, and 201% compared to the bubbles in free field, when measured in hydrogels with storage moduli of 7, 40, and 110 kPa, respectively. The elasticity of the hydrogels measured at MHz frequencies using microbubbles was significantly higher compared to values obtained using a rotational rheometer at low strain rates. Furthermore, we also show that bubble stability in re-

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sponse to ultrasound driving is greatly enhanced in stiffer hydrogels. These findings provide insights into microbubble dynamics within viscoelastic media, contributing to the optimization of contrast-based imaging and therapeutic ultrasound modalities.

4.1 INTRODUCTION

Ultrasound contrast microbubbles are part of the daily clinical routine in medical imaging [88] and diagnostics for their ability to enhance blood-to-tissue contrast. The key to their unique scattering capabilities lies in their compressible gas core, and in a resonant behavior that grants them a scattering cross-section that is up to nine orders of magnitude larger than that of red blood cells [9]. These microbubbles, therefore, facilitate perfusion quantification of organs, such as heart, liver, kidney and the brain [67], [88], microvascular perfusion imaging of tumors [89], ultrasound molecular imaging [90] ultrafast ultrasound imaging for blood flow quantification [69], and super-resolution ultrasound imaging [91].

Contrast microbubble agents are typically 1-5 μm in diameter, and are stabilized by a biocompatible phospholipid monolayer. They also contain a heavy inert gas core with low solubility to maximize their stability [53]. These phospholipid shells have been shown to have an important effect on bubble dynamics and therefore on the scattered pressure and contrast generation. More specifically, phospholipid shells increase the resonance frequency of bubbles at low driving pressures [57], [92], increase bubble nonlinear responses used in contrast-specific imaging schemes [88], and dominate the overall damping of the bubble dynamics. Characterizing the response of contrast bubbles is therefore crucial for optimizing their performance in medical applications [26], [67].

These bubbles have been modeled extensively [8], providing detailed insights into the linear and nonlinear bubble dynamics and the impact of viscoelastic shells. Similarly, a range of experimental techniques have been developed to measure coated microbubble dynamics, each with its specific strength. High-speed optical imaging was used to provide direct time-resolved visualization of the radial response of single microbubbles [20] but is difficult to setup in the free field [93], [94]. Acoustic attenuation measurements [95] allow for the measurement of the ensemble response of a cloud of microbubbles. It thus offers high SNR measurements of free bubble suspensions, but an indirect measure of the underlying bubble dynamics. High-frequency acoustic scattering measurements [92], [96]–[98] provide direct information on the radial dynamics of single, freely-floating bubbles, but the measurement requires careful calibration and provide a low signal-to-noise ratio (SNR).

The bubble response to ultrasound also depends strongly on the physical prop-

erties of the surrounding medium, as well as on vessel geometry and confinement [22]. This has been demonstrated through a number of analytical and numerical studies. Prosperetti et al. [43] has developed a theory elucidating bubble behavior in non-Newtonian liquids. Others, including Allen and Roy [35], [80], have integrated viscoelastic constitutive equations into the bubble dynamics equations, clarifying the role of linear and nonlinear elasticity. Gaudron et al. [17] have investigated the effects of nonlinear elasticity on microbubble collapse, employing large deformation elastic models. Recently, Oratis et al. [36] have proposed a unifying approach to modeling bubble dynamics in linear viscoelastic media based on relaxation functions.

Experimental techniques have been developed to study the interaction between cavitation bubbles and viscoelastic media [99]–[102]. For instance, Wilson et al. [103] have conducted a comprehensive examination of the collapse of laser-generated and acoustically-generated bubbles in gels with varying stiffness, utilizing a high-speed camera method to capture bubble collapse dynamics as a function of stiffness. Non-inertial bubble oscillations in viscoelastic materials, most relevant for ultrasound imaging, have been explored using high-speed cameras by Jamburidze et al. [33], who suspended 100–200 μm uncoated air bubbles in agar gels. They subsequently fitted their experimental results to a Rayleigh-Plesset-type model that incorporates a linear Kelvin-Voigt viscoelastic component. However, experimental studies on clinically relevant bubbles in viscoelastic materials, especially at relevant pressures and frequencies, remain scarce.

In this study, we use acoustic scattering experiments to investigate the behavior of microbubbles suspended within viscoelastic polyacrylamide hydrogels. These measurements provide a direct measure of the microbubble activity without hindrance from boundaries. Furthermore, we retain a high SNR by measuring a suspension of monodisperse microbubbles. We quantify the pressure scattered by microbubbles using a transducer array controlled by an ultrasound research scanner to perform spectroscopic measurements. From these measurements, we quantify the resonance behavior of the microbubble suspension, including resonance frequency and damping, and assess the stability of the suspensions in hydrogels with 3 different stiffness in the range 10 to 100 kPa.

4.2 THEORY

4.2.1 Bubble oscillations

In the limit of a narrow-frequency band, most viscoelastic materials can be locally approximated by a Kelvin-Voigt model. We therefore use the Kelvin-Voigt description as a first approximation and denote the viscosity and elastic modulus of the surrounding medium by μ and G [22], respectively (for more info refer to Ch. 2). In the linear regime of deformation of a viscoelastic medium, the Rayleigh- Plesset (RP) equation of motion for a spherically symmetric gas bubble in an incompressible and uniform Kelvin-Voigt elastic medium can be expressed as [22]:

$$\rho_m\left(R\ddot{R}+\frac{3}{2}\dot{R}^2\right)=\left(P_0+\frac{2\sigma(R_0)}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\gamma}-P_0-P_{ac}- \frac{2\sigma(R)}{R}-\frac{4\kappa_s\dot{R}}{R^2}-\frac{4\mu\dot{R}}{R}-\frac{4G}{3}\left(\frac{R^3-R_0^3}{R^3}\right). \tag{4.1}$$

Equation 4.1 accounts for the polytropic behavior of the gas core, the size-dependent surface tension and shell viscous effects caused by the phospholipid shell [6], and the properties of the viscoelastic medium. ρ_m is the density of the surrounding medium. The bubble radius is denoted by R, with $\dot{R}$ and $\ddot{R}$ its first and second derivatives with respect to time. R_0 is the radius of the bubble at rest. P_{ac} is the acoustic driving pressure, and P_0 is the ambient pressure. γ is the polytropic exponent of the gas. The effective surface tension of the phospholipid coating is represented by $\sigma(R)$ [6], where the initial surface tension is denoted by $\sigma(R_0)$. The dilatational viscosity of the shell is represented by κ_s.

Assuming small oscillation amplitudes around the equilibrium radius $R = R_0(1+x(t))$, where $x(t) << 1$, and neglecting higher-order terms, yields the classic equation of a driven harmonic oscillator: $\ddot{x}+2\zeta\omega_0\dot{x}+\omega_0^2x=-\frac{P_{ac}}{\rho_m R_0^2}\sin(\omega t)$ where $\omega_0=2\pi f_0$ is the natural angular frequency and ζ is the total damping. The resulting expression for the natural frequency is:

$$f_0=\frac{\omega_0}{2\pi}=\frac{1}{2\pi R_0}\sqrt{\frac{1}{\rho_m}\left(3\gamma P_0+(3\gamma-1)\frac{2\sigma(R_0)}{R_0}+\frac{4\chi}{R_0}+4G\right)}, \tag{4.2}$$

with χ the shell elasticity [31]. The damping ratio is:

$$\zeta = \frac{1}{2\rho_m \omega_0 R_0^2}\left(\frac{4\kappa_s}{R_0} + 4\mu\right). \tag{4.3}$$

Note that the damping contributions from acoustic re-radiation and thermodynamic losses are not accounted for in Eq. 4.1. Using the complex notation where $x = Re(\underline{x})$ The amplitude of oscillation $|\underline{x}|$ of the microbubble as a function of frequency obeys the relation:

$$|\underline{x}| = \frac{\frac{P_{ac}}{\rho_m R_0^2}}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \left(2\zeta\omega\omega_0\right)^2}}. \tag{4.4}$$

4.2.2 Scattered pressure

The scattered pressure from an oscillating microbubble can be obtained using the momentum conservation equation in a weakly compressible medium [24]:

$$P_{s,RP}\left(r, t + \frac{r}{c}\right) = \frac{\rho_m R(t)}{r}\left(2\dot{R}^2 + R\ddot{R}\right), \tag{4.5}$$

with r the distance from the bubble. Note that this expression neglects the rapidly vanishing near-field term which is negligible at the transducer surface. Using the complex notation $P_{s,RP} = Re(\underline{P_{RP}})$ and linearizing in the context of harmonic oscillations yields:

$$\left|\underline{P_{RP}}(r)\right| = \frac{\rho_m R_0^3}{r}\omega^2 \left|\underline{x}\right|, \tag{4.6}$$

with $|\underline{x}|$ from Eq. 4.4. A more comprehensive derivation on the resonance behavior and the role of the associated damping can be found in Sec. 3.2. Sample simulations of the resonance behavior can also be found in the Results section of Ch. 3.

4.3 Materials and Methods

4.3.1 Monodisperse bubble production

Monodisperse bubble populations were produced as described in [15]. In short, monodisperse phospholipid microbubble suspensions are prepared using an in-house-built machine. This system utilizes a microfluidic flow-focusing chip to gen-

erate microbubbles of a target size [15]. In the chip, a gas inflow is focused through a narrow orifice using two orthogonal liquid flows containing phospholipids, which generate bubbles through hydrodynamic instability [2]. The phospholipid mixture, composed of a 9:1 molar ratio of DSPC and DPPE-PEG 5000, was prepared with a concentration of 12.5 mg/mL in phosphate-buffered saline solution (Beckman Coulter, ISOTON II Diluent). To enhance the stability and longevity of the bubbles, the bubbles were filled with a gas mixture of 15 v% of C_4F_{10} gas in CO_2 gas as detailed by Segers et al. [53]. The bubble size distribution and concentration in the original vial were measured using a Coulter counter (Multisizer 3, 50 μm aperture tube, Beckman Coulter, Brea, CA, USA), as illustrated in Fig. 4.1a with a median size of R_0 = 2.7 μm. This characterization method ensured the consistency and reliability of the microbubble suspension.

4.3.2 Viscoelastic sample preparation

The viscoelastic samples were produced using polyacrylamide, a hydrogel that is known for its acoustic transparency [58]. Acrylamide (acrylamide/bis solution 19:1, 40% w/v, SERVA) was mixed with APS (Ammonium Persulfate, ACS reagent $\geq$ 98.0%, Sigma-Aldrich) for polymerization and the mixture was then diluted to reach weight concentrations of 6%, 12%, and 20% in MilliQ water. These are hereafter referred to as soft, semi-stiff, and stiff hydrogels, respectively. Finally TEMED (N,N,N′,N′ - Tetramethylethylenediamine, ReagentPlus®, 99%) was added to the solution to initiate the polymerization. The exact amounts of materials used in the preparation of the hydrogel are listed in Table 4.1.

Table 4.1: Formulation for the preparation of a 100 mL polyacrylamide hydrogel.

Materials	Soft hydrogel (6% PAM)	Semi - stiff hydrogel (12% PAM)	Stiff hydrogel (20% PAM)
40% acrylamide (mL)	15	30	50
MilliQ water (mL)	85	70	50
Ammonium Persulfate (mg)	140	140	140
TEMED (μL)	30	30	30

Figure 4.1b presents the rheological characterization of the hydrogels, measured using a rotational rheometer (MCR 502e, Anton-Paar) in a frequency sweep at 0.1%

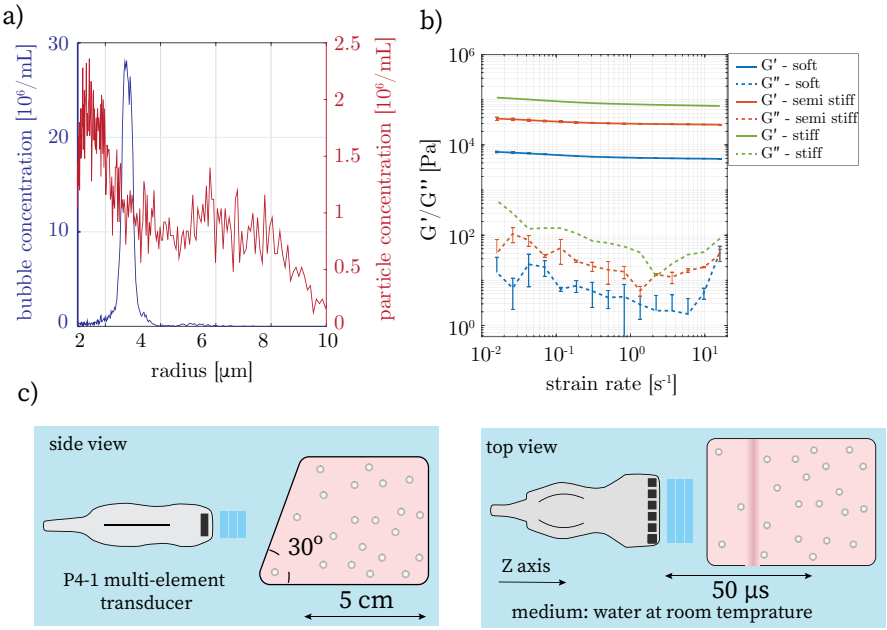


Figure 4.1: a) Size distribution of the microbubbles (with a mean radius of $R_0 = 2.7$ μm) and of the polydisperse particles. b) Frequency sweep measurement of elastic moduli of polyacrylamide hydrogel performed with a rotational rheometer. The stiff hydrogel moduli were measured once and the other two were measured 3 times. c) Schematic of the setup from the top and the side-view (microbubbles not drawn to scale). The transducer-facing side of the hydrogel was slanted at 30°with respect to the transmit direction, z-axis.

shear strain. The storage moduli ($G' = G$ as per Kelvin-Voigt description) were determined to be 7, 40, and 110 kPa for soft, semi-stiff, and stiff hydrogels, respectively. The loss moduli ($G'' = \mu\omega$) also increase with hydrogel stiffness, and at a comparable rate as G'. These measurements, conducted at a maximum frequency of 16 Hz, show that the storage moduli of the gel fabricated are within the same order of magnitude as for human tissues at low frequencies [52]. However, the shear rate applied by the microbubbles lies in the MHz range where the viscoelastic behavior of most materials is unknown.

Three samples were made for each stiffness. The first sample contained 30 mg of polydisperse glass microparticles (1-10 μm glass spheres, SUPELCO Analytical) in 100 mL of acrylamide solution. The size distribution of the glass particles is shown in Fig. 4.1a. This sample was used as a linear scattering reference. The second sample

contained 2 μL of monodisperse microbubble solution from the original vial (5×10^8 bubble/mL) into 100 mL of acrylamide solution. This concentration was chosen to have a mean bubble-to-bubble distance of at least 500 μm to minimize crosstalk and bubble-to-bubble interactions. The third sample was made from polyacrylamide alone, i.e., containing neither bubbles nor particles.

The sample size was $5 \times 5 \times 5$ cm^3 with one slanted side directly facing the transducer to minimize reflections in the recordings and to minimize interferences and standing waves within the sample as depicted in Fig. 4.1c.

4.3.3 Transducer array calibration

A P4-1 transducer (Philips ATL, 96 elements) was connected to a Vantage 256 ultrasound system (Verasonics, Kirkland WA, USA). The output pressure of the transducer P as a function of frequency f is given by $P(f) = H(f)V(f)$, where $V(f)$ is the driving voltage and $H(f)$ is the transmit characteristics of the transducer. To determine the transfer function $H(f)$, we performed a constant-voltage frequency sweep and measured the acoustic pressure output with a calibrated fiber-optic hydrophone (FP5461-09, Precision Acoustics, UK). The tip of the hydrophone was positioned in a water tank at a distance of 7.5 cm from the transducer surface along the z-axis and the transducer was aligned with the center of the viscoelastic samples in the scattering experiments. The output of the hydrophone was recorded with a PicoScope digital oscilloscope (Picoscope 5000 Series, Pico Technology).

To achieve a homogeneous pressure distribution across the region of interest (ROI) for scattering experiments, we simulated the pressure field of the P4-1 with the ultrasound simulator PROTEUS [104]. We applied the same transmit settings as in the scattering experiments (infinite focus, uniform apodization, 15 μs tristate voltage pulses) and simulated the pressure profile for each frequency in the frequency sweep from 1.4 MHz to 4.0 MHz in steps of 0.1 MHz. The simulated pressure field for frequencies of f = 1.5, 2.5, and 3.5 MHz both in the axial-lateral and elevational-axial planes are shown in Fig. 4.2. For the analysis, we chose the region of interest (ROI) depicted by the red dashed rectangle in Fig. 4.2a, where the pressure is nearly homogeneous. Within this ROI, which measures 2×1.5 cm^2, the standard deviation of the pressure is 5.7 kPa at a mean driving pressure of 60 kPa and a driving frequency of 2.5 MHz. However, somewhat stronger pressure gradients in the elevation direction cannot be avoided.

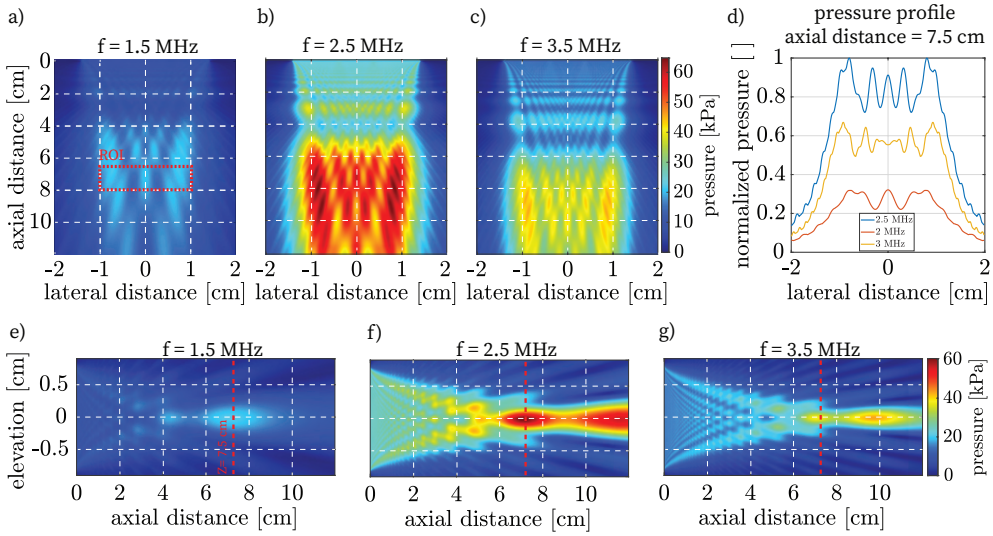


Figure 4.2: Simulated pressure fields in the axial-lateral plane at a) 1.5 MHz, b) 2.5 MHz, and c) 3.5 MHz. d) Pressure profile at a fixed distance of 7.5 cm from the surface of the transducer, corresponding to the middle of the sample normalized to the maximum pressure. e, f, g) Pressure fields for elevational-axial plane. The elevation axis is increased by a factor 3 for a better visualization.

4.3.4 Data acquisition

The acoustic signal scattered by the microbubbles was recorded using the calibrated P4-1 multi-element ultrasound transducer array transmitting narrowband plane waves. We performed a frequency sweep from 1.4 to 4.0 MHz in increments of 0.1 MHz for acoustic pressures ranging from 60 to 120 kPa in increments of 10 kPa, following a pressure calibration in water [104]. For each frequency, the driving voltage was adjusted according to the inverse of the transducer characteristic $H(f)$ to maintain a constant pressure output. The duration of the transmit signals was set to the integer number of periods closest to 15 μs, corresponding to a constant bandwidth of approximately 66 kHz to comply with the desired frequency sampling. The microbubbles in the hydrogels are stationary and the experiments were repeated ten times for each ultrasound pressure, then averaged to increase the signal-to-noise ratio.

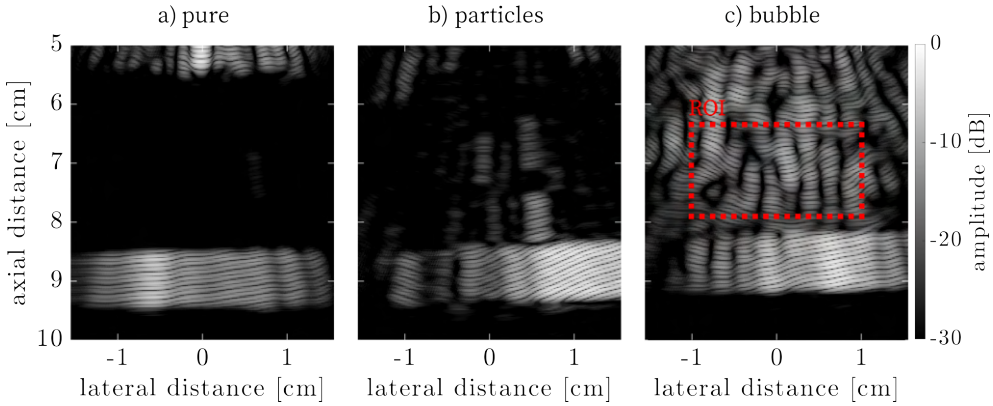


Figure 4.3: Reconstructed images in a semi-stiff hydrogel a) pure, b) loaded with particles, and c) loaded with bubbles. The images were acquired at an acoustic pressure of 60 kPa and a frequency of 2.0 MHz. In all images, the reflection of pressure waves from the back wall of the samples is visible at an axial distance of 9 cm.

4.3.5 Signal analysis

The received averaged scatter signals were band-pass filtered around the ultrasound driving frequency with a bandwidth of 260 kHz, i.e. four times the signal bandwidth. The images were then reconstructed using an in-house written delay-and-sum code with an angle restriction of 45° and a time-to-peak of 8 μs. Figure 4.3 shows examples of reconstructed images of semi-stiff hydrogels, of a pure sample of hydrogel (a), a microparticle-loaded hydrogel (b) and microbubble-loaded hydrogel (c).

Given the long acoustic pulse length, these images have an intrinsically low resolution, which is, however not a limitation here. Reconstructing the images allows for spatial filtering by selecting a region of interest (2.0×1.5 cm^2 ROI) within the hydrogels, see Fig. 4.3c, in a position distal from the reflection of the front interface, and proximal to that of the back interface.

The relation between the signal amplitude in the RF line and the speckle intensity in the reconstructed images is a priori non-trivial and frequency-dependent. However, this relation which results from delay-and-sum (DAS) beamforming is identical for all samples. The beamformed data can therefore be quantitatively compared across the different samples.

The scatter amplitude of the fundamental response, P_F, within the ROI was extracted using a Fast Fourier Transform (FFT) applied in the wave propagation direc-

tion for each driving frequency. The resonance curve then was obtained by dividing the pressure amplitude scattered by the bubbles $P_{F,RP}(f)$ and the pressure amplitude scattered by the linear reference sample with particles $P_{F,part}(f)$ which, as discussed above, eliminates the effect of image reconstruction on each individual frequency. Therefore:

$$\frac{P_{bub}(f)}{P_{part}(f)} = \frac{P_{F,RP}(f)}{P_{F,part}(f)}, \tag{4.7}$$

where P_{bub} and P_{part} are the amplitudes of the pressures scattered by the bubbles and by the particles, respectively, (see Eq. 4.5). For monodisperse bubbles, $P_{bub} \propto P_{RP}$, which is described by Eq. 4.6. Furthermore, the pressure scattered from the particles P_{part} lies in the Rayleigh regime, and scales with the square of the frequency f [105], [106]. Therefore:

$$\frac{P_{F,RP}(f)}{P_{F,part}(f)} \propto \frac{P_{s,RP}(f)}{f^2}. \tag{4.8}$$

Note that the implicit prefactor in Eq. 4.8 also accounts both for the unknown particle concentration in the ROI, and for the particle size distribution. This expression was used to fit the experimental data to extract the resonance frequency and the damping ratio.

4.3.6 Local pressure inside the hydrogel

Polyacrylamide hydrogels are known for their good ultrasound transparency. While this is in general true for soft hydrogels, it is less so for the stiffer hydrogel [58]. In particular, in this study, we have observed non-negligible acoustic reflections originating from the hydrogel-water interface, which is evidence of a significantly higher acoustic impedance in the semi-stiff and stiff hydrogels. This section details the procedure to estimate the pressure amplitude in each hydrogel based on the amplitude spectra of the pressures scattered by both the particles and the bubbles.

We define $\tilde{P}_i(f)$ the frequency spectrum of the pressure wave scattered by the particles in the hydrogel labeled i. As before, these spectra are extracted from the beamformed data. In the frequency domain, scattering can described by a simple product that translates the convolution in the time domain [107]:

$$\tilde{P}_i(f) = S_{p,i}(f)\tilde{P}_0(f)T_{w-i}T_{i-w}, \tag{4.9}$$

where $\tilde{P}_0(f)$ is the amplitude spectrum of the ultrasound wave without hydrogel, i.e., as calibrated in water. T_{w-i} is the pressure transmission coefficient from the water to the hydrogel and T_{i-w} is the pressure transmission coefficient from the hydrogel to the water. $S_{p,i}$ is the function representing the scattering in the hydrogel. This expression assumes no spectral distortion, i.e., the spectrum of the wave in the hydrogel is identical to that in water, albeit scaled in amplitude. This assumption is justified since attenuation is neglected, and because the reflection coefficients do not depend on frequency. Taking the mean value over frequency:

$$< \tilde{P}_i > = < S_{p,i}(f)\tilde{P}_0(f) > T_{w-i}T_{i-w} = < S_{p,i}(f)\tilde{P}_0(f) > T_{E,i}, \tag{4.10}$$

where <> is the mean value, and $T_{E,i}$ is the reflection coefficient in intensity (or energy) between the hydrogel and water.

Since the particle concentration and size distributions are the same in all hydrogels, $< S_{p,soft}(f)\tilde{P}_0(f) > \approx < S_{p,i}(f)\tilde{P}_0(f) >$, where the subscript soft specifically refers to the softest hydrogel which we will use as reference. This assumes that the particles scatter in the same way in all hydrogels, which is an approximation. We can then write:

$$\frac{< \tilde{P}_i >}{< \tilde{P}_{soft} >} \approx \frac{T_{E,i}}{T_{E,soft}}. \tag{4.11}$$

Using the relation between the transmission coefficients in pressure and intensity:

$$\frac{< \tilde{P}_i >}{< \tilde{P}_{soft} >} \approx \frac{Z_{soft}}{Z_i}\frac{T_{w-i}^2}{T_{w-soft}^2} = \frac{Z_{soft}}{Z_i}\frac{P_i^2}{P_{soft}^2}, \tag{4.12}$$

with P_i and P_{soft} are the local driving pressure amplitudes in the hydrogel i under consideration and in the softest hydrogel, respectively. Z_i and Z_{soft} are the acoustic impedances of both hydrogels. Furthermore,

$$\frac{T_{E,i}}{T_{E,soft}} = \frac{T_{w-i}T_{i-w}}{T_{w-soft}T_{soft-w}} = \frac{T_{i-w}}{T_{soft-w}}\frac{P_i}{P_{soft}} = \frac{Z_{soft}+Z_w}{Z_i+Z_w}\frac{P_i}{P_{soft}}. \tag{4.13}$$

Rewriting Eq. 4.12 and inserting Eq. 4.11:

$$\frac{Z_{soft}}{Z_i} = \frac{< \tilde{P}_i >}{< \tilde{P}_{soft} >}\frac{P_{soft}^2}{P_i^2} = \frac{T_{E,i}}{T_{E,s}}\frac{P_{soft}^2}{P_i^2}, \tag{4.14}$$

Using Eq. 4.13 then yields:

$$1 = \frac{1 + \frac{Z_w}{Z_{soft}}}{1 + \frac{Z_w}{Z_i}} \frac{P_{soft}}{P_i}, \tag{4.15}$$

We can then use Eq. 4.12 to remove Z_i:

$$1 + \frac{Z_w}{Z_i} = 1 + \frac{Z_w}{Z_{soft}} \frac{Z_{soft}}{Z_i} = 1 + \frac{Z_w}{Z_{soft}} \frac{< \tilde{P}_i >}{< \tilde{P}_{soft} >} \frac{P_{soft}^2}{P_i^2} = \left(1 + \frac{Z_w}{Z_{soft}}\right) \frac{P_{soft}}{P_i}, \tag{4.16}$$

Rewriting:

$$y^2 - y \left(1 + \frac{Z_{soft}}{Z_w}\right) \frac{< \tilde{P}_{soft} >}{< \tilde{P}_i >} + \frac{< \tilde{P}_{soft} >}{< \tilde{P}_i >} \frac{Z_{soft}}{Z_w} = 0. \tag{4.17}$$

where $y = \frac{P_{soft}}{P_i}$. Finally, assuming the acoustic impedance of the soft hydrogel to be close to that of water, namely that $y \approx \frac{P_{ac}}{P_i}$ and $Z_{soft} \approx Z_w$:

$$y^2 - 2\frac{< \tilde{P}_{soft} >}{< \tilde{P}_i >} y + \frac{< \tilde{P}_{soft} >}{< \tilde{P}_i >} = 0, \tag{4.18}$$

which allows for calculating local pressure due to the loss across the hydrogel/water interfaces. This quadratic equation has two solutions. We selected the solution for $y < 1$ because polyacrylamide hydrogel has a higher speed of sound c_s and density compared to water. Consequently, the impedance $Z = \rho_m c_s$ is larger, resulting in higher pressure inside the hydrogel [108], as the transmission coefficient is given by $T = \frac{2Z_i}{Z_i + Z_{soft}}$.

4.4 RESULTS

4.4.1 Resonance curves

The measured resonance curves are shown in Figs. 4.4a-c. The color bar represents the local driving pressure. Based on the correction described in Sec. 4.3.6, the local pressure inside the soft hydrogel is assumed to be the same as the pressure measured in water during the transducer calibration, whereas the pressure inside the semi-stiff and stiff hydrogels is increased by approximately 46% and 70%, respectively. The

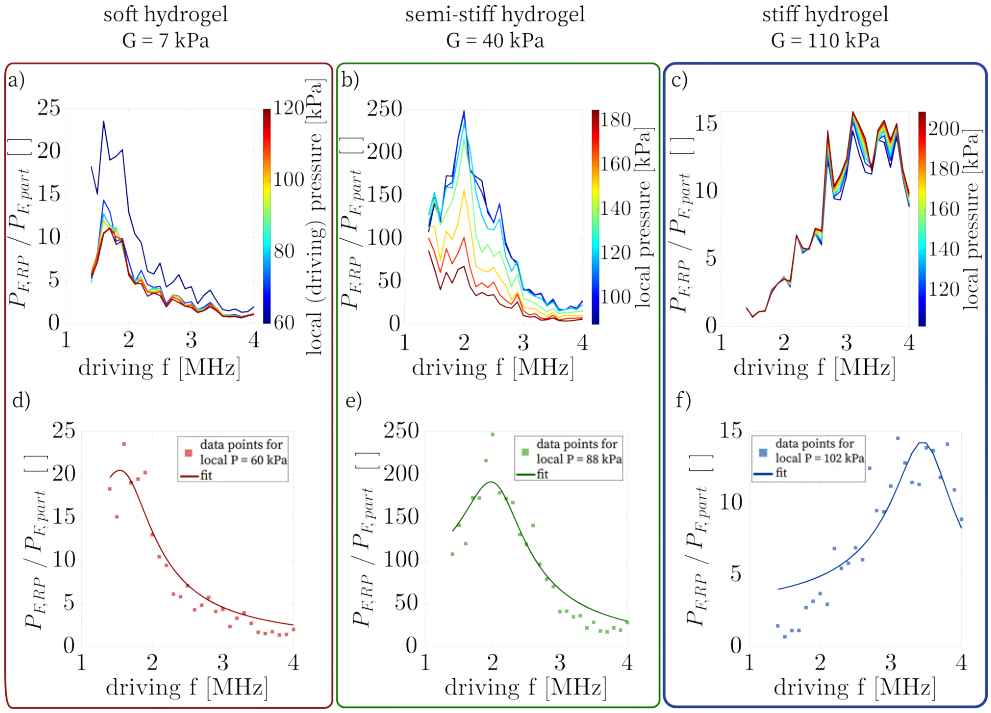


Figure 4.4: a, b, c) Resonance curves of bubbles in soft, semi-stiff, and stiff hydrogels, respectively, for various acoustic pressures. d, e, f) Fit of the resonance curves acquired with the lowest local pressure, i.e., 60, 88, and 102 kPa in the soft, semi-stiff, and stiff hydrogels.

variations in the scattered pressure in Figs. 4.4a-c is a consequence of small interferences by internal reflections within the sample, even if these are minimized by the slanted sample geometry.

Equation 4.8 was used to fit the resonance curves at the lowest driving pressure for which the bubbles remained stable. The result is shown in Figs. 4.4e, f, and g. The results of the fitting in terms of resonance parameters (resonance frequency and damping) are detailed in Table 4.2. The resonance frequencies for the soft, semi-stiff, and stiff hydrogels are 1.7 MHz, 2.1 MHz, and 3.5 MHz, respectively. This is approximately (42 $\pm$ 18)%, (77 $\pm$ 12)%, and, (201 $\pm$ 8)% higher than the eigenfrequencies of the same microbubbles in water, depending on the initial surface tension $\sigma(R_0) \in [20, 72]$ mN/m, as predicted by Eq. 4.2. This estimate uses $\chi = 0$ N/m since the shell elasticity is expected to have a small effect for the driving pressures

considered [25].

Table 4.2: Fitting results for bubbles in soft, semi-stiff, and stiff hydrogels driven at lowest local pressure.

Fitting parameter	Soft hydrogel	Semi - stiff hydrogel	Stiff hydrogel
Resonance frequency [MHz]	1.7	2.1	3.5
Damping ratio ζ	0.066(*)	0.050	0.10

(*) value with limited accuracy.

The amplitude of the resonance curves is not reported as it is a function of the exact bubble to particle concentration ratio, which is of minor interest here. The damping ratios are also reported in Table 4.2. A coated bubble in water experiences a damping ratio of about 0.1 [8]. The damping ratio is roughly 5-10 times larger than that of coated bubbles in water for the semi-stiff and stiff hydrogels, respectively. In the Kelvin-Voigt model, the loss modulus is directly linked to viscosity through $G'' = \mu\omega$. The increased damping ratio is therefore consistent with the trend observed in Fig. 4.1b for low strain rates. Note, however, that the damping ratio estimated for the soft hydrogel is unexpectedly high. However, this value is unreliable because of the incomplete resonance curve, which we therefore mark with an asterisk in Table 4.2.

4.4.2 Microbubbles stability

Figures 4.4a and b show a rapid decrease in the amplitude of the resonance curves with increasing driving pressure, suggesting that the microbubbles are destroyed by the higher driving pressures. Interestingly, while decreasing the pressure amplitude, the curves retain the original resonance frequency, which suggests that a fraction of the population is destroyed while another fraction remains unaffected: we do not observe a gradual shift of the resonance frequency that one would expect for bubbles that are slowly changing in size or shell properties.

To better quantify this effect, Fig. 4.5 shows the scattered energy, i.e., the area under the square of the curves presented in Fig. 4.4, as a function of local pressure in the hydrogel. The same procedure was also repeated for the pure hydrogels (red curves in Fig. 4.5). As expected, these reference integrated spectra show no significant variation as a function of the driving pressure. The blue curves in Fig. 4.5, which correspond to bubble scattering, show that destabilization already occurs at 60 kPa (the lowest pressure) for the soft gel, while it occurs at about 120 kPa in the semi-

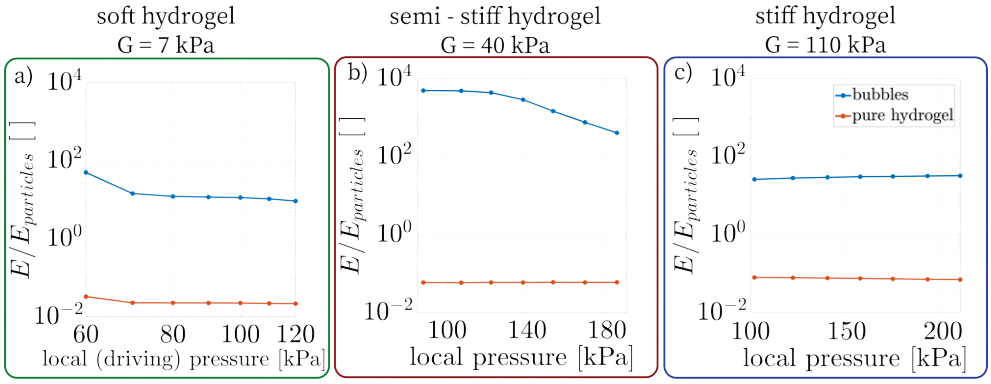


Figure 4.5: Scattered energy from the pure and bubble-containing hydrogels relative to the energy scattered from the particle-containing hydrogels for the a) soft, b) semi-stiff, and c) stiff hydrogels.

stiff gel, and does not occur at all in the stiff gel. The bubbles are thus increasingly more stable as the hydrogel become stiffer. This is most likely due to a lower oscillation amplitude of the bubbles in the stiffer hydrogels which is itself due to both the increased viscosity (and thus damping) and elasticity of the hydrogel [34].

4.4.3 Relation to hydrogel rheology

The resonance frequency and the damping ratio extracted from the fittings are used to calculate G and μ from Eq. 4.2 and 4.3. Once again considering $\chi = 0$ and rewriting Eq. 4.2 gives:

$$G = \frac{\rho_m \left(2\pi f_0 R_0\right)^2 - 3\gamma P_0 - (3\gamma - 1)\, 2\sigma(R_0)/R_0}{4}. \tag{4.19}$$

Now considering $\kappa_s = 6 \times 10^{-9}$ N/m [25] and rewriting Eq. 4.3 gives:

$$\mu = \frac{2\rho_m \omega_0 \zeta R_0^2 - 4\kappa_s / R_0}{4} \tag{4.20}$$

The results of Eqs. 4.19 and 4.20 are provided in Table 4.3, where the uncertainty provided for G is calculated for $\sigma(R_0) \in [20, 72]$ mN/m.

Table 4.3: Rheological properties of soft, semi-stiff, and stiff hydrogels.

Parameter	**soft**	**semi-stiff**	**stiff**
rheometer G (kPa)	7	40	110
bubble - probed G (kPa)	$97 \pm 11\%$	$206 \pm 5\%$	$787 \pm 2\%$
rheometer μ (Pa.S)	0.68	2.46	8.75
bubble - probed μ (mPa.S)	NA	0.1 (*)	5.8

(*) damping dominated by the shell

The value of the storage modulus G measured with bubbles at high strain rate is significantly higher than that measured with the rheometer. Since the damping ratio for the bubbles obtained from the soft hydrogel is inaccurate, we do not provide the corresponding viscosity as the value would not be meaningful. For the semi-stiff hydrogel, we find a relatively low value for the viscosity, which implies that the damping is dominated by the shell. The value of the viscosity is thus mostly determined by the accuracy of the shell viscosity, which is not well known. For the stiff hydrogel, the dissipation is high enough to extract a meaningful viscosity value of 5.8 mPa.s, which is in good agreement with the viscosity determined in Ch. 2.

4.5 Discussion

In summary, we aimed to clarify the interaction between monodisperse microbubbles and hydrogel elastic modulus in the megahertz frequency range by measuring scattered pressure from the microbubbles. We used a Rayleigh-Plesset type model to interpret this interaction and introduced a method for determining the resonance frequency of microbubbles by normalizing their captured signal against that of particles.

Some discussion on the limitation of the presented measurement technique are in order. Each bubble was driven 10 times at each driving pressure to average the scattered signals and improve the signal-to-noise ratio. Resonance curves obtained from each repeat for the soft and semi-stiff hydrogels at local pressures of 60 kPa and 138 kPa, respectively, are shown in Fig. 4.6. It can be observed that the bubbles in the soft hydrogel start to dissolve from the second repeat. Yet, the dissolution is not immediate, and the scattered pressure gradually decays during the 10 repeats at this pressure. For the semi-stiff hydrogel, the decay in the amplitude of resonance

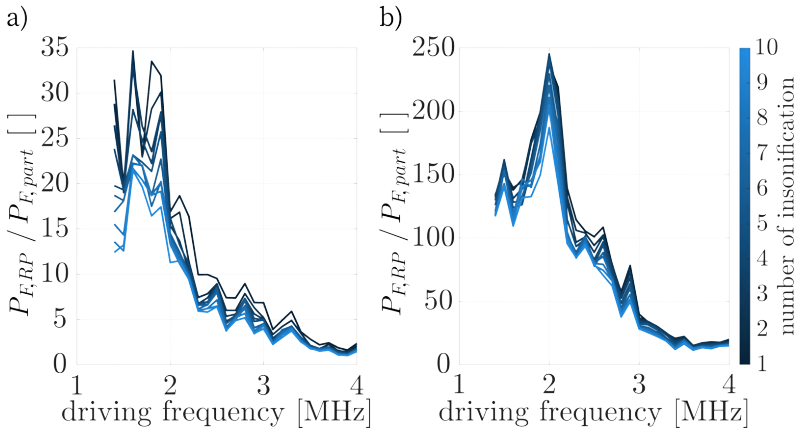


Figure 4.6: a) Resonance frequency as a function of the driving pressure for a microbubble with $R_0 = 2.7$ μm. Resonance curves obtained over 10 repeats b) for a local pressure of 60 kPa in the soft hydrogel, c) for a local pressure of 138 kPa in the semi-stiff hydrogel.

curves starts from the 5th repeat. The fittings in Sec. 4.4, where always performed on the first curve of Figs. 4.4d-f, corresponding to the lowest pressure. They remain therefore representative of the bubble behavior.

Higher harmonics also bear interest, and can, in principle, be extracted from a Fourier filtering analysis. However, despite the narrow bandwidth of the transmitted pulses, the signal-to-noise ratio of the harmonic signal is too low to be able to reliably separate them from the fundamental spectrum. We suggest utilizing the pulse inversion (PI) technique to recover the nonlinear scattering in future investigations [109].

The theory presented in Sec. 4.2 is designed for a single microbubble. Experimentally, the presented method utilizes monodisperse bubbles that have a uniform narrowband acoustic response. The polydispersity index, calculated as the ratio of the standard deviation of the size distribution and the mean radius of the suspension, was 0.08 for the bubble sample (cf. Fig. 4.1a). While this number indicates a well-controlled monodispersity, the method may be further improved by taking into account the size distribution of the sample of microbubbles.

The polymerization of PAM is an exothermic reaction, especially for high acrylamide concentrations. For example, the temperature can rise by up to 35°C for a 20% acrylamide solution. These temperatures, however, are lower than the production temperature of the microbubbles (60°C), which corresponds to the phase transi-

tion of the phospholipids. Furthermore, these bubble can endure heat treatments of up to 60 degrees for an hour without deleterious effects [15]. We therefore do not expect a measurable impact of the temperature rise experienced on the bubble stability during polymerization.

Attenuation of the acoustic transmit signal within the hydrogel can decrease the pressure received by the bubbles. The attenuation can be measured by comparing the amplitude of the pressure wave passing through the sample to a reference such as water, and was found to be approximately 0.9 dB/cm in the pure stiff hydrogel, which remains low [58], [92].

4.6 Conclusion

This work introduces a spectroscopic method for capturing the acoustic signals scattered from a cloud of monodisperse phospholipid-coated microbubbles within viscoelastic hydrogels, utilizing a calibrated multi-element transducer. We have experimentally measured the resonance curves of monodisperse microbubbles in bulk viscoelastic media with three different elastic moduli. The results show the expected increase in resonance frequency with increasing medium elasticity. The elastic moduli extracted at the MHz rates are significantly higher than those obtained using a rotational rheometer at strain rates of several Hz. Additionally, a procedure was introduced to calculate the local pressure inside the hydrogel solely based on the scattered spectra, revealing that the pressure in semi-stiff and stiff hydrogels was higher than in soft hydrogels, by approximately 45% and 70%, respectively. These findings provide valuable insights for refining bubble-media interaction models and optimizing microbubble contrast agents for biomedical applications. They also open an interesting avenue for characterizing the viscoelastic properties of opaque soft tissue in the megahertz frequency regime.

4.7 Acknowledgment

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CHAPTER 5

Acoustic cavitation of microbubbles in microchannels[1]

ABSTRACT

Ultrasound contrast agents (UCAs) are routinely used in medical ultrasound imaging and have promising potential for drug and gene delivery applications using ultrasound. Their practical use heavily relies on microbubble resonance, which entails both excellent acoustic backscattering properties and biomechanical actuation. Upon intravascular injection, the bubbles are confined within the vascular network. However, the effects of this confinement on bubble dynamics are currently unclear. Here, we experimentally investigate the dynamics of bubbles confined in viscoelastic microcapillaries. We utilize ultra-high-speed imaging at 10 million frames per second to measure the nonlinear oscillations of UCAs in microchannels within a silicone polymer block. The channels have diameters ranging from 15 to 100 μm and a length of 1.5 cm. Microbubble resonance curves were extracted from ultra-high-speed recordings of the oscillating bubbles. We measure a decrease in resonance frequency with decreasing microchannel diameter, a behavior that is in agreement

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with prior numerical studies. The frequency spectra of the radial dynamics show clear nonlinear second-harmonic oscillations when the bubble is driven at its resonance frequency and at half its resonance frequency. Notably, the latter is enhanced as compared to bubbles in the free field. Moreover, our data show that the viscoelasticity of the surrounding medium increases the damping by 50% on average. The fact that resonance still exist in this system indicates that even for relatively stiff materials such as PDMS, with an elastic modulus of 500 kPa, the wall elasticity plays a role in the dynamics of confined bubbles. This is a unique study, as it examines both phospholipid coating and elastic microchannel effects in the regime of small oscillations. These experimental findings will aid the development of more accurate models of microbubbles oscillating in confinement for more precise use of UCAs in future clinical practice.

5.1 INTRODUCTION

Ultrasound (US) is the most widely used medical imaging modality: it is used for diagnostic imaging [110], patient monitoring [111], functional imaging [112], image-guided therapies [113], and therapy [114] itself. It is inherently safe, portable, real-time, and cost-effective compared to other modalities such as computed tomography (CT) or magnetic resonance imaging (MRI). US imaging relies on the scattering from tissue inhomogeneities to form pulse-echo images. However, while organ perfusion and blood flow quantification has increasingly high clinical value, it remains a challenge for ultrasound since blood is a poor ultrasound scatterer. An effective solution to this issue involves the use of ultrasound contrast agents (UCA) [26], [67].

The most common ultrasound contrast agents consist of a suspension of coated microbubbles (diameter R_0= 1 - 5 μm) filled with a heavy gas core and stabilized by a phospholipid shell. When exposed to ultrasound, the microbubbles undergo volumetric oscillations, that generate a strong scattered pressure field, or echoes [88]. Due to their size, the bubbles remain confined within the blood pool, which in combination with their unique acoustic scattering properties, is the basis for contrast-enhanced perfusion imaging with ultrasound. Furthermore, the pressure scattered by the bubbles is highly nonlinear, which can be used in contrast-specific modes to further enhance the contrast to tissue ratio [88], [115]. In the contest of ultrafast ultrasound plane wave imaging, these highly (non)linear signatures further enable vector flow imaging [116] at several thousand frames per second, and boost the performance of super-resolution imaging [91], [117], [118]. Ultrasound-driven microbubbles also present a unique potential for localized drug and gene delivery [119] where microbubble induce mechanical stress on neighboring cells, leading to a transient increase in perfusion through, so-called, sonoporation [120]–[123], sonoprinting[124], [125], or sonopermeation [126].

Whether it is to optimize imaging or to control bioeffects, a detailed understanding of microbubble dynamics is essential. The dynamics of a bubble oscillating spherically in an unbounded fluid has been extensively investigated and is well understood [8], [24], [127], [128], However, for both imaging and therapeutic applications, the interest lies in the bubbles that travel through capillary vessels. The capillary network is essential, for example, in (early) tumor diagnosis, organ perfusion quantification, functional imaging, and in quantifying inflammatory responses. Capillary vessels have diameters comparable to that of the microbubbles, i.e., down

to a diameter of 10 μm and thus constitute a viscoelastic boundary around the microbubbles.

The pioneering work by Oguz and Prosperetti and Geng et al. [129], [130] has shown that the confinement of a bubble in a rigid tube has a dramatic impact on its dynamics. More specifically, they have demonstrated that bubble oscillations are impossible in an infinitely long rigid tube filled with an incompressible liquid, and that the resonance frequency of the bubble decreases with decreasing tube diameter (in a finite length pipe). Subsequently, Qin et al. [131] have proposed a computational model that incorporates the elasticity of the vasculature wall and an effective dynamic thickness of the surrounding tissue. They have demonstrated that the resonance frequency of a bubble in an elastic tube can be either lower or higher than that of the bubble in the unbounded field depending on the stiffness of the tube, a result that contrasts with the description of the bubble dynamics in a rigid tube. Furthermore, a number of theoretical and numerical studies have investigated the behavior of bubbles near a boundary, demonstrating the effect of the presence of a rigid or elastic wall [132]–[134] and the occurence of non-spherical microbubble oscillations at low pressures ($\sim$200 kPa) and shape instabilities that can arise, depending on bubble radius and acoustic pressure [56], [135], [136].

While considerable work has been performed on investigating the rich phenomena of bubble oscillation in both in vivo [65], [137], [138] and excised ex-vivo tissue [139], few experimental studies have focused on delivering input for a quantitative comparison with confined bubble dynamics model, especially at lower driving pressures. Zhao et al. [140] have acoustically investigated the spectral response of microbubbles confined in an agarose microchannel as a function of microchannel diameter and acoustic driving pressure. They have discovered that the threshold for inertial cavitation (while this classification does not strictly hold [141]), the level of subharmonics, and the level of ultra-harmonics increases as the microchannel diameter decreases. In parallel, using a high-speed camera, they have visualized the accumulation and coalescence of microbubbles due to secondary acoustic radiation forces, which ultimately impacts acoustic emission spectra [142]. Nonetheless, there is a continuing scarcity of experimental data in the regime of small oscillation amplitudes of coated microbubbles within viscoelastic microchannels of various elasticities and diameters.

The objective of this study is to experimentally explore the changes in microbubble dynamics in the range of ultrasound imaging pressures, as a function of mi-

crochannel diameter. We use an ultra-high-speed camera operating at 10 million frames per second to record the radial oscillations of phospholipid-coated microbubbles in microchannels made in polydimethylsiloxane (PDMS). We extract the resonance curves of oscillating phospholipid-coated microbubbles and quantify their change in linear and nonlinear bubble response as compared to bubbles of the same size in an unbounded medium.

5.2 THEORY

The dynamics of a bubble in an incompressible, unbounded medium are governed by the Rayleigh-Plesset (RP) equation [8], [44]:

$$\rho_L \left(R\ddot{R} + \frac{3}{2}\dot{R}^2 \right) = \left(P_0 + \frac{2\sigma(R_0)}{R_0} \right) \left(\frac{R_0}{R} \right)^{3\gamma} - P_0 - P_{ac} - \frac{2\sigma(R)}{R} - \frac{4\kappa_s \dot{R}}{R^2} - \frac{4\mu_L \dot{R}}{R}. \tag{5.1}$$

Here, ρ_L is the density of the surrounding liquid. R represents the radius of the bubble, $\dot{R}$ and $\ddot{R}$ represent the bubble wall velocity and acceleration, respectively. R_0 is the initial bubble radius. P_0 is the atmospheric pressure and P_{ac} is the acoustic driving pressure. The left-hand term arises from the momentum of the liquid generated by the bubble wall motion. On the right-hand side, the first term represents the pressure in the bubble. It has been derived using a polytropic law with an exponent γ. The capillary pressure term, $\frac{2\sigma(R)}{R}$, reflects the Laplace pressure exerted on the bubble surface with a size-dependent surface tension $\sigma(R)$ as described by the shell buckling model of Marmottant et al [6]. The pressure term, $\frac{4\kappa_s \dot{R}}{R^2}$, describes the dissipation occurring in the shell through molecular friction where κ_s represents the dilatational surface viscosity of the bubble shell. Finally, $\frac{4\mu_L \dot{R}}{R}$, is the damping term induced by viscous dissipation in the liquid [25].

Equation 5.1 can be linearized for small bubble oscillations using $R = R_0(1 + x(t))$, where $x(t) \ll 1$ is the relative excursion and $r_\varepsilon = xR_0$ is the amplitude of oscillation. The resulting equation is that of a second-order harmonic oscillator:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = -\frac{P_{ac}}{\rho_L R_0^2}, \tag{5.2}$$

The right-hand side of the equation represents the driving pressure term. ζ and

ω_0 correspond to the dimensionless damping ratio, and angular eigenfrequency of the bubble, respectively. The damping ζ comprises contributions from liquid viscosity and shell viscosity:

$$\begin{aligned} \zeta_{vis} &= \frac{2\mu_L}{\rho_L \omega_0 R_0^2}, \\ \zeta_{shell} &= \frac{2\kappa_s}{\rho_L \omega_0 R_0^3}. \end{aligned} \tag{5.3}$$

While Eq. 5.3 only contains damping contributions from the shell and from the surrounding liquid, the total damping for an oscillating bubble also comprises a components related to sound re-radiation and thermal damping. For micrometer-sized bubbles, the magnitude of thermal damping is of the same order as that of the viscous damping in water [49]. As a result, the medium's viscosity is typically increased by a factor of 2 to account for thermal damping in a phenomenological approach [20]. Acoustic re-radiation is small and not accounted for in this study. Ultimately, shell dissipation accounts for approximately 75% of the total damping [8].

In this study, we experimentally investigate the resonance of coated bubbles in a microchannel, while the expressions presented above were developed for a bubble in a free field. We assume that bubbles will also resonate in microchannels, as was observed for bubbles in a gel in Ch.2, and that this resonance, at linear order, can also be described by the second-order harmonic oscillator equation, Eq. 5.2:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = -\omega_0^2 P_{ac} K, \tag{5.4}$$

where the gain K replaces the unconfined bubble gain $\frac{1}{\rho_L \omega_0^2 R_0^2}$. The expressions for the eigenfrequency and damping ratio also change, and will be extracted from experiment. This approach also allows for a direct comparison of the resonance of confined coated bubbles to that of coated bubbles in the free field. The normalized amplitude of oscillation can then be expressed as [31]:

$$\frac{|\underline{r_\varepsilon}|}{|\underline{P_{ac}}|} = \frac{K}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \left(2\zeta\omega_0\omega\right)^2}}. \tag{5.5}$$

where $r_\varepsilon = Re(\underline{r_\varepsilon})$ and $P_{ac} = Re(\underline{P_{ac}})$. This expression can then be fitted to the

experimental resonance curves to provide ω_0, ζ, and K.

5.3 Material and methods

5.3.1 Microbubble fabrication

Microbubbles were produced using a custom-built bubble generator equipped with a flow-focusing device [15]. This method allows us to prepare monodisperse microbubble suspensions at the target size. In short, the flow-focusing technique consists in focusing a gas jet through a narrow orifice using two perpendicular liquid flows that contain the phospholipid solution. The focusing leads to the formation of uniformly sized microbubbles. The phospholipid mixture was created using a 9:1 molar ratio of DSPC and DPPE-PEG 5000, with a concentration of 12.5 mg/mL in a phosphate-buffered saline solution (Beckman Coulter, ISOTON II Diluent). The bubbles were produced using a gas mixture of CO_2 and C_4F_{10}, an inert high molecular-weight gas with a low solubility in water of 1.5 mg/L. The gas mixture ratio allows the rapid stabilization of C_4F_{10} bubbles after diffusion of the CO_2 into the surrounding medium [53].

5.3.2 Microchannel fabrication

The viscoelastic microchannels were created in a silicone polymer block (polydimethylsiloxane, PDMS) by molding. Two syringe needles facing each other were positioned in a 3D-printed mold. A fishing thread or tungsten wire was passed from one needle to the other before pouring PDMS into the mold. The PDMS consisted of a Sylgard 184 silicone elastomer base and curing agent mixed with a 10:1 ratio. The mixture was cured in an oven at 80 °C for three hours. Subsequent extraction of the wire produced the desired microchannel with diameters of 15, 25, 60, or 100 μm. Since the behavior of viscoelastic materials in general strongly depends on strain rate (here ultrasound frequency), the PDMS was characterized using a rotational rheometer (MCR 502e, Anton-Paar) and the data was fitted to a Chasset-Thirion model of which its validity has been demonstrated for PDMS [143]. The fit provides a storage modulus (G') of 510 kPa at low strain rates and predicts a storage modulus of approximately 4.3 and 7.6 MPa for frequencies of 1 MHz and 10 MHz, respectively. Note that, as of yet, little is known on the high-frequency rheology of PDMS,

but existing data suggests a comparable behavior as for soft tissues [61], [144], see also Ch. 2.

The relative position of the bubble with respect to the microchannel walls is another crucial factor in our experiments. To ensure uniform interaction with the channel walls, we have diluted the bubbles in a soft polyacrylamide hydrogel with a shear modulus of $G \approx 10$ kPa, as measured by a rotational rheometer, and perfused them into the microchannel. The viscoelastic behavior of such a soft hydrogel, which was also evaluated at MHz frequencies (see Ch. 2), can be assumed to have a negligible effect on the bubble dynamics since $G \ll P_0$. Consequently, curing the gel allows for maintaining the bubbles in place within the channel, thereby preventing them from floating up against the wall. For the optical experiments, we selected bubbles positioned near the center of the channel by focusing on the mid-plane of the microchannel and studying only those bubbles that were in the focus of the imaging plane. This procedure offers sufficient precision since the depth of focus of the high-NA microscope objectives used is in the order of a few micrometers. Figure 5.1c shows a typical frame from a recording of a 2.2 μm-radius microbubble placed in a channel with diameter $D = 15$ μm.

5.3.3 Experimental procedure

Figure 5.1 shows the schematics of the complete setup. The optical configuration of the setup in depicted in Fig. 5.1a and a schematic of the microchannel in shown in Fig. 5.1b. The dynamics of acoustically driven phospholipid-coated microbubbles in PDMS microchannels contained in the soft hydrogel were recorded using ultra-high-speed imaging with a Shimadzu HPV-X2 camera operated at a frame rate of 10 million frames per second. Imaging was performed through a microscope (Modular Focus System BXFM, Olympus) equipped with a 2× magnifying block together with a 60× water-immersion objective (LUMPLFLN60XW, Olympus).

The ultrasound wave was generated with a programmable arbitrary waveform generator (AWG model 8026, Tabor Electronics Ltd.) and transmitted to a high-power RF amplifier (Vectawave VBA100-200, 10kHz - 100MHz 200W). The amplifier then drives a single-element focused transducer (C305 SU, Olympus). The driving pulse was 10 cycles long, with tapering over two cycles at the start and end of the signal to suppress transient responses of the bubble. The pressure was maintained at a low enough pressure level ($P_{ac} < 100$ kPa) to keep the bubble in the small oscil-

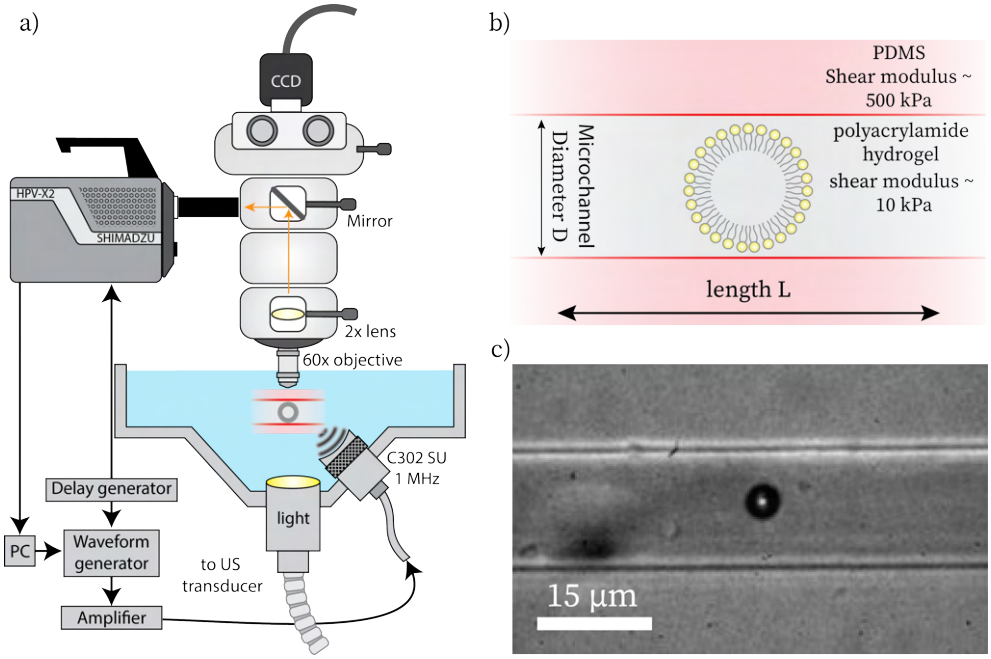


Figure 5.1: a) Optical characterization setup b) Schematic of a PDMS microchannel containing a phospholipid-coated microbubble c) Microscopy image of a bubble in a microchannel of diameter D = 15 μm.

lations regime and to prevent ultrasound-induced dissolution.

5.3.4 Analysis

The ultra-high-speed videos were analyzed using a custom-made edge detection script written in MATLAB®. Figure 5.2a show snapshots of a typical high-speed recording of a 2.2 μm-radius bubble. A typical radius-time curve $R(t)$ is shown in Fig. 5.2b, also for a 2.2 μm radius bubble. The Fast Fourier Transform (FFT) algorithm was then used to obtain the frequency content of the microbubble oscillations using the FFT function in MATLAB, see Fig 5.2c. A resonance curve was subsequently constructed from the amplitude of the fundamental response, A_1, for each driving frequency f. Figure 5.2d shows the maximum fundamental oscillation amplitude of bubbles with a radius of $R_0 = 2.2$ μm as a function of driving frequency in three different microchannels. The markers indicate the maximum fundamental amplitude. The solid lines represent the fit of Eq. 5.5 to each resonance curve, which

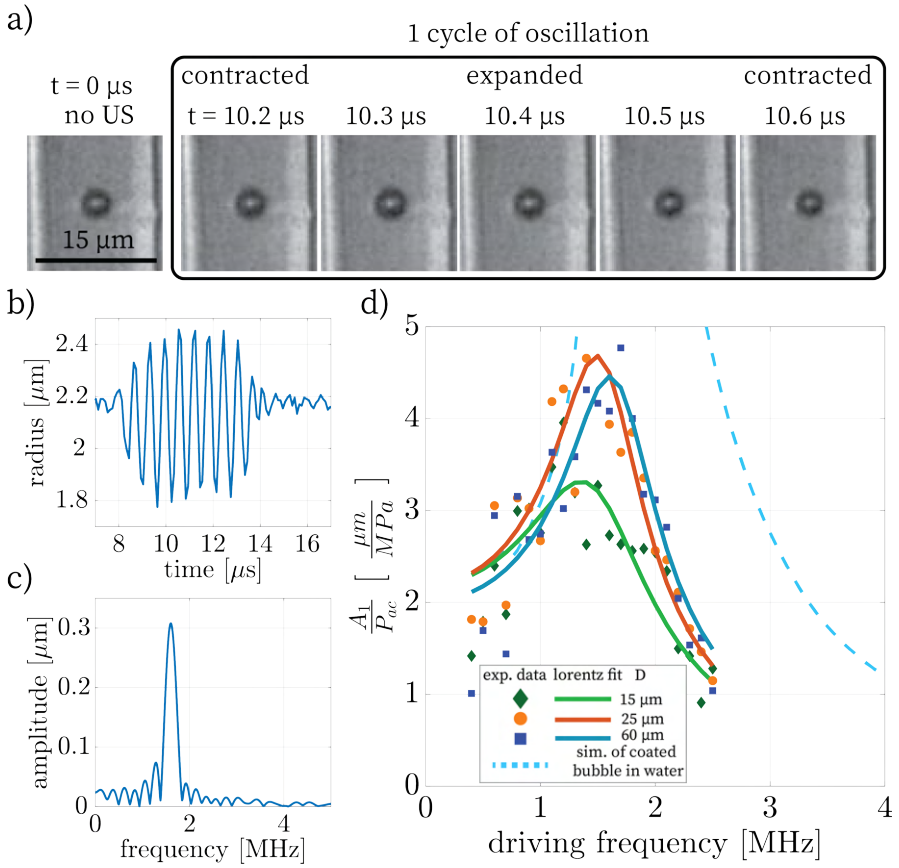


Figure 5.2: a) Snapshots of a high-speed recording of a 2.2 μm bubble driven with a 50 kPa, 1.8 MHz ultrasound pulse, and recorded at 10 million frames per second. b) Radius versus time curve extracted from an ultra-high-speed recording. c) Fast Fourier Transform (FFT) of the radius-time curve in (b). d) Resonance curves of microbubbles with $R_0 = 2.2\ \mu m$ in D= 15, 25 and 60 μm microchannels normalized to the driving pressure P_{ac}. The driving pressure used to obtain these resonance curves are $P_{ac} = 50$, 80 and 80 kPa, respectively.

then provide the resonance frequency, the gain, and the damping ratio.

5.3.5 Comparison to free-field bubbles

The resonance behavior of phospholipid-coated microbubbles is known to depend on the driving pressure. At low pressure, this is a consequence of the dependency of

the surface tension on the bubble surface area. At higher pressures, this is a consequence of the skewing of the resonance curve by the nonlinearity of the gas pressure with respect to the size of the bubble. Both are directly related to the amplitude of oscillation of the bubble [6], [57]. To provide a better comparison, we therefore chose to compare the measured resonance frequency of confined bubbles to that of a free-field bubble oscillating with the same oscillation amplitude. For this study this is more meaningful than comparing the response at equal driving pressure. In addition, the local acoustic pressure that the bubble experiences in the capillary is not precisely known due to attenuation and reflection in the PDMS and its surfaces, the presence of the capillary itself, and uncertainties inherent to the hydrophone calibrations. To that end, Eq. 5.1 was computed for a large range of acoustic pressures and frequencies, see Table 5.1. The reference free-field resonance curve for each experimental resonance curve measured in the capillary was then taken to be the one with an oscillation amplitude closest to that of the measured resonance curve.

Table 5.1: Simulation parameters for solving Eq. 5.1 for a microbubble in the free field.

Parameter	value or range (min:step:max)
R_0 (μm)	R_0 (experiment)
P_{ac} (kPa)	20:1:100
f (MHz)	0.4:0.1:2.5
P_0 (kPa)	100
σ_0 (mN/m)	20
μ_L (mPas)	3:0.1:10
γ	1.07
κ_s (kg/s)	6×10^{-9}

5.4 RESULTS

5.4.1 Bubble resonance frequency

Figure 5.3a shows the ratio of the measured resonance frequency in the microchannel to that of the bubble in free space, see Sec. 5.3.5. The graph shows that the resonance frequency decreases with decreasing microchannel diameter. This observation is consistent with previous theoretical and numerical studies on the resonance frequency of bubbles in rigid tubes [131], [145]. Note that we observe a slight depen-

dency of resonance frequency on bubble size where, interestingly, smaller bubbles are more strongly affected. This observation, however, does not follow prior predictions[131], [145].

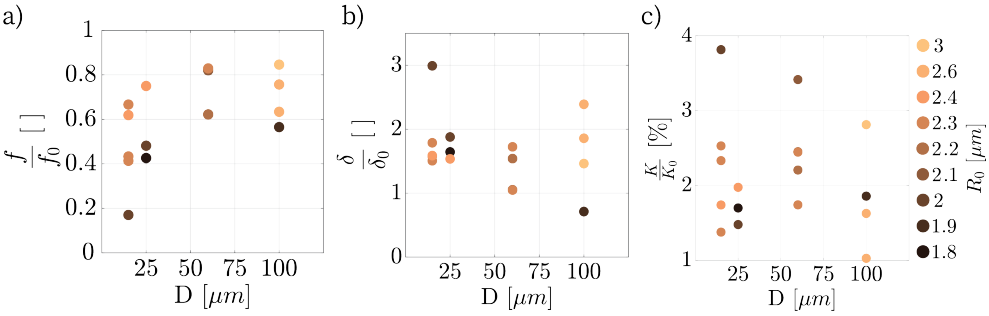


Figure 5.3: a) Ratio of the resonance frequency of the bubble in the tube to that of the same bubble in free field, plotted against the diameter of the microchannel. b) Damping coefficient in the microchannel normalized by the damping in an unconstrained medium as a function of the diameter of the microchannel. c) Ratio between the gain in the microchannel and the free-field gain versus the microchannel diameter.

5.4.2 Gain and damping

Figure 5.3b shows the ratio of the measured damping in the microchannel to the reference damping in an unconstrained liquid see Sec. 5.2. The measured damping in the microchannel is larger than that of the unconstrained liquid by 50% on average. The increased damping experienced by bubbles is likely caused, partly by the interaction with PDMS, and partly by the confined flow. PDMS indeed has an effective viscosity of approximately 0.27 Pa.s at 1 MHz (G'' =1.71 MPa according to the Appendix, Sec. 5.8), which is two orders of magnitude larger than the viscosity of water. Furthermore, confined flow increases shear and thus viscous dissipation in the fluid. Fig. 5.3b, however, shows large bubble-to-bubble variations which does not allow for further quantifying the increase in damping imparted by the channel. These variations might be caused by slightly off-centered bubble positions within the microchannel or by the acoustic field disturbances indiced by the microchannel itself like interference and standing waves, see for example ref. [146] since 100 μm is on the order of half the acoustic wavelength in PDMS, which is 420 μm at 2.5 MHz, assuming a sound speed of 1050 m/s [147].

The gain of the system corresponds to the low-frequency amplitude of oscillation

of the bubble in response to a unit driving amplitude. Fig. 5.3c, shows the ratio between the gain of the microbubbles extracted from the fit, K, and the gain expected for coated microbubbles in a free field, K_0. It can be seen that the gain in the microchannel is less than 5% of the gain in the free-field. It suggests a very strong decrease in scattering strength both on and off resonance. This, in turn, has important consequences for the sensitivity of perfusion imaging.

5.4.3 Nonlinear bubble response

Contrast in ultrasound imaging can be further improved by utilizing the nonlinear second harmonic response of UCAs, e.g., by using pulsing schemes such as pulse-inversion [88], [109]. There is therefore great interest in quantifying the nonlinear response of bubbles in viscoelastic microchannels. In Figs. 5.4a–c, we plot the frequency spectra of bubble oscillations as a function of the driving frequency. Figures 5.4a–c clearly show the presence of second harmonics, with even a visible third harmonic in Figs. 5.4b and c. No clear subharmonics can be observed.

Figures 5.4d–f compare the measured amplitude of the second harmonic response with that of the simulated bubble in free field which display the same fundamental response amplitude. As before, for comparison, we have adapted the driving pressure in the simulated free-field response so as to obtain the same fundamental amplitude as for the experimental data. The 25 μm and 60 μm microchannels display an amplitude of the second-harmonic response that is 1.9 and 2.3 times smaller than that of the bubble in free space. The same observation has also been reported by Zhao et al. [140]. Interestingly, the microbubble in the 15-μm microchannel displays a similar harmonic amplitude as its free-field counterpart.

Besides the change in harmonic amplitude, there is also an evident shift in the frequencies at which the harmonics are most efficiently generated by the microbubbles. This shift is expected and in agreement with the shift in resonance frequency displayed in Fig. 5.4. More specifically, the first peak in the harmonic emissions corresponding to driving the bubbles at half their resonance frequency, and the second is a result of driving the bubbles at resonance. The shifts in the peak harmonic emission thus directly follows the shift in resonance frequency. In all three cases, the free field bubble emits stronger harmonics at the resonance frequency than at half the resonance frequency. There is, however, a key difference for the confined bubbles: the harmonics are strongest when driving the bubbles at half the resonance fre-

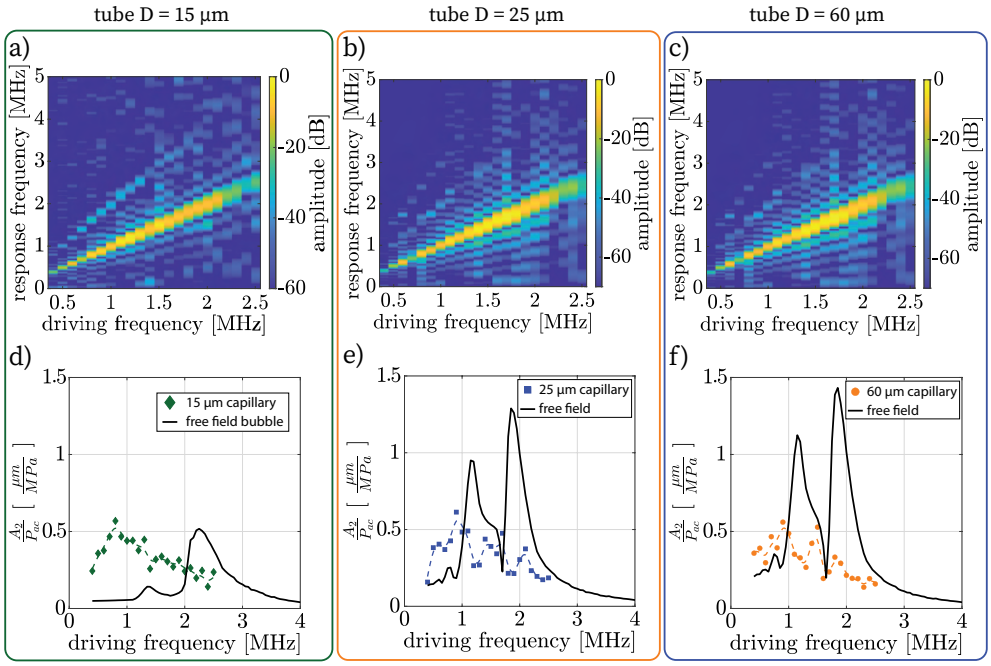


Figure 5.4: a, b, c) 2D spectral maps of bubbles with a radius of $R_0 = 2.2 \pm 0.05$ μm in microchannels with diameters of 15, 25, and 100 μm, respectively. d, e, f) show the amplitude of the 2nd-harmonic visible in panels a, b, and c. The solid black lines correspond to the free-field simulation that have the same maximum fundamental response amplitude as the experimental resonance curve. The acoustic pressures for D = 15, 25, and 60 μm are $P_{ac} = 50$, 80, and 80 kPa, respectively.

quency. This effect becomes increasingly important as the capillary becomes smaller. Notably, the lower frequency peak is nearly 50 % larger than the resonant harmonic peak for the bubble confined in the 15 μm channel. This observation suggests that, in small capillaries, harmonic imaging or nonlinear pulsing schemes can be performed more efficiently by driving the bubbles at half the resonance frequency, rather than at the resonance frequency itself as is traditionally done.

5.4.4 Nonspherical oscillations

Microbubbles are known to undergo shape oscillations either when driven with long ultrasound pulses, or in the presence of a boundary that breaks the symmetry of the system. The specific shapes, or modes of oscillation, depend on bubble size, bub-

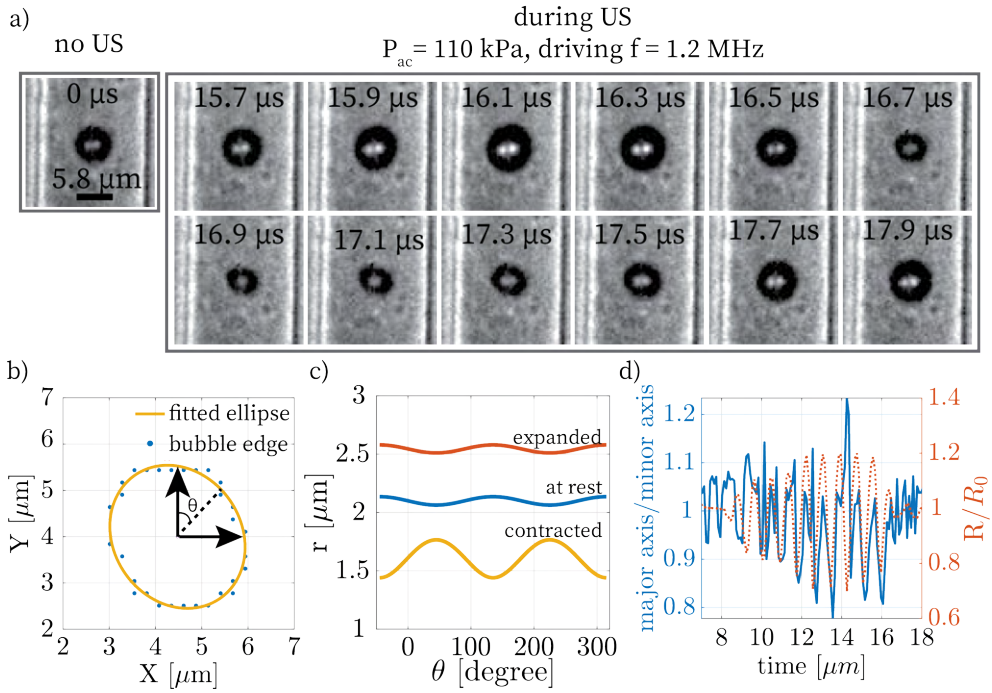


Figure 5.5: a) Frames from an ultra-high-speed recording showing the elliptical shape assumed by a $R_0 = 2$ μm microbubble in a D = 15 μm channel during ultrasound driving $P_{ac} = 100$ kPa. b) Elliptical fit to the microbubble boundary. c) Radius of the bubble versus angle in polar coordinates. d) Normalized radius of bubble plotted together with the ratio of major and minor axis of bubble as a function of time.

ble shell, ultrasound pressure, ultrasound frequency, and pulse length [135], [148]. Numerical studies have predicted that non-spherical oscillations could arise from confinement in a tube [134], [149]. During our experiments, we have observed that, at high pressures (> 100 kPa), bubble oscillations indeed become nonspherical, especially around resonance.

Figure 5.5a shows an example of non-spherical oscillations in a 15 μm microchannel. We quantify this change in the bubble shape by fitting an ellipse to the bubble contour, as demonstrated in Fig. 5.5b. The resulting shape, in polar coordinates, is shown in Fig. 5.5c. According to Jang et al. [149], when the bubble contracts, the major axis is orthogonal to the channel axis while, during expansion, the long axis aligns with the channel axis. This is in line with Fig. 5.5a and c, albeit with a small but measurable tilt of $\sim 45°$ between the axis of the bubble and the microchannel

axis. For a more quantitative analysis, Fig. 5.5d shows the ratio between the minor and major axis for each frame, together with the radius-time curve of the bubble oscillations. As expected, the change in aspect ratio is synchronized with the volumetric oscillations of the bubble and that the change in the orientation of the long axis of the bubble is cyclic. The reader is referred to the supplementary material for the corresponding videos.

5.5 Discussion

Oguz and Prosperetti et al. [129], along with other researchers [150], [151], have developed theoretical equations to determine the resonance frequency of both large and small bubbles in rigid tubes. These theories require that the tubes have a finite length. Consequently, the resonance frequency converges to zero for very long microchannels [152] which is thus physically different from a bubble trapped in a long but flexible capillary blood vessel. Furthermore, the oscillation of a bubble in a rigid tube is influenced by its position relative to the open ends of the tube, as well as to the axis of the tube, since volumetric oscillations require a displacement of the surrounding liquid. We have not investigated these effects here, but they warrant more attention in future endeavors.

The resonance observed in our experiments using long PDMS tubes (1.5 cm) suggests that either the surrounding water (or hydrogel) is compressible or that the microchannel itself is deformable. Considering the shear modulus on the order of 500 kPa for PDMS, and 50 kPa for soft tissue, as compared to a bulk modulus of over 2 GPa for water, it is likely that bubble oscillations are predominantly enabled by the elasticity of the PDMS. This conclusion thus holds a fortiori for soft tissue. Optical imaging of the deformation of the capillary walls at low pressures ($P_{ac} < 100$ kPa) is challenging due to the limited optical resolution. Higher pressures were therefore applied to the microbubbles in order to assess the presence of wall deformation. The result is shown in Fig. 5.6b for an acoustic pressure $P_{ac} = 3.0$ MPa, and a frequency of 1.0 MHz. Here, wall deformations are indeed clearly visible as the bubble expands and compresses.

In the experimental setup, Fig. 5.1, the sample and ultrasound wave are at a 45° angle with respect to each other. This angle leads to a reflection from the water/PDMS interface due to the acoustic impedance mismatch, which then results in a loss of pressure amplitude. The acoustic attenuation within PDMS on the other

hand can be neglected over the 1 mm distance needed to reach the bubble from the bottom of the PDMS slab. In addition, the low speed of sound in PDMS (1050 m/s [147]) leads to wave refraction and thus to a shift in the location of the ultrasound focus. However, with a 2 mm focal width (measured with a PVDF hydrophone), the effect of refraction is also negligible.

While the optical characterization of microbubbles provides valuable insight into resonance behavior, this approach is limited by resolution and a short recording duration of 25.6 μs at the necessary frame rate of ten million frames per second. In contrast, acoustic characterization methods such as an acoustical camera [97], [98] provide an interesting alternative, offering insight into nonlinear scattered pressure from microbubbles in viscoelastic microchannels over longer driving pulses. On the other hand, such an acoustical camera approach cannot capture the non-spherical oscillations which, so far, are critical details observable only with optical means.

5.6 Conclusion

In this study, we have experimentally characterized the dynamics of ultrasound contrast agents in viscoelastic microchannels with diameters ranging from 15 to 100 μm using ultra-high-speed imaging. Our findings confirm that the microchannel diameter and viscoelasticity of the confinement significantly impact the resonance behavior of microbubbles. The resonance frequency decreases by up to 60% and the damping increases by about 50% as the diameter of the microchannel decreases to 15 μm. As the capillary diameter decreases, the harmonic spectra also show a shift from the second harmonic being dominantly generated at resonance toward a second harmonic response dominated by bubbles driven at half their resonance frequency. This suggests that harmonic imaging or nonlinear pulsing schemes could drive bubbles more efficiently at half the resonance frequency rather than directly at resonance. The existence of a resonance indicates that even for relatively stiff materials the wall elasticity plays a key role in the dynamics of confined bubbles. These experimental results contribute to the development of models that could accurately describe microbubble oscillations in the confinement of capillaries, enabling more precise predictions and improved clinical applications.

5.7 Acknowledgments

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5.8 Appendix

The viscoelastic parameters of PDMS were characterized using a rheometer in a parallel plate setup. The frequency was swept between ∼0.001 Hz and ∼16 Hz at 0.1% strain. The data was then fitted to the Chasset-Thirion model [153], see Fig. 5.6a. The Chasset-Thirion power law is rewritten so as to separate its storage and loss components:

$$\begin{aligned} G = G' + jG'' = G(1 + (\tau\omega)^n) \\ = \left[G + (\tau\omega)^n cos(\frac{n\pi}{2})\right] + j\left[(\tau\omega)^n sin(\frac{n\pi}{2})\right]. \end{aligned} \tag{5.6}$$

Here, G represents the shear modulus, G' and G'' denote the storage and loss moduli, respectively, τ stands for the material's response time, and ω is the strain rate. To maximize the fitting precision, the analysis was carried out in two steps. Initially, only the loss modulus was fitted using the nonlinear least squares solver `lsqcurvefit` in MATLAB. In the subsequent step, the power law exponent n was retained from the first step, while both G and τ were refitted for both the storage and loss moduli simultaneously using the same solver [143].

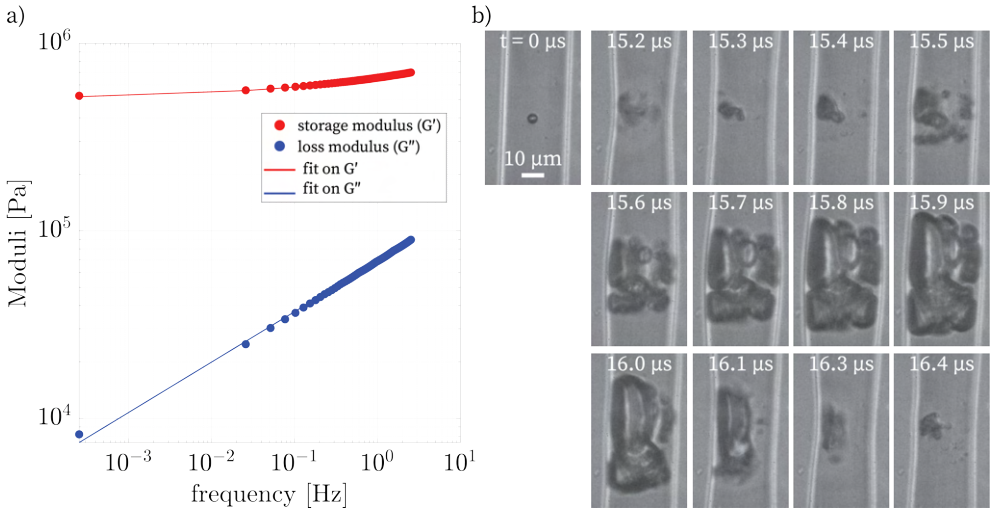


Figure 5.6: a) Rheometer frequency sweep measurement of PDMS with a shear strain of 0.1 %. The data points are fitted with a Chasset-Thirion model b) Frames from an ultra-high-speed recording. A $R_0 = 2.2\ \mu m$ bubble in a $D = 25\ \mu$m channel, driven with an acoustic wave with f= 1 MHz, $P_{ac} = 3$ MPa, and a duration of 10 cycles.

CHAPTER 6

Conclusions and future directions

Abstract

In this thesis, we have investigated the interaction between phospholipid-coated microbubbles and viscoelastic media. We began with a simple configuration a microbubble in bulk hydrogel and used the microbubble's response to measure the elastic and loss moduli of the hydrogels at MHz strain rates. We then simulated the impact of hydrogel elasticity on the scattered signal from microbubbles using a linear viscoelastic model, laying the groundwork for the next chapter, where we measured scattered pressures and extracted resonance curves of microbubbles in media with different shear moduli. With a better understanding of microbubble interactions in bulk viscoelastic media, we finally moved on to exploring microbubble dynamics within microchannels. We have investigated the effects of microchannel diameter and viscoelasticity on resonance curves and second harmonics generation and have compared these results to the simulation of coated bubble in the free-field. These studies aim to provide a clearer understanding of the interplay between microbubbles and viscoelastic media, contributing to the optimization of ultrasound-mediated therapies for therapeutic applications.

6.1 Conclusions

In Chapter 2, we have extracted the rheological properties of polyacrylamide hydrogels under shear rates of several MHz induced by microbubbles. We have compared these to results obtained at shear rates of several Hz using a rotational rheometer. The microbubble-probed measurements indicated an elastic modulus of 70 kPa for soft hydrogels and 270 kPa for stiff hydrogels. The findings showed comparable storage moduli at both high and low frequencies, while the loss modulus increased by three orders of magnitude at high frequencies, consistent with the minimum viscous loss that can be expected from water. This trend aligns with predictions from the linear Kelvin-Voigt viscoelastic model.

In Chapter 3, we have modeled the scattered pressure of microbubbles in media with varying elastic moduli and observed that the resonance frequency increased with the storage modulus. For a 2.0 μm bubble, this increase is calculated to be 50% when the medium elasticity rises from 0 to 200 kPa for a driving pressure of 120 kPa. While the radial excursion of this bubble decreased by 45% in a medium with an elastic modulus of 200 kPa, the scattered pressure increased by approximately 50% as compared to a bubble in water. An interesting observation was the effect of elasticity on the subharmonic response: their amplitude increased as a function of the storage modulus when driven at the resonance frequency but dropped when driven at twice the resonance frequency, contrary to what is observed in water.

In Chapter 4, building on the simulations from Ch. 3, we have experimentally measured the resonance curves of monodisperse microbubbles in bulk viscoelastic media with three different elastic moduli. The results showed a trend consistent with the simulations: the resonance frequency increased as the medium elasticity increased. For a soft hydrogel with an elastic modulus of 7 kPa, the resonance frequency was 1.6 MHz; for a semi-stiff hydrogel with an elastic modulus of 40 kPa, it was 2.0 MHz; and for a stiff hydrogel with an elastic modulus of 110 kPa, the resonance frequency was 3.0 MHz. A higher elastic modulus was calculated at MHz strain rates compared to measurements made with a rotational rheometer at a few tens of Hertz. In addition a model was developed to calculate the local pressure inside the hydrogel, showing that the pressure inside the semi-stiff and stiff hydrogels was approximately 45% and 70% higher than that in water.

In Chapter 5, we have recorded the dynamics of microbubbles in viscoelastic microchannels with diameters ranging from 15 to 100 μm using an ultra-high-speed

camera and compared them to bubbles in free-field conditions. The resonance curves revealed a decrease in resonance frequency as the microchannel diameter decreased, accompanied by an on average 50% increase in damping when positioned within the channels. Viscoelasticity of the microchannel walls played a critical role in maintaining resonance in channels being 1000 times longer than the size of the bubbles. Additionally, the nonlinear behavior of the bubbles in the microchannels was investigated, showing that, in small microcapillaries, harmonic imaging or nonlinear pulsing schemes should be performed by driving the bubbles at half the resonance frequency rather than at the resonance frequency to increase the signal-to-noise ratio.

These chapters provide insights into the interaction between microbubbles and viscoelastic media, both in bulk and confined environments, which will enhance the understanding of microbubble behavior for imaging and therapeutic applications.

6.2 FUTURE DIRECTIONS

Our intial introduction in Ch. 1 outlined our objective to study the acoustic cavitation of microbubbles in viscoelastic materials at low acoustic driving pressures. This topic represents an area of cavitation that has been less studied, and the literature was not sufficiently developed. We aimed to address the limited availability of experimental data in this domain. Throughout the chapters, the focus remained on the physical aspects of microbubble cavitation and its interaction with viscoelastic media. In this future outlook we will now expand on establishing a connection between these physical insight and associated therapeutic applications. We will also discuss how microbubble physics can bridge to biomedical engineering in order to understand how the miscrovasculature and surrounding viscoelastic tissue material may influence not only the response of microbubbles, but to provide a more comprehensive insight how these physical mechanisms may be beneficial for an efficient ultrasound-driven and bubble-mediated drug delivery systems.

6.2.1 Bubble dynamics near a viscoelastic boundary

Cavitation induces biophysical effects on neighboring cells by pushing and pulling the cell membrane, thereby enhancing drug delivery through a mechanism called sonoporation [154]. The rupture of cell membranes via acoustic means creates new opportunities for the minimally invasive delivery of large molecules into cells. Un-

derstanding the physical mechanisms behind membrane rupture is crucial for increasing the efficiency of treated cells [122], [155], [156].

Bubble collapse occurs when a bubble in a fluid, exposed to an intense acoustic field, undergoes rapid expansion during low-pressure phases and sudden collapse during high-pressure phases. If the collapse is violent, it releases a large amount of energy. In a free field, the collapse is typically symmetrical, creating shock waves that spread outward [157]. However, near a solid wall, or in the presence of strong pressure gradients, the collapse becomes asymmetric. Upon expansion liquid is pushed away in all directions. Upon compression however, the suction of the surrounding liquid is hindered especially from the side of the wall. This imbalance causes the bubble to flatten toward the wall and generates a high-speed jet of liquid that penetrates through the bubble and is directed to the rigid wall [21], [158], [159].

Numerical models that describe bubble cavitation near rigid walls have some limitations. They rely on simplified physical models and therefore none encompass the full problem in its multiple dimensions, including biomechanics, fluid dynamics, viscoelasticity, etc. Experimental studies using laser-generated bubbles in solid and viscoelastic materials like agar have been extensively explored for applications such as surface cleaning [21], [160]–[162] and kidney stone fragmentation [62]. While these studies have advanced our understanding of cavitation near solid and viscoelastic boundaries, the dynamics at the microscale and under low acoustic pressures remain poorly investigated.

One interesting future avenue will be to investigate bubble dynamics near tissue phantoms. We present some of our preliminary work in this area. Fig. 6.1a, illustrates the schematic of an endothelial membrane of a blood vessel in proximity to a cancer cell. The dynamics of microbubbles, influenced by driving pressure, number of cycles, and tissue stiffness, are of particular interest in this context. The experimental setup proposed to capture these dynamics is depicted in Fig. 6.1b. A phospholipid-coated microbubble floating against a tissue phantom could be recorded under ultra-high-speed imaging. Figures 6.1c and d provide examples of jets captured in this setup. This novel side-view setup allows to observe the dynamics of microbubbles in more detail than the more conventional top-view approaches that may provide limited access to the relevant phenomena of jetting.

We also observed that it is easier to generate a jet near a stiffer hydrogel likely due to the higher pressure gradient between the free surface of the bubble and the wall. Additionally, increasing the number of cycles facilitates jet formation, as can

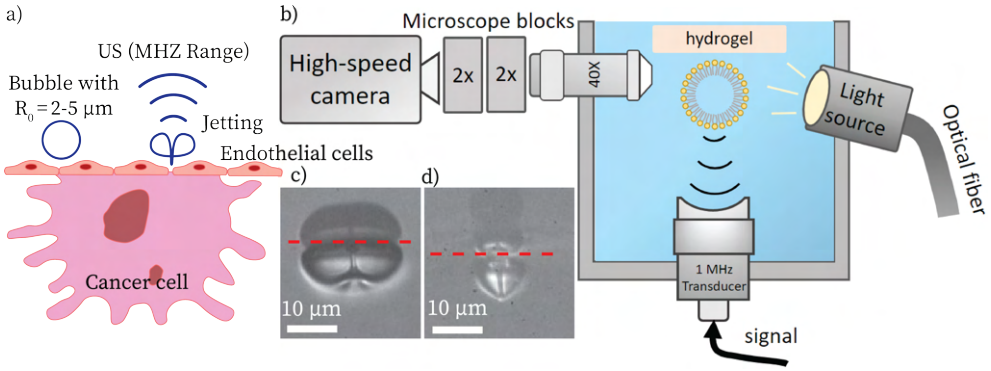


Figure 6.1: a) Schematic of a liquid jet penetrating the endothelial wall of a vessel and interacting with a cancer cell. b) Side-view setup for capturing liquid jets. c) Liquid jet directed toward a solid microscope slide with a thickness of 1 mm, driven by a 500 kPa pressure, 1 cycle, and 1 MHz frequency. d) Liquid jet moving outward from the surface of a gel with a shear modulus of 200 kPa, driven by 500 kPa pressure, 8 cycles, and 1 MHz frequency.

be seen in Fig. 6.2a, where the jet emanates during the fifth cycle. Jets can also form outward from the surface, particularly when bubbles are driven at a higher number of cycles. For example, in Fig. 6.2a, the bubble was driven with one cycle near a gel with a shear modulus of 70 kPa, and after 8 cycles (at t = 13.3 μs), a jet was generated directed towards the elastic wall, followed by an outward jet at t = 15.7 μs.

In Fig. 6.2b, the bubble was driven with a single cycle at a frequency of 1.0 MHz and a pressure of 500 kPa near a gel with an elastic modulus of 200 kPa. During oscillation, the bubble deformed the gel surface, and upon collapse, it generated a traveling wave on the surface. This phenomenon has potential applications in shear-wave elastography (SWE), offering an alternative method based on bubbles and jets, other than conventional SWE that is based on ultrasound alone, to probe the mechanical properties of soft tissues [163].

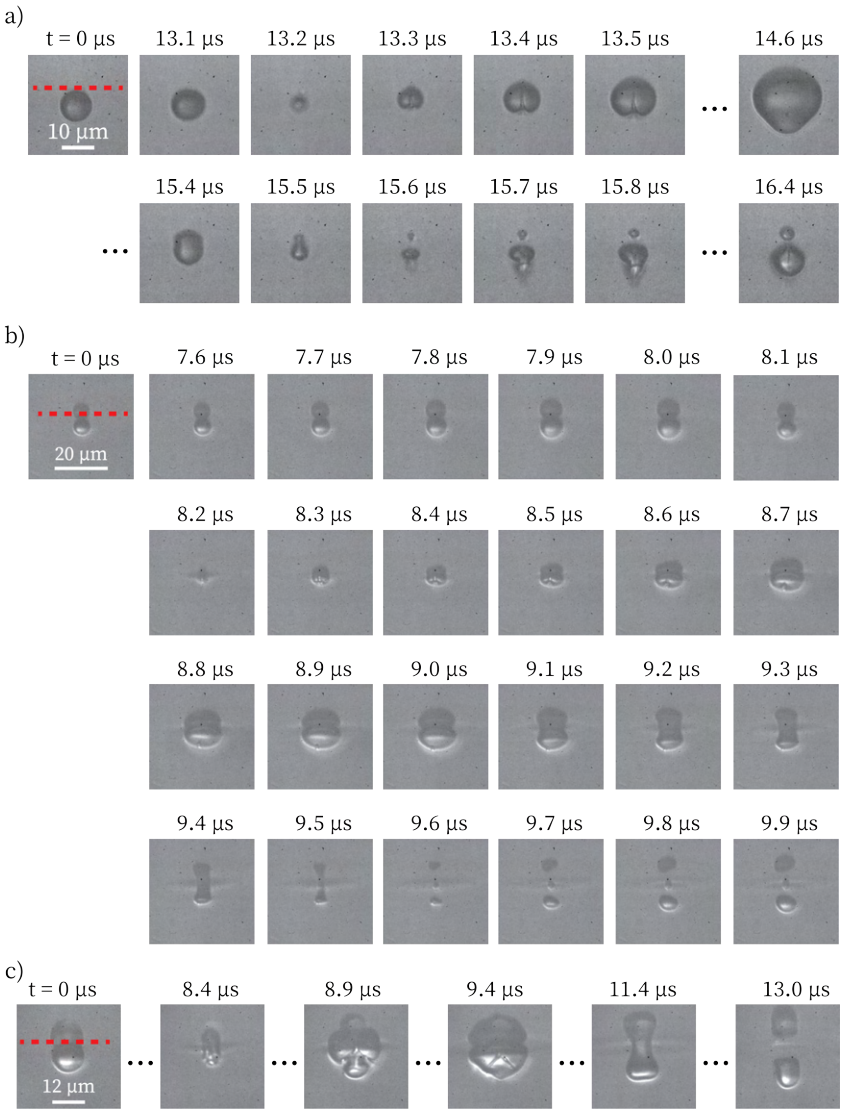


Figure 6.2: a) A bubble floating against a polyacrylamide hydrogel with a shear modulus of 70 kPa, driven by an 8-cycle pulse at 300 kPa pressure at a frequency of 1.0 MHz. b) A bubble floating against a polyacrylamide hydrogel with a shear modulus of 200 kPa, driven by a 1-cycle pulse at 500 kPa pressure and a frequency of 1.0 MHz. Note the surface deformation and the traveling wave in the hydrogel. c) Visible surface damage to a polyacrylamide hydrogel with a shear modulus of 200 kPa at t = 13.0 μs, driven by a 1-cycle pulse at 500 kPa pressure and a frequency of 1.0 MHz.

Liquid jets are fast and violent, and may cause surface damage or leave undesired defects, depending on the application [21]. An example of this can be observed in Fig. 6.2c at t = 13.0 μs, where signatures of rupture in the hydrogel are visible as the bubble moves away from the hydrogel surface. Here the elastic modulus of medium is 200 kPa and the bubble was driven with a one-cycle pressure wave with amplitude of 500 kPa and frequency of 1 MHz.

6.2.2 Sonoporation and drug delivery

Once the dynamics of microbubbles in bulk, near walls, and within viscoelastic capillaries are better understood, the underlying physical mechanisms of sonoporation may be elucidated. This knowledge then forms a foundation for translating these insights into biological applications. At this stage, the questions evolve from how microbubbles interact with their environment to how much they can effectively enhance drug delivery efficiency. We can ask ourselves the question, to what extent can ultrasound-driven microbubbles increase the efficacy of controlled local drug delivery?

Fig. 6.3a illustrates our experimental setup designed to study the propagation of fluorescent material in GelMA hydrogels with embedded tumor spheroids. Fig. 6.3b provides a schematic of the biological tissue prepared for this study, featuring microbubbles and drugs floating in a vessel surrounded by cancer cells. This setup is proposed to demonstrate the distribution rate of fluorescent material, acting as a model drug, in the surrounding medium, both in the absence and presence of cavitation.

As an example, Fig. 6.3c depicts a 100 μm microchannel with four tumor spheroids embedded in a GelMA hydrogel [164]. Microbubbles and fluorescent material were mixed and introduced in the microchannel. The fluorescent image shown in Fig. 6.3d illustrates the extent of dye permeation after 5 minutes into the surrounding medium under ultrasound exposure at 100 cycles and a pressure of 500 kPa at a driving frequency of 1.0 MHz, applied twice per minute. This study is highly complex, involving multiple physical factors in the experiments that need to be controlled, as well as biological factors that further challenge the setup. The propagation of dye over time can be analyzed, as well as the fluorescence intensity as a qualitative measure of the concentration.

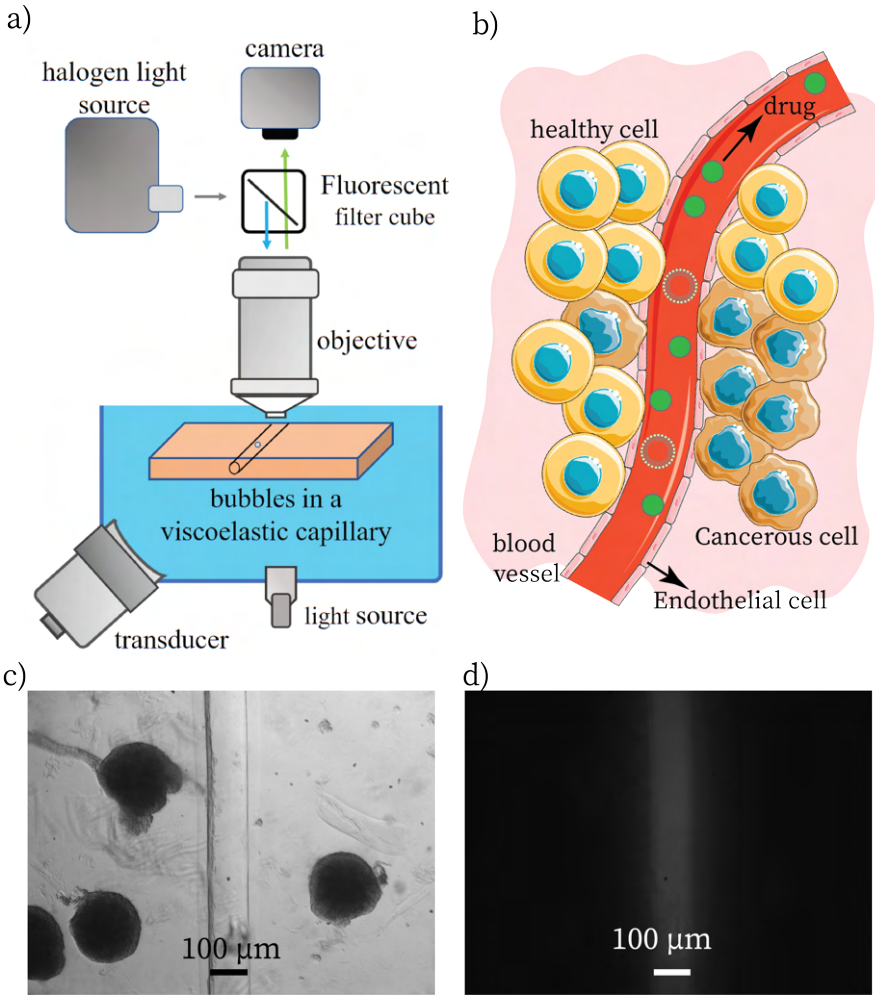


Figure 6.3: a) Experimental setup for investigating the efficacy of sonoporation through fluorescent imaging. b) Schematic representation of cancerous cells with a blood vessel containing drugs and microbubbles. c) Bright-field image of a sample featuring a microchannel with a diameter of 100 μm and four tumor spheroids. d) Fluorescent image corresponding to (c), illustrating the extent of dye permeation into the surrounding medium after 13 minutes of ultrasound exposure at 100 cycles, 500 kPa, and 1.0 MHz, at a pulse repetition rate of two times per minute.

One intriguing avenue for exploration is training a neural network to predict the distribution of a drug. By leveraging proper input data, such as the excitation frequency of microbubbles, anatomical images of the tumor location, and contrast images of the region of interest, a neural network could be designed and trained to predict both the location of microbubbles and the potential distribution of the drug [12].

The training dataset could be collected using dual-frequency transducers, which operate at lower frequencies (e.g., 2 MHz) for bubble excitation and higher frequencies (e.g. 8 MHz) for contrast imaging and monitoring bubble movements. Additionally, a 3D-printed vascularized tumor model could serve as an anatomical model, providing a controlled environment for initiating such a line of research.

6.2.3 Laser generated cavitation in viscoelastic media

One interesting area of research is laser-generated cavitation bubbles, a topic that has been extensively studied yet still holds potential for further exploration. To describe the radial dynamics of these bubbles, theoretical models of varying complexity have been developed [165]. Experimentally, cavitation bubbles have been investigated in diverse configurations, such as in free space in water [157], near rigid boundaries [166], inside and adjacent to viscoelastic media [161], and even near granular boundaries [167].

The schematic of such a setup is shown in Fig. 6.4. A laser capable of emitting short pulses with durations on the order of a few nanoseconds or less, and energies per pulse of a few millijoules with a tight focus, can be used. The laser pulse induces heating of impurities and/or dielectric breakdown via avalanche ionization, creating a plasma spot that expands to form the cavity. Fig. 6.4 a and b illustrates bubble generation by a Nd:YAG laser with a 8 ns pulse and the accompanying shock-wave emission (see Fig. 6.4d at t= 4 μs) inside a polyacrylamide hydrogel with elasticity of 200 kPa. Bubble dynamics occur at extremely high speeds, with the collapse of a bubble observable only through high-speed cameras.

In Fig. 6.4a, a bubble is formed in hydrogel that undergoes a rapid first collapse, emitting a shock wave (not shown in the pictures). Starting from its maximum radius, the bubble shrinks at an accelerating rate due to ambient pressure and elastic stress, compressing its contents (gas and vapor). This steep pressure increase stops the collapse, causing the bubble to rebound like a spring.

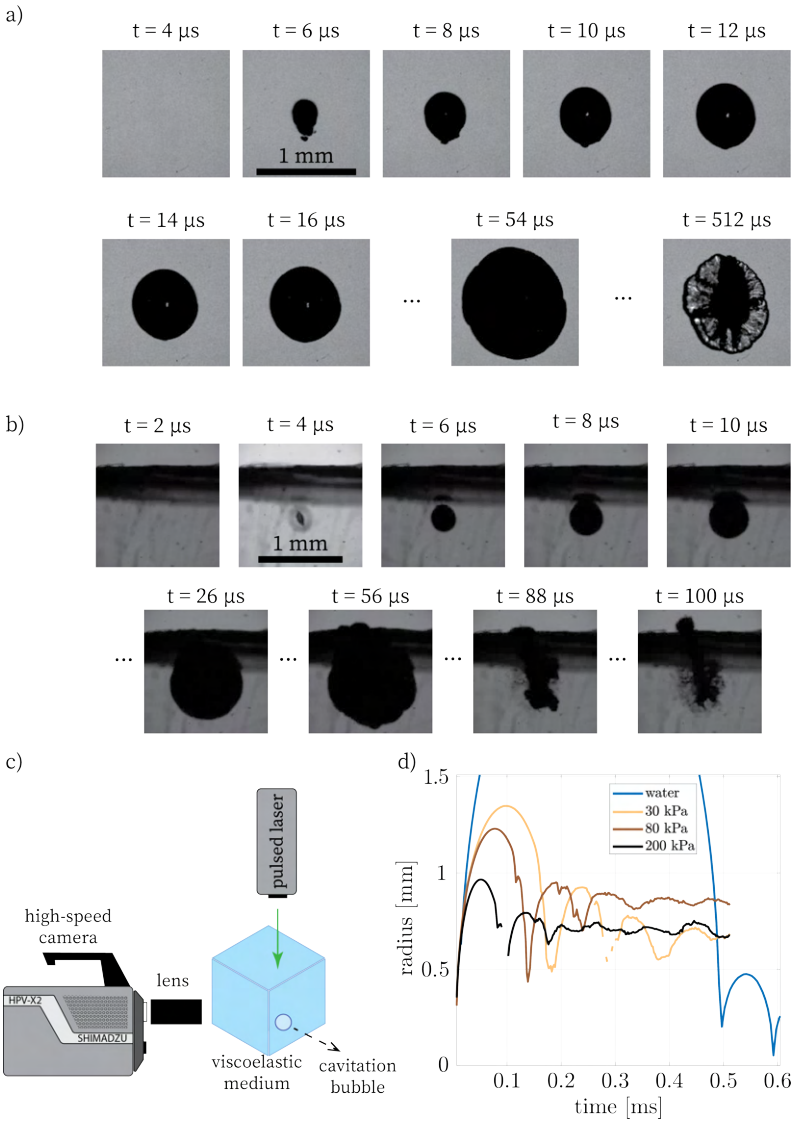


Figure 6.4: Laser-generated cavitation bubbles were created using a pulsed Nd:YAG laser with an 8 ns pulse width inside a polyacrylamide hydrogel with an elastic modulus of 200 kPa. The dynamics were recorded using a high-speed camera operating at 0.5 million frames per second. a) Bubble dynamics inside the hydrogel, far from boundaries. b) Bubble dynamics near the boundary with water. c) Schematic of the experimental setup. d) Radius-time curve extracted from the high-speed recordings.

Figure 6.4b shows the radius-time curves of laser-generated bubbles in hydrogels with elastic moduli of 30, 80, and 200 kPa. The blue curve corresponds to a bubble in water excited with the same laser pulse. The maximum bubble amplitude is unknown as the bubble expands out of the frame. It can be observed that as the stiffness of the hydrogel increases, the amplitude of oscillation decreases, and the collapse time becomes shorter in the stiffer hydrogels [103].

One avenue to explore is the study of bubble dynamics inside a viscoelastic medium near a boundary (interface with water). Examples of such configurations are shown in Fig. 6.4b. The inverse configuration, where the cavity is in water near a viscoelastic boundary, has been extensively studied [100], [161].

6.2.4 Cavitation in microchannels and modeling

Although in Chapter 5 we aimed to determine the resonance frequency of microbubbles in microchannels, several simplifications were made that could be addressed in future studies. For instance, the experiments were conducted in microchannels made of PDMS with a single stiffness, whereas bubble response depends on the viscoelasticity of the medium. Additionally, future experiments could rely on acoustic techniques and analyze the scattered signals from microbubbles, which are closer to practical applications, rather than using high-speed imaging. Exciting the bubbles with chirp waves could also facilitate the acquisition of resonance curves.

The bubble dynamics were measured at the center of the channel, where the effect of the walls was symmetric. However, modeling off-center bubble dynamics by considering the phospholipid shell would be more informative, as it could better mimic practical conditions [168]. Ultimately, the cavitation data obtained could enhance the training of neuronal networks used for predicting bubble location and optimizing sonoporation.

In this section, we present some noteworthy studies that, while not quantified or suitable for standalone chapters, provide valuable insights for future studies. These topics bridge the gap between complex mathematical modeling and physical experiments, ultimately guiding us toward practical applications aimed at enhancing ultrasound-mediated therapies in real-life scenarios.

References

[1] R. Gramiak and P. M. Shah, "Echocardiography of the aortic root," Investigative Radiology, vol. 3, no. 5, pp. 356–366, 1968.

[2] T. Segers, L. De Rond, N. De Jong, M. Borden, and M. Versluis, "Stability of monodisperse phospholipid-coated microbubbles formed by flow-focusing at high production rates," Langmuir, vol. 32, no. 16, pp. 3937–3944, 2016.

[3] A. M. Gañán-Calvo and J. M. Gordillo, "Perfectly monodisperse microbubbling by capillary flow focusing," Physical Review Letters, vol. 87, no. 27, p. 274 501, 2001.

[4] S. L. Anna, N. Bontoux, and H. A. Stone, "Formation of dispersions using flow focusing in microchannels," Applied Physics Letters, vol. 82, no. 3, pp. 364–366, 2003.

[5] D. Chatterjee and K. Sarkar, "A newtonian rheological model for the interface of microbubble contrast agents," Ultrasound in Medicine & Biology, vol. 29, no. 12, pp. 1749–1757, 2003.

[6] P. Marmottant, S. Van Der Meer, M. Emmer, et al., "A model for large amplitude oscillations of coated bubbles accounting for buckling and rupture," Journal of the Acoustical Society of America, vol. 118, no. 6, pp. 3499–3505, 2005.

[7] T. Leighton and R. E. Apfel, "The acoustic bubble," Journal of Acoustical Society of America, vol. 96, no. 4, p. 2616, 1994.

[8] M. Versluis, E. Stride, G. Lajoinie, B. Dollet, and T. Segers, "Ultrasound contrast agent modeling: A review," Ultrasound in Medicine & Biology, vol. 46, no. 9, pp. 2117–2144, 2020.

[9] N. De Jong, F. Ten Cate, C. Lancee, J. Roelandt, and N. Bom, "Principles and recent developments in ultrasound contrast agents," Ultrasonics, vol. 29, no. 4, pp. 324–330, 1991.

[10] G. T. Taneva, H. Mirgolbabaee, E. Groot Jebbink, M. M. Reijnen, and K. P. Donas, "Systematic review of the current in vitro experience of the endovascular treatment of juxtarenal abdominal aortic aneurysms by fenestrated and parallel endografting," Journal of Endovascular Therapy, vol. 30, no. 1, pp. 8–17, 2023.

[11] A. Coudert, L. Denis, A. Chavignon, et al., "3D transcranial ultrasound localization microscopy reveals major arteries in the sheep brain," bioRxiv, pp. 2024–02, 2024.

[12] N. Blanken, J. M. Wolterink, H. Delingette, C. Brune, M. Versluis, and G. Lajoinie, "Super-resolved microbubble localization in single-channel ultrasound RF signals using deep learning," IEEE Transactions on Medical Imaging, vol. 41, no. 9, pp. 2532–2542, 2022.

[13] J. A. Feshitan, C. C. Chen, J. J. Kwan, and M. A. Borden, "Microbubble size isolation by differential centrifugation," Journal of Colloid and Interface Science, vol. 329, no. 2, pp. 316–324, 2009.

[14] T. Segers and M. Versluis, "Acoustic bubble sorting for ultrasound contrast agent enrichment," Lab on a Chip, vol. 14, no. 10, pp. 1705–1714, 2014.

[15] B. van Elburg, G. Collado-Lara, G.-W. Bruggert, T. Segers, M. Versluis, and G. Lajoinie, "Feedback-controlled microbubble generator producing one million monodisperse bubbles per second," Review of Scientific Instruments, vol. 92, no. 3, p. 035 110, 2021.

[16] M. B. De Matos, R. Deckers, B. van Elburg, et al., "Ultrasound-sensitive liposomes for triggered macromolecular drug delivery: Formulation and in vitro characterization," Frontiers in Pharmacology, vol. 10, p. 1463, 2019.

[17] R. Gaudron, M. Warnez, and E. Johnsen, "Bubble dynamics in a viscoelastic medium with nonlinear elasticity," Journal of Fluid Mechanics, vol. 766, pp. 54–75, 2015.

[18] F. Forsberg, I. Gupta, P. Machado, et al., "Contrast-enhanced subharmonic aided pressure estimation (SHAPE) using ultrasound imaging with a focus on identifying portal hypertension," Journal of Visualized Experiments: JoVE, no. 166, pp. 10–3791, 2020.

[19] W. van Hoeve, M. de Vargas Serrano, L. Te Winkel, et al., "Improved sensitivity of ultrasound-based subharmonic aided pressure estimation using monodisperse microbubbles," Journal of Ultrasound in Medicine, vol. 41, no. 7, pp. 1781–1789, 2022.

[20] S. M. Van der Meer, B. Dollet, M. M. Voormolen, et al., "Microbubble spectroscopy of ultrasound contrast agents," Journal of the Acoustical Society of America, vol. 121, no. 1, pp. 648–656, 2007.

[21] C.-D. Ohl, M. Arora, R. Dijkink, V. Janve, and D. Lohse, "Surface cleaning from laser-induced cavitation bubbles," Applied Physics Letters, vol. 89, no. 7, 2006.

[22] B. Dollet, P. Marmottant, and V. Garbin, "Bubble dynamics in soft and biological matter," Annual Review of Fluid Mechanics, vol. 51, pp. 331–355, 2019.

[23] J. S. Allen and R. A. Roy, "Dynamics of gas bubbles in viscoelastic fluids. II. Nonlinear viscoelasticity," Journal of the Acoustical Society of America, vol. 108, no. 4, pp. 1640–1650, 2000.

[24] J. B. Keller and M. Miksis, "Bubble oscillations of large amplitude," Journal of the Acoustical Society of America, vol. 68, no. 2, pp. 628–633, 1980.

[25] T. Segers, E. Gaud, M. Versluis, and P. Frinking, "High-precision acoustic measurements of the nonlinear dilatational elasticity of phospholipid coated monodisperse microbubbles," Soft Matter, vol. 14, no. 47, pp. 9550–9561, 2018.

[26] E. Stride, T. Segers, G. Lajoinie, et al., "Microbubble agents: New directions," Ultrasound in Medicine & Biology, vol. 46, no. 6, pp. 1326–1343, 2020.

[27] S. Roovers, T. Segers, G. Lajoinie, et al., "The role of ultrasound-driven microbubble dynamics in drug delivery: From microbubble fundamentals to clinical translation," Langmuir, vol. 35, no. 31, pp. 10 173–10 191, 2019.

[28] C. S. Lee and K. W. Leong, "Advances in microphysiological blood-brain barrier (BBB) models towards drug delivery," Current Opinion in Biotechnology, vol. 66, pp. 78–87, 2020.

[29] K. F. Timbie, B. P. Mead, and R. J. Price, "Drug and gene delivery across the blood-brain barrier with focused ultrasound," Journal of Controlled Release, vol. 219, pp. 61–75, 2015.

[30] S. M. van der Meer, B. Dollet, M. M. Voormolen, et al., "Microbubble spectroscopy of ultrasound contrast agents," Journal of the Acoustical Society of America, vol. 121, no. 1, pp. 648–656, 2007.

[31] J. Sijl, M. Overvelde, B. Dollet, et al., "Compression-only behavior: A second-order nonlinear response of ultrasound contrast agent microbubbles," Journal of the Acoustical Society of America, vol. 129, no. 4, pp. 1729–1739, 2011.

[32] T. Segers, P. Kruizinga, M. P. Kok, G. Lajoinie, N. De Jong, and M. Versluis, "Monodisperse versus polydisperse ultrasound contrast agents: Non-linear response, sensitivity, and deep tissue imaging potential," Ultrasound in Medicine & Biology, vol. 44, no. 7, pp. 1482–1492, 2018.

[33] A. Jamburidze, M. De Corato, A. Huerre, A. Pommella, and V. Garbin, "High-frequency linear rheology of hydrogels probed by ultrasound-driven microbubble dynamics," Soft Matter, vol. 13, no. 21, pp. 3946–3953, 2017.

[34] E. Vlaisavljevich, K.-W. Lin, M. T. Warnez, et al., "Effects of tissue stiffness, ultrasound frequency, and pressure on histotripsy-induced cavitation bubble behavior," Physics in Medicine & Biology, vol. 60, no. 6, p. 2271, 2015.

[35] J. S. Allen and R. A. Roy, "Dynamics of gas bubbles in viscoelastic fluids. I. Linear viscoelasticity," Journal of the Acoustical Society of America, vol. 107, no. 6, pp. 3167–3178, 2000.

[36] A. T. Oratis, K. Dijs, G. Lajoinie, M. Versluis, and J. H. Snoeijer, "A unifying Rayleigh-Plesset-type equation for bubbles in viscoelastic media," Journal of the Acoustical Society of America, vol. 155, no. 2, pp. 1593–1605, 2024.

[37] J. B. Estrada, C. Barajas, D. L. Henann, E. Johnsen, and C. Franck, "High strain-rate soft material characterization via inertial cavitation," Journal of the Mechanics and Physics of Solids, vol. 112, pp. 291–317, 2018.

[38] H. S. Fogler and J. D. Goddard, "Collapse of spherical cavities in viscoelastic fluids," Physics of Fluids, vol. 13, no. 5, pp. 1135–1141, 1970.

[39] A. Niederquell, A. C. Völker, and M. Kuentz, "Introduction of diffusing wave spectroscopy to study self-emulsifying drug delivery systems with respect to liquid filling of capsules," International Journal of Pharmaceutics, vol. 426, no. 1-2, pp. 144–152, 2012.

[40] Z. Xing, A. Caciagli, T. Cao, et al., "Microrheology of DNA hydrogels," Proceedings of the National Academy of Sciences, vol. 115, no. 32, pp. 8137–8142, 2018.

[41] V. Bertin, Z. Zhang, R. Boisgard, et al., "Contactless rheology of finite-size air-water interfaces," Physical Review Research, vol. 3, no. 3, pp. L0–32 007, 2021.

[42] S. Catheline, J.-L. Gennisson, G. Delon, et al., "Measurement of viscoelastic properties of homogeneous soft solid using transient elastography: An inverse problem approach," Journal of the Acoustical Society of America, vol. 116, no. 6, pp. 3734–3741, 2004.

[43] A. Prosperetti, "A generalization of the Rayleigh–Plesset equation of bubble dynamics," Physics of Fluids, vol. 25, no. 3, pp. 409–410, 1982.

[44] T. Leighton, "The acoustic bubble," Academic press, 2012.

[45] T. Leighton, "The RayleighPlesset equation in terms of volume with explicit shear losses," Ultrasonics, vol. 48, no. 2, pp. 85–90, 2008.

[46] A. Prosperetti, L. A. Crum, and K. W. Commander, "Nonlinear bubble dynamics," Journal of the Acoustical Society of America, vol. 83, no. 2, pp. 502–514, 1988.

[47] E. Brujan, "Cavitation in non-newtonian fluids: With biomedical and bioengineering applications," Springer Science & Business Media, 2010.

[48] H. A. Barnes, "A handbook of elementary rheology," University of Wales, Institute of Non-Newtonian Fluid Mechanics, 2000.

[49] C. Devin Jr, "Survey of thermal, radiation, and viscous damping of pulsating air bubbles in water," Journal of the Acoustical Society of America, vol. 31, no. 12, pp. 1654–1667, 1959.

[50] J. Sijl, B. Dollet, M. Overvelde, et al., "Subharmonic behavior of phosp - holipid-coated ultrasound contrast agent microbubbles," Journal of the Acoustical Society of America, vol. 128, no. 5, pp. 3239–3252, 2010.

[51] J. Liu, H. Zheng, P. S. Poh, H.-G. Machens, and A. F. Schilling, "Hydrogels for engineering of perfusable vascular networks," International Journal of Molecular Sciences, vol. 16, no. 7, pp. 15 997–16 016, 2015.

[52] P. A. Janmey and R. T. Miller, "Mechanisms of mechanical signaling in development and disease," Journal of Cell Science, vol. 124, no. 1, pp. 9–18, 2011.

[53] T. Segers, E. Gaud, G. Casqueiro, A. Lassus, M. Versluis, and P. Frinking, "Foam-free monodisperse lipid-coated ultrasound contrast agent synthesis by flow-focusing through multi-gas-component microbubble stabilization," Applied Physics Letters, vol. 116, no. 17, 2020.

[54] A. S. Mikhail, A. H. Negussie, C. Graham, M. Mathew, B. J. Wood, and A. Partanen, "Evaluation of a tissue-mimicking thermochromic phantom for radiofrequency ablation," Medical Physics, vol. 43, no. 7, pp. 4304–4311, 2016.

[55] A. H. Negussie, A. Partanen, A. S. Mikhail, et al., "Thermochromic tissue-mimicking phantom for optimisation of thermal tumour ablation," International Journal of Hyperthermia, vol. 32, no. 3, pp. 239–243, 2016.

[56] G. Lajoinie, Y. Luan, E. Gelderblom, et al., "Non-spherical oscillations drive the ultrasound-mediated release from targeted microbubbles," Communications Physics, vol. 1, no. 1, p. 22, 2018.

[57] M. Overvelde, V. Garbin, J. Sijl, et al., "Nonlinear shell behavior of phospholipid-coated microbubbles," Ultrasound in Medicine & Biology, vol. 36, no. 12, pp. 2080–2092, 2010.

[58] A. F. Prokop, S. Vaezy, M. L. Noble, P. J. Kaczkowski, R. W. Martin, and L. A. Crum, "Polyacrylamide gel as an acoustic coupling medium for focused ultrasound therapy," Ultrasound in Medicine & Biology, vol. 29, no. 9, pp. 1351–1358, 2003.

[59] J. B. Estrada, C. Barajas, D. L. Henann, E. Johnsen, and C. Franck, "High strain-rate soft material characterization via inertial cavitation," Journal of the Mechanics and Physics of Solids, vol. 112, pp. 291–317, 2018.

[60] C. A. Sennoga, J. S. Yeh, J. Alter, et al., "Evaluation of methods for sizing and counting of ultrasound contrast agents," Ultrasound in Medicine & Biology, vol. 38, no. 5, pp. 834–845, 2012.

[61] M. Bruning, M. Costalonga, J. Snoeijer, and A. Marin, "Turning drops into bubbles: Cavitation by vapor diffusion through elastic networks," Physical Review Letters, vol. 123, no. 21, p. 214 501, 2019.

[62] A. J. Coleman, J. E. Saunders, L. A. Crum, and M. Dyson, "Acoustic cavitation generated by an extracorporeal shockwave lithotripter," Ultrasound in Medicine & Biology, vol. 13, no. 2, pp. 69–76, 1987.

[63] G. ter Haar and C. Coussios, "High intensity focused ultrasound: Physical principles and devices," International Journal of Hyperthermia, vol. 23, no. 2, pp. 89–104, 2007.

[64] K. Hynynen, N. McDannold, N. Vykhodtseva, and F. A. Jolesz, "Noninvasive MR imaging–guided focal opening of the blood-brain barrier in rabbits," Radiology, vol. 220, no. 3, pp. 640–646, 2001.

[65] Y. Guo, H. Lee, C. Kim, et al., "Ultrasound frequency-controlled microbubble dynamics in brain vessels regulate the enrichment of inflammatory pathways in the blood-brain barrier," Nature Communications, vol. 15, no. 1, p. 8021, 2024.

[66] Y. Engelen, D. V. Krysko, I. Effimova, et al., "Optimizing high-intensity focused ultrasound-induced immunogenic cell-death using passive cavitation mapping as a monitoring tool," Journal of Controlled Release, vol. 375, pp. 389–403, 2024.

[67] P. Frinking, T. Segers, Y. Luan, and F. Tranquart, "Three decades of ultrasound contrast agents: A review of the past, present and future improvements," Ultrasound in Medicine & Biology, vol. 46, no. 4, pp. 892–908, 2020.

[68] H. Mirgolbabaee, L. van de Velde, R. H. Geelkerken, M. Versluis, E. Groot Jebbink, and M. M. Reijnen, "Ultrasound particle image velocimetry to investigate potential hemodynamic causes of limb thrombosis after endovascular aneurysm repair with the anaconda device," Journal of Endovascular Therapy, 2023. DOI: 10.1177/15266028231219988.

[69] O. Couture, S. Bannouf, G. Montaldo, J.-F. Aubry, M. Fink, and M. Tanter, "Ultrafast imaging of ultrasound contrast agents," Ultrasound in Medicine & Biology, vol. 35, no. 11, pp. 1908–1916, 2009.

[70] P. Sidhu, S. Ellis, J. Karani, and S. Ryan, "Hepatic artery stenosis following liver transplantation: Significance of the tardus parvus waveform and the role of microbubble contrast media in the detection of a focal stenosis," Clinical Radiology, vol. 57, no. 9, pp. 789–799, 2002.

[71] D. Daye and T. G. Walker, "Complications of endovascular aneurysm repair of the thoracic and abdominal aorta: Evaluation and management," Cardiovascular Diagnosis and Therapy, vol. 8, p. 138, 2018.

[72] C. R. Anderson, X. Hu, H. Zhang, et al., "Ultrasound molecular imaging of tumor angiogenesis with an integrin targeted microbubble contrast agent," Investigative Radiology, vol. 46, no. 4, pp. 215–224, 2011.

[73] N. Hosseinkhah, H. Chen, T. Matula, P. Burns, and K. Hynynen, "Mechanisms of microbubble–vessel interactions and induced stresses: A numerical study," Journal of the Acoustical Society of America, vol. 134, no. 3, pp. 1875–1885, 2013.

[74] V. Daeichin, J. G. Bosch, A. Needles, F. S. Foster, A. van der Steen, and N. de Jong, "Subharmonic, non-linear fundamental and ultraharmonic imaging of microbubble contrast at high frequencies," Ultrasound in Medicine & Biology, vol. 41, no. 2, pp. 486–497, 2015.

[75] P. Shankar, P. D. Krishna, and V. Newhouse, "Advantages of subharmonic over second harmonic backscatter for contrast-to-tissue echo enhancement," Ultrasound in Medicine & Biology, vol. 24, no. 3, pp. 395–399, 1998.

[76] N. De Jong and L. Hoff, "Ultrasound scattering properties of Albunex microspheres," Ultrasonics, vol. 31, no. 3, pp. 175–181, 1993.

[77] C. C. Church, H. B. Levene, and J. M. Keene, "Recent theoretical and experimental results with Albunexő, a contrast agent for diagnostic ultrasonography," Journal of the Acoustical Society of America, vol. 93, no. 4, pp. 2330–2331, 1993.

[78] K. Tsiglifis and N. A. Pelekasis, "Nonlinear radial oscillations of encapsulated microbubbles subject to ultrasound: The effect of membrane constitutive law," Journal of the Acoustical Society of America, vol. 123, no. 6, pp. 4059–4070, 2008.

[79] A. Sojahrood, H. Haghi, R. Karshafian, and M. Kolios, "Nonlinear dynamics and bifurcation structure of ultrasonically excited lipid coated microbubbles," Ultrasonics Sonochemistry, vol. 72, p. 105 405, 2021.

[80] J. S. Allen and R. A. Roy, "Dynamics of gas bubbles in viscoelastic fluids. II. Nonlinear viscoelasticity," Journal of the Acoustical Society of America, vol. 108, no. 4, pp. 1640–1650, 2000.

[81] C. W. Macosko, "Rheology principles," Measurements and Applications, 1994.

[82] K. B. Bader, E. Vlaisavljevich, and A. D. Maxwell, "For whom the bubble grows: Physical principles of bubble nucleation and dynamics in histotripsy ultrasound therapy," Ultrasound in Medicine & Biology, vol. 45, no. 5, pp. 1056–1080, 2019.

[83] C. C. Coussios and R. A. Roy, "Applications of acoustics and cavitation to noninvasive therapy and drug delivery," Annu. Rev. Fluid Mech., vol. 40, no. 1, pp. 395–420, 2008.

[84] J. Chomas, P. Dayton, D. May, and K. Ferrara, "Nondestructive subharmonic imaging," IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, vol. 49, no. 7, pp. 883–892, 2002.

[85] Y. Sun, D. E. Kruse, P. A. Dayton, and K. W. Ferrara, "High-frequency dynamics of ultrasound contrast agents," IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, vol. 52, no. 11, pp. 1981–1991, 2005.

[86] A. Katiyar and K. Sarkar, "Excitation threshold for subharmonic generation from contrast microbubbles," Journal of the Acoustical Society of America, vol. 130, no. 5, pp. 3137–3147, Nov. 2011.

[87] A. Katiyar and K. Sarkar, "Effects of encapsulation damping on the excitation threshold for subharmonic generation from contrast microbubbles," Journal of the Acoustical Society of America, vol. 132, no. 5, pp. 3576–3585, Nov. 2012.

[88] M. A. Averkiou, M. F. Bruce, J. E. Powers, P. S. Sheeran, and P. N. Burns, "Imaging methods for ultrasound contrast agents," Ultrasound in Medicine & Biology, vol. 46, no. 3, pp. 498–517, 2020.

[89] S. Turco, P. Frinking, R. Wildeboer, et al., "Contrast-enhanced ultrasound quantification: From kinetic modeling to machine learning," Ultrasound in Medicine & Biology, vol. 46, no. 3, pp. 518–543, 2020.

[90] A. Kosareva, L. Abou-Elkacem, S. Chowdhury, J. R. Lindner, and B. A. Kaufmann, "Seeing the invisibleultrasound molecular imaging," Ultrasound in Medicine & Biology, vol. 46, no. 3, pp. 479–497, 2020.

[91] K. Christensen-Jeffries, O. Couture, P. A. Dayton, et al., "Super-resolution ultrasound imaging," Ultrasound in Medicine & Biology, vol. 46, no. 4, pp. 865–891, 2020.

[92] M. Emmer, H. J. Vos, D. E. Goertz, A. van Wamel, M. Versluis, and N. de Jong, "Pressure-dependent attenuation and scattering of phospholipid-coated microbubbles at low acoustic pressures," Ultrasound in Medicine & Biology, vol. 35, no. 1, pp. 102–111, 2009.

[93] V. Garbin, D. Cojoc, E. Ferrari, et al., "Changes in microbubble dynamics near a boundary revealed by combined optical micromanipulation and high-speed imaging," Applied Physics Letters, vol. 90, no. 11, 2007.

[94] B. L. Helfield, B. Y. Leung, and D. E. Goertz, "The effect of boundary proximity on the response of individual ultrasound contrast agent microbubbles," Physics in Medicine & Biology, vol. 59, no. 7, p. 1721, 2014.

[95] T. Segers, N. De Jong, and M. Versluis, "Uniform scattering and attenuation of acoustically sorted ultrasound contrast agents: Modeling and experiments," Journal of the Acoustical Society of America, vol. 140, no. 4, pp. 2506–2517, 2016.

[96] J. Sijl, E. Gaud, P. J. A. Frinking, et al., "Acoustic characterization of single ultrasound contrast agent microbubbles," Journal of the Acoustical Society of America, vol. 124, no. 6, pp. 4091–4097, 2008.

[97] G. Renaud, J. Bosch, A. Van der Steen, and N. de Jong, "Chirp resonance spectroscopy of single lipid-coated microbubbles using an acoustical camera," Journal of the Acoustical Society of America, vol. 132, no. 6, EL470–EL475, 2012.

[98] S. Spiekhout, B. van Elburg, J. Voorneveld, et al., "Are monodisperse phospholipid-coated microbubbles mono-acoustic?" Applied Physics Letters, vol. 124, no. 23, 2024.

[99] O. Vincent, P. Marmottant, S. R. Gonzalez-Avila, K. Ando, and C.-D. Ohl, "The fast dynamics of cavitation bubbles within water confined in elastic solids," Soft Matter, vol. 10, no. 10, pp. 1455–1461, 2014.

[100] A. Sieber, D. Preso, and M. Farhat, "Cavitation bubble dynamics and microjet atomization near tissue-mimicking materials," Physics of Fluids, vol. 35, no. 2, p. 027 101, 2023.

[101] G. T. Bokman, O. Supponen, and S. A. Mäkiharju, "Cavitation bubble dynamics in a shear-thickening fluid," Physical Review Fluids, vol. 7, no. 2, p. 023 302, 2022.

[102] S. Izak Ghasemian, F. Reuter, and C. Ohl, "High-speed ultrasound imaging of laser-induced cavitation bubbles," Applied Physics Letters, vol. 119, no. 11, 2021.

[103] C. T. Wilson, T. L. Hall, E. Johnsen, et al., "Comparative study of the dynamics of laser and acoustically generated bubbles in viscoelastic media," Physical Review E, vol. 99, no. 4, p. 043 103, 2019.

[104] N. Blanken, B. Heiles, A. Kuliesh, et al., "PROTEUS: A physically realistic contrast-enhanced ultrasound simulator—Part I: Numerical methods," IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, 2024. DOI: 10.1109/TUFFC.2024.3427850.

[105] J. J. Faran, "Sound scattering by solid cylinders and spheres," Journal of the Acoustical Society of America, vol. 23, no. 4, pp. 405–418, 1951.

[106] H. Medwin, "Counting bubbles acoustically: A review," Ultrasonics, vol. 15, no. 1, pp. 7–13, 1977.

[107] L. E. Kinsler, A. R. Frey, A. B. Coppens, and J. V. Sanders, "Fundamentals of acoustics," John Wiley & Sons, 2000.

[108] P. Chen, A. M. Pollet, A. Panfilova, et al., "Acoustic characterization of tissue-mimicking materials for ultrasound perfusion imaging research," Ultrasound in Medicine & Biology, vol. 48, no. 1, pp. 124–142, 2022.

[109] D. H. Simpson, C. T. Chin, and P. N. Burns, "Pulse inversion doppler: A new method for detecting nonlinear echoes from microbubble contrast agents," IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, vol. 46, no. 2, pp. 372–382, 1999.

[110] J. R. Lindner, "Microbubbles in medical imaging: Current applications and future directions," Nature reviews Drug discovery, vol. 3, no. 6, pp. 527–533, 2004.

[111] A. S. McQueen and K. S. Bhatia, "Thyroid nodule ultrasound: Technical advances and future horizons," Insights into imaging, vol. 6, pp. 173–188, 2015.

[112] J. R. Lindner, J. Song, F. Xu, et al., "Noninvasive ultrasound imaging of inflammation using microbubbles targeted to activated leukocytes," Circulation, vol. 102, no. 22, pp. 2745–2750, 2000.

[113] M. van Helvert, S. Engelhard, J. Voorneveld, et al., "High-frame-rate contrast-enhanced ultrasound particle image velocimetry in patients with a stented superficial femoral artery: A feasibility study," European Radiology Experimental, vol. 6, no. 1, p. 32, 2022.

[114] A. L. Klibanov, "Microbubble contrast agents: Targeted ultrasound imaging and ultrasound-assisted drug-delivery applications," Investigative Radiology, vol. 41, no. 3, pp. 354–362, 2006.

[115] E. Quaia, "Contrast media in ultrasonography: Basic principles and clinical applications," Springer, 2005.

[116] H. J. Vos, J. D. Voorneveld, E. Groot Jebbink, et al., "Contrast-enhanced high-frame-rate ultrasound imaging of flow patterns in cardiac chambers and deep vessels," Ultrasound in Medicine & Biology, vol. 46, no. 11, pp. 2875–2890, 2020.

[117] C. Errico, J. Pierre, S. Pezet, et al., "Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging," Nature, vol. 527, no. 7579, pp. 499–502, 2015.

[118] K. Christensen-Jeffries, R. J. Browning, M.-X. Tang, C. Dunsby, and R. J. Eckersley, "In vivo acoustic super-resolution and super-resolved velocity mapping using microbubbles," IEEE Transactions on Medical Imaging, vol. 34, no. 2, pp. 433–440, 2015.

[119] A. L. Klibanov, J. J. Rychak, W. Yang, et al., "Targeted ultrasound contrast agent for molecular imaging of inflammation in high-shear flow," Contrast Media & Molecular Imaging, vol. 1, no. 6, pp. 259–266, 2006.

[120] J. R. Lindner, "Microbubbles in medical imaging: Current applications and future directions," Nature Reviews Drug Discovery, vol. 3, no. 6, pp. 527–533, 2004.

[121] Z. Fan, R. E. Kumon, and C. X. Deng, "Mechanisms of microbubble-facilitated sonoporation for drug and gene delivery," Therapeutic Delivery, vol. 5, no. 4, pp. 467–486, 2014.

[122] I. Lentacker, I. De Cock, R. Deckers, S. De Smedt, and C. Moonen, "Understanding ultrasound induced sonoporation: Definitions and underlying mechanisms," Advanced Drug Delivery Reviews, vol. 72, pp. 49–64, 2014.

[123] K. Kooiman, H. J. Vos, M. Versluis, and N. de Jong, "Acoustic behavior of microbubbles and implications for drug delivery," Advanced Drug Delivery Reviews, vol. 72, pp. 28–48, 2014.

[124] I. De Cock, G. Lajoinie, M. Versluis, S. C. De Smedt, and I. Lentacker, "Sonoprinting and the importance of microbubble loading for the ultrasound mediated cellular delivery of nanoparticles," Biomaterials, vol. 83, pp. 294–307, 2016.

[125] S. Roovers, G. Lajoinie, I. De Cock, et al., "Sonoprinting of nanoparticle-loaded microbubbles: Unraveling the multi-timescale mechanism," Biomaterials, vol. 217, p. 119 250, 2019.

[126] S. Snipstad, E. Sulheim, C. de Lange Davies, et al., "Sonopermeation to improve drug delivery to tumors: From fundamental understanding to clinical translation," Expert Opinion on Drug Delivery, vol. 15, no. 12, pp. 1249–1261, 2018.

[127] M. S. Plesset and A. Prosperetti, "Bubble dynamics and cavitation," Annual Review of Fluid Mechanics, vol. 9, no. 1, pp. 145–185, 1977.

[128] M. P. Brenner, S. Hilgenfeldt, and D. Lohse, "Single-bubble sonoluminescence," Reviews of Modern Physics, vol. 74, no. 2, p. 425, 2002.

[129] H. Oguz and A. Prosperetti, "The natural frequency of oscillation of gas bubbles in tubes," Journal of the Acoustical Society of America, vol. 103, no. 6, pp. 3301–3308, 1998.

[130] X. Geng, H. Yuan, H. Oguz, and A. Prosperetti, "The oscillation of gas bubbles in tubes: Experimental results," Journal of the Acoustical Society of America, vol. 106, no. 2, pp. 674–681, 1999.

[131] S. Qin and K. W. Ferrara, "The natural frequency of nonlinear oscillation of ultrasound contrast agents in microvessels," Ultrasound in Medicine & Biology, vol. 33, no. 7, pp. 1140–1148, 2007.

[132] A. A. Doinikov, S. Zhao, and P. A. Dayton, "Modeling of the acoustic response from contrast agent microbubbles near a rigid wall," Ultrasonics, vol. 49, no. 2, pp. 195–201, 2009.

[133] A. A. Doinikov, L. Aired, and A. Bouakaz, "Acoustic scattering from a contrast agent microbubble near an elastic wall of finite thickness," Physics in Medicine and Biology, vol. 56, no. 21, pp. 6951–6967, 2011.

[134] T. A. Hay, Y. A. Ilinskii, E. A. Zabolotskaya, and M. F. Hamilton, "Model for the dynamics of a spherical bubble undergoing small shape oscillations between parallel soft elastic layers," Journal of the Acoustical Society of America, vol. 134, no. 2, pp. 1454–1462, 2013.

[135] B. Dollet, S. M. van Der Meer, V. Garbin, N. de Jong, D. Lohse, and M. Versluis, "Nonspherical oscillations of ultrasound contrast agent microbubbles," Ultrasound in Medicine & Biology, vol. 34, no. 9, pp. 1465–1473, 2008.

[136] M. Guédra, S. Cleve, C. Mauger, P. Blanc-Benon, and C. Inserra, "Dynamics of nonspherical microbubble oscillations above instability threshold," Physical Review E, vol. 96, no. 6, p. 063 104, 2017.

[137] S. B. Feinstein, P. M. Shah, R. J. Bing, et al., "Microbubble dynamics visualized in the intact capillary circulation," Journal of the American College of Cardiology, vol. 4, no. 3, pp. 595–600, 1984.

[138] P. T. Yemane, A. K. Åslund, S. Snipstad, et al., "Effect of ultrasound on the vasculature and extravasation of nanoscale particles imaged in real time," Ultrasound in Medicine & Biology, vol. 45, no. 11, pp. 3028–3041, 2019.

[139] H. Chen, A. A. Brayman, M. R. Bailey, and T. J. Matula, "Blood vessel rupture by cavitation," Urological research, vol. 38, pp. 321–326, 2010.

[140] X. Zhao, A. Wright, and D. E. Goertz, "An optical and acoustic investigation of microbubble cavitation in small channels under therapeutic ultrasound conditions," Ultrasonics Sonochemistry, vol. 93, p. 106 291, 2023.

[141] Q. Wu, M. Gray, C. A. B. Smith, et al., "Challenges in classifying cavitation: Correlating high-speed optical imaging and passive acoustic mapping of cavitation dynamics," Journal of the Acoustical Society of America, vol. 156, no. 5, pp. 3608–3620, 2024.

[142] Y. Xie, J. Hu, W. Lei, and S. Qian, "Prediction of vascular injury by cavitation microbubbles in a focused ultrasound field," Ultrasonics Sonochemistry, vol. 88, p. 106 103, 2022.

[143] M. van Gorcum, S. Karpitschka, B. Andreotti, and J. H. Snoeijer, "Spreading on viscoelastic solids: Are contact angles selected by neumann's law?" Soft Matter, vol. 16, no. 5, pp. 1306–1322, 2020.

[144] S. A. Lippert, E. M. Rang, and M. J. Grimm, "The high frequency properties of brain tissue," Biorheology, vol. 41, no. 6, pp. 681–691, 2004.

[145] E. Sassaroli and K. Hynynen, "Resonance frequency of microbubbles in small blood vessels: A numerical study," Physics in Medicine & Biology, vol. 50, no. 22, p. 5293, 2005.

[146] I. Beekers, T. van Rooij, M. D. Verweij, et al., "Acoustic characterization of a vessel-on-a-chip microfluidic system for ultrasound-mediated drug delivery," IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, vol. 65, no. 4, pp. 570–581, 2018.

[147] G. Xu, Z. Ni, X. Chen, et al., "Acoustic characterization of polydimethylsiloxane for microscale acoustofluidics," Physical Review Applied, vol. 13, no. 5, p. 054 069, 2020.

[148] M. Versluis, D. E. Goertz, P. Palanchon, et al., "Microbubble shape oscillations excited through ultrasonic parametric driving," Physical Review E, vol. 82, no. 2, p. 026 321, 2010.

[149] N. W. Jang, S. M. Gracewski, B. Abrahamsen, T. Buttaccio, R. Halm, and D. Dalecki, "Natural frequency of a gas bubble in a tube: Experimental and simulation results," Journal of the Acoustical Society of America, vol. 126, no. 1, EL34–EL40, 2009.

[150] E. Sassaroli and K. Hynynen, "Forced linear oscillations of microbubbles in blood capillaries," Journal of the Acoustical Society of America, vol. 115, no. 6, pp. 3235–3243, 2004.

[151] C. F. Caskey, D. E. Kruse, P. A. Dayton, T. K. Kitano, and K. W. Ferrara, "Microbubble oscillation in tubes with diameters of 12, 25, and 195 microns," Applied Physics Letters, vol. 88, no. 3, 2006.

[152] T. Leighton, "The inertial terms in equations of motion for bubbles in tubular vessels or between plates," Journal of the Acoustical Society of America, vol. 130, no. 5, pp. 3333–3338, 2011.

[153] P. Thirion and R. Chasset, Pure and Applied Chemistry, vol. 23, no. 2-3, pp. 183–200, 1970.

[154] C.-D. Ohl, M. Arora, R. Ikink, et al., "Sonoporation from jetting cavitation bubbles," Biophysical Journal, vol. 91, no. 11, pp. 4285–4295, 2006.

[155] K. Kooiman, S. Roovers, S. A. Langeveld, et al., "Ultrasound-responsive cavitation nuclei for therapy and drug delivery," Ultrasound in Medicine & Biology, vol. 46, no. 6, pp. 1296–1325, 2020.

[156] J. Deprez, G. Lajoinie, Y. Engelen, S. De Smedt, and I. Lentacker, "Opening doors with ultrasound and microbubbles: Beating biological barriers to promote drug delivery," Advanced Drug Delivery Reviews, vol. 172, pp. 9–36, 2021.

[157] C.-.-D. Ohl, T. Kurz, R. Geisler, O. Lindau, and W. Lauterborn, "Bubble dynamics, shock waves and sonoluminescence," Philosophical Transactions of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences, vol. 357, no. 1751, pp. 269–294, 1999.

[158] W. Lauterborn and C.-D. Ohl, "Cavitation bubble dynamics," Ultrasonics Sonochemistry, vol. 4, no. 2, pp. 65–75, 1997.

[159] B. Boyd and S. Becker, "Numerical modelling of an acoustically-driven bubble collapse near a solid boundary," Fluid Dynamics Research, vol. 50, no. 6, p. 065 506, 2018.

[160] A. Vogel and V. Venugopalan, "Mechanisms of pulsed laser ablation of biological tissues," Chemical Reviews, vol. 103, no. 2, pp. 577–644, 2003.

[161] E.-A. Brujan, K. Nahen, P. Schmidt, and A. Vogel, "Dynamics of laser-induced cavitation bubbles near elastic boundaries: Influence of the elastic modulus," Journal of Fluid Mechanics, vol. 433, pp. 283–314, 2001.

[162] G. Sankin, F. Yuan, and P. Zhong, "Pulsating tandem microbubble for localized and directional single-cell membrane poration," Physical Review Letters, vol. 105, no. 7, p. 078 101, 2010.

[163] R. M. Sigrist, J. Liau, A. El Kaffas, M. C. Chammas, and J. K. Willmann, "Ultrasound elastography: Review of techniques and clinical applications," Theranostics, vol. 7, no. 5, p. 1303, 2017.

[164] M. A. Heinrich, R. Bansal, T. Lammers, Y. S. Zhang, R. Michel Schiffelers, and J. Prakash, "3D-bioprinted mini-brain: A glioblastoma model to study cellular interactions and therapeutics," Advanced Materials, vol. 31, no. 14, p. 1 806 590, 2019.

[165] W. Lauterborn and C.-D. Ohl, "The peculiar dynamics of cavitation bubbles," in In Fascination of Fluid Dynamics: A Symposium in Honour of Leen van Wijngaarden, Springer, 1998, pp. 63–76.

[166] A. Vogel, W. Lauterborn, and R. Timm, "Optical and acoustic investigations of the dynamics of laser-produced cavitation bubbles near a solid boundary," Journal of Fluid Mechanics, vol. 206, pp. 299–338, 1989.

[167] A. Sieber, D. Preso, and M. Farhat, "Dynamics of cavitation bubbles near granular boundaries," Journal of Fluid Mechanics, vol. 947, A39, 2022.

[168] Z. Heidary, C.-D. Ohl, and A. Mojra, "Numerical analysis of ultrasound-mediated microbubble interactions in vascular systems: Effects on shear stress and vessel mechanics," Physics of Fluids, vol. 36, no. 8, 2024.

Summary

The aim of this thesis is to advance our understanding of the dynamics of ultrasound-driven microbubbles, with a particular focus on their interaction with viscoelastic media and confined environments. Through the investigation of both theoretical models and experimental data, we investigate to deepen knowledge of microbubble resonance, scattering, and oscillation in various configurations, including bulk media and microchannels. The insights gained from these studies are intended to ultimately optimize the use of microbubbles in clinical practice, enhancing the accuracy of ultrasound imaging and the effectiveness of ultrasound-mediated therapies.

After a brief introduction to bubble dynamics and viscoelastic tissue models, Chapter 2 investigates how microbubble oscillation affects the rheological properties of the surrounding medium. The mechanical properties of tissues at the high strain rates ($10^6\ s^{-1}$) typically driven by microbubbles are largely unknown. To address this, the chapter explores the rheology of polyacrylamide (PAM) hydrogels which are used as tissue-mimicking materials, using a microbubble as a probe. By analyzing ultra-high-speed imaging data and fitting them to simulations of coated bubbles, the shear moduli and viscosities of soft and stiff PAM hydrogels were extracted and compared to their low-frequency counterparts. The strain rate strongly affects the viscosity but the elastic modulus does not show a big deviation from low-frequency measurements.

In Chapter 3, building on the understanding that the medium's elastic modulus of these hydrogels remains the same order in the $10^6\ s^{-1}$ range as low frequency range, the focus shifts to the interplay between the microbubble's phospholipid coating and the shear modulus of the viscoelastic medium. The scattered pressure from coated microbubbles in the viscoelastic medium is modeled using a Rayleigh-Plesset

type equation, the chapter examines the effects of elastic modulus on resonance frequency, harmonics generations, and sub-harmonics oscillations of microbubbles.

These simulations set the stage for Chapter 4, where the effect of the surrounding medium is further explored experimentally. We introduce a microbubble spectroscopy scattering method. With this method, we measure the correlation between the elastic modulus of hydrogels and the resonance frequency of phospholipid-coated microbubbles, quantify damping experienced by microbubbles and finally show how the gels affect the stability of the microbubbles.

Chapter 5 explores the behavior of microbubbles when confined within small blood vessels which is a more complex but more realistic configuration to represent bubbles flowing in blood capillaries. Using ultra-high-speed imaging, the chapter examines the dynamics of microbubbles in silicone polymer microchannels with diameters ranging from 15 to 100 μm. We measure the decrease in resonance frequency as the diameter of the confining channel becomes smaller as well as the damping experienced by the bubbles. The optical recording also allows us to directly have the information on bubble radial oscillation and compare it with the theoretical models.

These insights enhance our understanding of how ultrasound-driven microbubbles interact with soft matter, paving the road toward more refined clinical models and greater precision in future therapeutic applications.

Samenvatting

Het onderwerp van dit proefschrift is het verbeteren van ons begrip van de dynamica van door ultrageluid aangedreven microbellen, met een special focus op hun veranderende gedrag in weefsel en in kleine bloedvaten. Daartoe gebruiken we zowel theoretische modellen als experimenten om onze kennis te verdiepen met betrekking tot het resonantiegedrag, de verstrooiingseigenschappen en het trilgedrag van microbellen in verschillende configuraties, waaronder viscoelastische media en microcapillairen. De inzichten die uit deze studies voortkomen, zijn bedoeld om uiteindelijk het gebruik van microbellen in de klinische praktijk te optimaliseren, de nauwkeurigheid van medische beeldvorming met ultrageluid te verbeteren en de effectiviteit van door ultrageluid aangestuurde therapieën te vergroten.

Na een korte inleiding over beldynamica en visco-elastische modellen van weefsel onderzoeken we in hoofdstuk 2 hoe het trilgedrag van microbellen de rheologie van het omringende medium kan beïnvloeden. De mechanische eigenschappen van weefsels bij de hoge vervormingssnelheden (10^6 s^{-1}) die typisch door microbellen worden geïnduceerd, zijn grotendeels onbekend. Hier verkennen we de rheologie van polyacrylamide (PAM) hydrogels, die we gebruiken als modelmateriaal voor weefsel, door een enkele microbel te gebruiken als sensor. Door hogesnelheidsbeelden, opgenomen op tien miljoen beelden per sconde, te analyseren en deze te vergelijken met simulaties van gecoate bellen in water, kunnen we de elasticiteit en en viscositeit van zachte en harde PAM-hydrogels bepalen en vergelijken met waardes verkregen bij veel lagere vervormings-snelheden. We zien dat de vervormingssnelheid een sterke invloed heeft op de viscositeit, maar dat de elasticiteit niet veel afwijkt van de metingen bij lagere snelheden.

In hoofdstuk 3, voortbouwend op het inzicht dat de elasticiteit van deze hydrogels in het bereik van 10^6 s^{-1} van dezelfde orde van grootte is als in het laagfre-

quente bereik, verschuift onze focus naar de wisselwerking tussen de elasticitet van de belschil en die van het visco-elastische medium. De druk die verstrooid wordt door de gecoate microbellen in het visco-elastische medium wordt gemodelleerd door een speciale vorm van de Rayleigh-Plesset vergelijking. Dit hoofdstuk onderzoekt de effecten van de elasticiteit op de resonantiefrequentie en de vorming van een harmonische en subharmonische respons van de microbellen.

Deze simulaties vormen verder de basis voor hoofdstuk 4, waar het effect van het omliggende medium verder experimenteel wordt onderzocht. We introduceren microbel-spectroscopie, een methode die is gebaseerd op de verstrooiingseigenschappen bij verschillende aandrijffrequenties. Met deze methode meten we de correlatie tussen de elasticiteit van hydrogels en de resonantiefrequentie van fosfolipide-gecoate microbellen, kwantificeren we de demping die bij microbellen optreedt en laten we uiteindelijk zien hoe de gels invloed heeft op de stabiliteit van de microbellen.

Hoofdstuk 5 onderzoekt het gedrag van microbellen wanneer ze zich in de kleine bloedvaten van de microcirculatie bevinden. Dit is een meer complexe en tegelijk meer realistische configuratie om bellen die zich in de haarvaten bevinden te bestuderen. Met behulp van hogesnelheidsopnamen onderzoeken we in dit hoofdstuk de dynamica van microbellen in microkanalen gemaakt in een polymeer met diameters variërend van 15 tot 100 μm. We meten een afname van de resonantiefrequentie naarmate de diameter van het kanaal kleiner wordt, alsook de veranderende demping van de bellen. De tijds-opgeloste opnamen bieden ons ook directe informatie over het trilgedrag van de bellen, die we vervolgens vergelijken met onze theoretische modellen.

De inzichten bijeengebracht in dit proefschrift vergroten ons begrip van hoe door ultrageluid aangedreven microbellen interactie hebben met zachte complexe materie en helpen in de ontwikkeling van betere klinische modellen en hogere precisie in toekomstige therapeutische toepassingen.

Acknowledgements

Don't waste your time reading this. I'm writing it here so that one day, when I open this thesis, I can read the acknowledgment and remember the good memories. Let's be honest, I won't open it to look at old graphs. I'll open it to remember my advisors and colleagues.

So, let me start by saying how I ended up here. I found this position on the Glassdoor website. I had no clue what the project was about, but I applied because the word "acoustic" was in the title, and that sounded cool. I had no idea it would take over five years of my life! But even now, if I had to choose again, I'd still go for it.

Somehow, I got an interview invite. Michel sent the email. I was so stressed - it was my first ever interview! I had never spoken English with foreigners before, only with classmates in language courses. One hour before the interview, I was chilling with my dad and younger sister in the living room, drinking tea in my pajamas. Then my laptop rang! I ran, changed into something half-decent, and joined the meeting. Surprise! There were three people there, I was only expecting Michel. One of them had an accent that I was not familiar with, I didn't understand a word of him. I have no clue what he asked; I just replied with whatever I could grasp. Later, in October 2019, I met him, turns out he was my daily supervisor, Guillaume! (Yeah, his name isn't pronounced how it's written.) I had seen him on the POF website but never in the office, until I spotted him walking into Michel's room. The other person was David. He recognized me later and said, I interviewed you! Why don't you remember me?

Michel: Charismatic, scientific, patient, and fun! That's how I'd describe you. I always enjoyed our chats whether it was a casual hello or a deep scientific discussion. You're a great listener. You'd let me talk, then share your thoughts. You're one of

my best teachers. I'll never forget the first time we met. You said, "Ali, here in the Netherlands, there is no hierarchy. We call each other by first names, and we're all on the same altitude (literally, ground level too)... except when it comes to offices. New PhD students go to the cold room, ME212, and when you become a professor like me, you get a private one." Iconic.

Guillaume: I'll never forget those Monday 8:30 a.m. meetings during the 2020 corona times. Cold mornings, dark rooms, tired faces - you still showed up, even with a newborn at home. You always had brilliant solutions to my experimental problems, and when I couldn't make them work, you'd come to the rescue. But man, you were hard to reach! Weeks would pass before I could talk to you, and when I finally did, you'd tell me the project was lagging behind. You're an amazing scientist with many intelligent ideas, but without enough time to implement them. Thank you for all your supervision. I truly appreciate it.

David: You always started meetings with so much positive energy, it instantly relaxed me. I was never stressed in our discussions. I'll never forget when you told me, "Keep up the good work". That single sentence gave me such a boost. Thanks for helping me polish my thesis and for your kind feedback.

Next up: my lab buddies! The A-team. We worked together, joked around, traveled, visited cities and zoos, and even did crazy Dutch stuff like jumping over canals with a stick (Fierljeppen). Of course, once they found partners, they all left me behind. The only one still guarding the pack is Dawid, but we'll get to him later.

Saeed: It took some time for us to become friends, I wish we'd spent more time together during the pandemic. I'll never forget you coming with me to Groningen for my motorcycle examand how happy we were to skip our weekly cavitation meeting on that day. I'll never forgive whoever stole your motorcycle. It crushed your spirit - you quit the gym and left me hanging. We were such dedicated gym bros. Honestly, I shouldn't even acknowledge you! You always stole optical equipment (especially Thorlab poles), and your experiments turned the lab into a disaster zone. But our coffee breaks were legendary. I award you the title of Greatest MC Ever.

Benjamin: Your title: My BC-loyal friend. Yeah, it was awkward when you opened the toilet door, but hey, that's how friendships start, right? I was very happy when you stole Nathan's music keyboard, at least Sophie has something to play now. You're always checking in on your friends, and I really appreciate that. You were a big help in the lab - whenever the bubble machine or acoustic setup didn't work, I knew you'd fix it. We didn't write chapters together, but you were always there till

the end.

Dawid: Your title: The Best Friend. I could write a book about us! I remember our walks through the woods and farms around Witbreukseweg, jumping over canals and discussing NVIDIA stock prices and GPU release dates back in 2020! We were ahead of the curve - if only we had invested instead of just talking! Corona was tough, but you made it easier. We spent hours on Telegram, talking about physics, computers, metal music, and guitars. We even cooked together during the lockdown and made that animation for our PhD funding consortium. You're a computer geek, a guitar wizard, and a brilliant physicist.

Nathan: Your title: The First Friend. I just recently found out who my secret Santa was in 2019. Guess who spilled the beans? Saeed! Never trust him. Remember how I had half of my qualifier at your place? Guillaume told me to leave the meeting temporarily, we will add you to the meeting again, but I didn't understand the last part and left! I left to pick up a rental car for our first trip, and he called me back, like, Where the hell are you? You saved me by letting me continue the meeting at your place. Thanks for being our Netherlands tour guide and helping me understand all the theory and simulations for this thesis. I forward a kiss to you (just a cultural note - men in the Middle East kiss each other, and it's totally normal!).

Hadi: I learned weekend work ethics from you! I'll never forget that Saturday I saw you in the biolab, blocking the PDMS mold holes with your finger and your nose while talking on the phone. You needed more fingers! We used to grab beers after long days - man, I miss that. I also have to acknowledge the MATLAB codes you gave me, with epic version names like V_final, V_finallyfinal, etc. I wish you and Ava all the happiness and success!

Alessia and Kay: I couldn't have finished this thesis without your help. Working with you was so much fun. Alessia, your energy was contagious. I still remember how you'd shout "Eccole! Eccole, Aliiiiii" whenever something cool happened. Kay, thank you for helping with Chapter 2. You put a lot of time and effort into this chapter. I still haven't read the book you gave me when you graduated, I haven't forgotten it!

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I went through all the alumni on the POF website and realized I should thank everyone... but that's impossible here. A big thank you to Joanita, Gert-Wim, Martin, Thomas, and Dennis. I deeply appreciate your support - completing this thesis would've been impossible without you.

I also want to thank my motorcycle! It wasn't just a toy - it was my companion. It helped me fight loneliness and brought joy and adrenaline. I traveled to Switzerland in 2022 and many other countries during my PhD. I'm glad I spent my holidays riding. Check out my Instagram and YouTube channel: @unpavedasphalt. Don't forget to like and subscribe.

I've been away from my family for so long. When I left home, I took a mental photo of my father and sisters -Ahmad, Fati, and Mobina. We've all changed since then. I'm sorry I wasn't there when you needed me. Thank you for every sacrifice. Words cant describe how much I've missed you. Maman Zahra, I wish I could hug you again. All that's left now are tears. Now, Shirin is here to calm me down - she's like a ray of light breaking through the clouds, brightening my heart.

And finally, to my home country: I hope that one day, peace, joy, and pride return to you. You were a good home... I was not brave enough to protect you from the darkness.

In this part, imagine "The River" music from Bruce Springsteen is playing.